AL4000 DMX512 Processors



Revision 5.0 September 2010

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C O N T E N T S

DMX512 Processor Range Overview..... 8

AL4002 Three Phase Dimmer 11

AL4005 Demux Controller..... 22

AL4006/9 Analogue & Relay Controller 34

AL4007 Demux Controller..... 41

AL4008 Relay Controller 48

AL4016 Single Phase Dimmer..... 54

AL4020 Multiplex Controller..... 63

AL4100 Servo Motion Controller 70

AL4400 PWM Dimmer Processor..... 80

AL4800 Moving Lamp Processor 85

AL2001/AL3001 Three Phase Dimmer 98

I N T R O D U C T I O N

Overview

Artistic Licence offers a range of products tailored to the specific requirements of product manufacturers in the Lighting Industry.

The AL4000 DMX512 Processor (aka Dimmer Chips) range include microprocessor solutions for a wide selection of lighting control products. The range is usually supplied at chip level for integration at the design phase. We can, however, supply board level solutions or indeed entire turnkey product design.

Originally aimed at small companies without the R&D facilities to develop microprocessor equipment, we now supply major lighting manufacturers around the world.

We guarantee that a Dimmer Chip will simplify your next design.

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AL4002

Three Phase Dimmer

The AL4002 is a 68 pin PLCC device. It provides all the control features required for a medium specification three phase digital dimmer.
Both analogue and DMX512 inputs are supported. Features include four dimmer curves, preheat, consecutive or alternate phase outputs, good data indication and fan speed control.
DMX base address is set by external thumbwheel inputs.

AL4005

Demux Controller

The AL4005 is a 40 pin DIL device. It provides all the control features required to develop a 96 channel DMX512 to analogue converter.
DMX base address is set by external thumbwheel inputs.

AL4006/9

Analogue & Relay Controller

The AL4006 and AL4009 are 40 pin DIL devices. They both provide two analogue and sixteen switched outputs. Excellent for automating smoke machines etc.
DMX base address is set by external thumbwheel inputs.

AL4007

Demux Controller

The AL4007 is a 40 pin DIL device. It provides all the control features required to develop a 72 channel DMX512 to analogue converter.
DMX base address is set by external thumbwheel inputs.

AL4008

Relay Controller

The AL4008 is a 40 pin DIL device. It provides all the control features required to develop a 24 channel DMX512 to switched relay converter. Individual DMX channels are used to set the state of each relay.
DMX base address is set by external thumbwheel inputs.

AL4016

Single Phase Dimmer

The AL4016 is a 40 pin DIL device. It provides all the control features required for a medium specification, sixteen channel, single phase digital dimmer.
DMX512 input only is supported.
DMX base address is set by external thumbwheel inputs.

AL4020

Multiplex Controller

The AL4020 is a 40 pin DIL device. It provides a DMX output that represents the htp merge of received DMX512 and up to 72 analogue inputs. The device can also be used as a simple analogue to DMX512 converter.
DMX base address is set by external thumbwheel inputs.

AL4100

Dual Servo Controller

The AL4100 is a 40 pin dip device. It receives DMX512 and uses the information to position control two DC servo motors. Perfect for colour scrollers, colour wheels, moving yokes, moving lamps and animatronics.

AL4200

Up Down Relay Controller

The AL4200 provides all the control features required for a four channel, forward - reverse relay driver.
Applications include drape release and general purpose animatronic effects.
The AL4200 is the core technology used in the DMX-Kabuki.

AL4400

PWM Dimmer Controller

The AL4400 provides all the control features required for a high specification DC - PWM dimmer.
Applications include low voltage lamp dimming, LED, LEP, EL and cold cathode colour mixing.
The AL4400 is the core technology used in the Power-Pipe and Colour-Fill.

AL4800

PWM Moving Lamp Controller

The AL4800 provides all the control features required for a medium specification moving lamp or moving yoke.
The basic configuration provides micro-stepping (high resolution) outputs for pan and tilt. Pin inputs allow 2 settings for pan motor type and 4 settings for tilt motor type.
The colour wheel and gobo wheel motors are also micro-stepping motors and assumed to be continuously rotating with inputs for a home sensor.
The iris motor is also a wave drive stepper motor channel with end stop sensor.
Applications include moving lamps, moving yokes, colour wheels and general purpose animatronics.
Custom variations of this part can be produced subject to order quantity.

AL2001 / AL3001

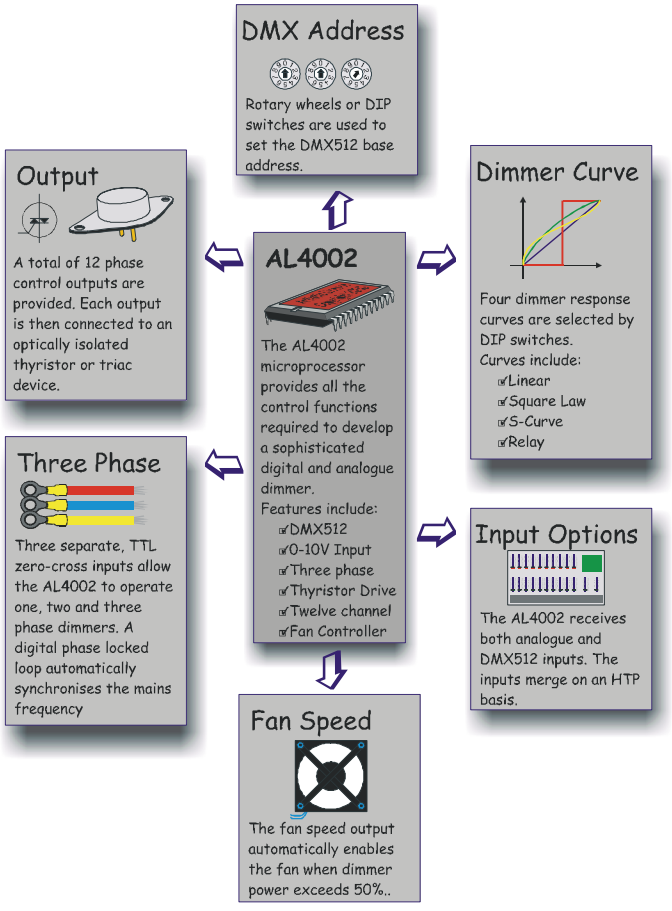
Three Phase Dimmer

This two chip solution provides the engine for a powerful three phase DMX and analogue digital dimmer. Features include LCD drive, keypad input, analogue and DMX512 input, dimmer curves, channel preheat and all data stored in flash ram.
It provides all the control features required for a high specification three phase digital dimmer.

Overview

The AL4002 is a 68 pin PLCC device. It provides all the control features required for a medium specification three phase digital dimmer.
Both analogue and DMX512 inputs are supported.
Key features:

- 12 Phase position modulated outputs
- Automatic mains frequency and phase tracking
- Four dimmer curves
- Preheat
- Consecutive or alternate phase outputs
- Good data indication
- Fan speed control
- DMX base address is set by external thumbwheel inputs
- Individual channel test at 50% & 100%



Pin Functions

Pin	Function	In / Out
1	NC	-
2	VCC	I
3	Output 1 (PH1-1)	O
4	BCD input (HUN1)	I
5	BCD input (HUN2)	I
6	BCD input (HUN4)	I
7	BCD input (HUN8)	I
8	Output 12 (PH3-4)	O
9	Output 11 (PH3-3)	O
10	Output 10 (PH3-2)	O
11	Output 9 (PH3-1)	O
12	BCD input (TEN8)	I
13	BCD input (TEN4)	I
14	BCD input (TEN2)	I
15	BCD input (TEN1)	I
16	RESET (Active low)	I
17	TEST (Connect to 0V)	I
18	NC	-
19	GND	I
20	VCC	I
21	NC	-
22	NC	-
23	Analogue Bus D (AN-D)	I
24	Analogue Bus C (AN-C)	I
25	NC	-
26	NC	-
27	NC	-
28	NC	-
29	VCC (Analogue)	I
30	GND (Analogue)	I
31	Analogue Bus A (AN-A)	I
32	Analogue Bus B (AN-B)	I
33	Analogue Address (ANAD1)	O
34	Analogue Address (ANAD0)	O
35	NC	-
36	GND	I
37	VCC	I
38	Zero Cross Input. Active Low (ZC3)	I
39	NC	-
40	Curve select (CURVE1)	I

Pin Functions

Pin	Function	In / Out
41	Curve select (CURVE0)	I
42	NC	-
43	TTL DMX512 Input (RX)	I
44	Phase Output Sequence (ORDER)	I
45	NC	-
46	NC	-
47	BCD input (UNIT2)	I
48	BCD input (UNIT1)	I
49	VCC	I
50	Crystal Connection (OSC1)	I
51	Crystal Connection (OSC2)	O
52	NC	-
53	GND	I
54	Good Data, Lo=Good (GOOD)	O
55	BCD input (UNIT4)	I
56	BCD input (UNIT8)	I
57	Fan drive, Hi=Fast (HIFAN)	O
58	Zero Cross Input. Active Low (ZC2)	I
59	Zero Cross Input. Active Low (ZC1)	I
60	AUX	I
61	Output 8 (PH2-4)	O
62	Output 7 (PH2-3)	O
63	Output 6 (PH2-2)	O
64	Output 5 (PH2-1)	O
65	Output 4 (PH1-4)	O
66	Output 3 (PH1-3)	O
67	Output 2 (PH1-2)	O
68	GND	I

Phase Outputs

The AL4002 phase outputs are synchronised to the three zero cross input signals over a frequency range of 45 to 65Hz. The output phase trigger position is defined by the highest value of the DMX512 input, the analogue input and the preheat level. The output is additionally modified by the output curve selected.

Phase Sequence

The AL4002 is designed such that it can be used for both standard configurations of three phase dimmer. The two conventions differ in the sequence of the phase outputs and is selected by tying the ORDER pin either Hi or Low:

DMX, Analogue or Test Input Channel	Consecutive (ORDER=0) Controlled Output	Alternate (ORDER=1) Controlled Output
1	PH1-1	PH1-1
2	PH1-2	PH2-1
3	PH1-3	PH3-1
4	PH1-4	PH1-2
5	PH2-1	PH2-2
6	PH2-2	PH3-2
7	PH2-3	PH1-3
8	PH2-4	PH2-3
9	PH3-1	PH3-3
10	PH3-2	PH1-4
11	PH3-3	PH2-4
12	PH3-4	PH3-4

Dimmer Curves

The phase output relationship to the control input can be globally modified to one of four response curves. The options are:

1. Linear - The output phase varies linearly with the input.
2. Square - The output power varies linearly with the input.
3. S curve - A modified form of Square with greater control in the centre of the range.
4. Relay - The output switches to full when the input exceeds 25%.

The curves are selected as follows:

CURVE1	CURVE0	Output
0	0	Linear
0	1	Square
1	0	S Curve
1	1	Relay

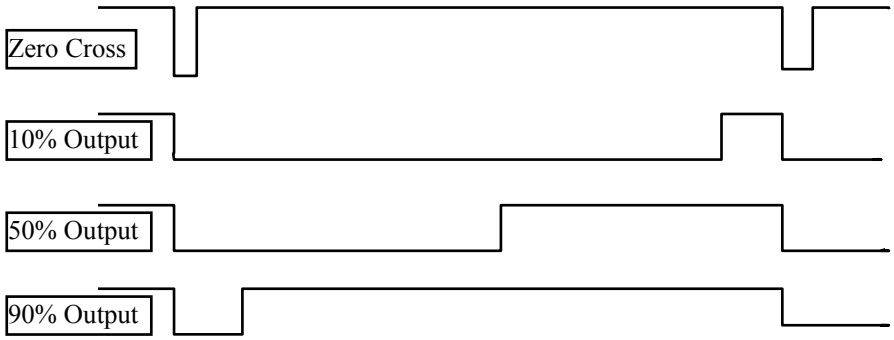
Zero Cross Inputs

The three zero cross inputs (ZC1-3) are used to provide the mains synchronisation. The inputs are active low, that is the signal should pulls low at the zero crossing point of each respective phase.
For single phase operation the three inputs should be commoned.

It is imperative that the rise time of the signal is greater that 1uS. Failure to adhere to this specification may cause output inaccuracy.
The minimum low period of the zero cross signal is 10uS.
The allowed frequency range is 90Hz to 125Hz. Frequencies outside this range will cause non linear output.

Output timing

The following examples show the relationship between Output, Zero cross for three input values.



DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.

Test Mode

The BCD inputs are also used to enter test mode.
When Test mode is active, the DMX512 input is disabled, but the Analogue inputs remain active.
When the Hundred inputs are set to either '6' or '7' Test mode is active. The Hundred input sets the intensity, while the Ten and Units inputs select the channel.
The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input	Analogue Input
0	0	0	All Off	Off	Off
0	0	1	Controlled by inputs	On	On
5	1	2	Controlled by inputs	On	On
7	0	1	Output 1 @ 100%	Off	On
6	0	1	Output 1 @ 50%	Off	On
7	0	9	Output 9 @ 100%	Off	On
6	1	2	Output 12 @ 50%	Off	On

Analogue Inputs

The Analogue inputs are decoded by the external multiplexers. Designs which do not require analogue inputs or preheat may ground the AN-X inputs. The Preheat input value is applied to all channels. No analogue input voltage may exceed 5V. The following table shows the decoding.

ANAD1	ANAD	AN-A	AN-B	AN-C	AN-D

	0				
0	0	Input 1	Input 5	Input 9	Spare 1
0	1	Input 2	Input 6	Input 10	Spare 2
1	0	Input 3	Input 7	Input 11	Spare 3
1	1	Input 4	Input 8	Input 12	Preheat

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Fan Speed

The HIFAN signal is driven Hi when any six outputs exceed 50% power. The signal can be used to increase the fan speed when the Dimmer is handling greater than 50% power.

Power Supply

The AL4002 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the Zero Cross and Crystal connections.

Crystal

The AL4002 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4002.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4002 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

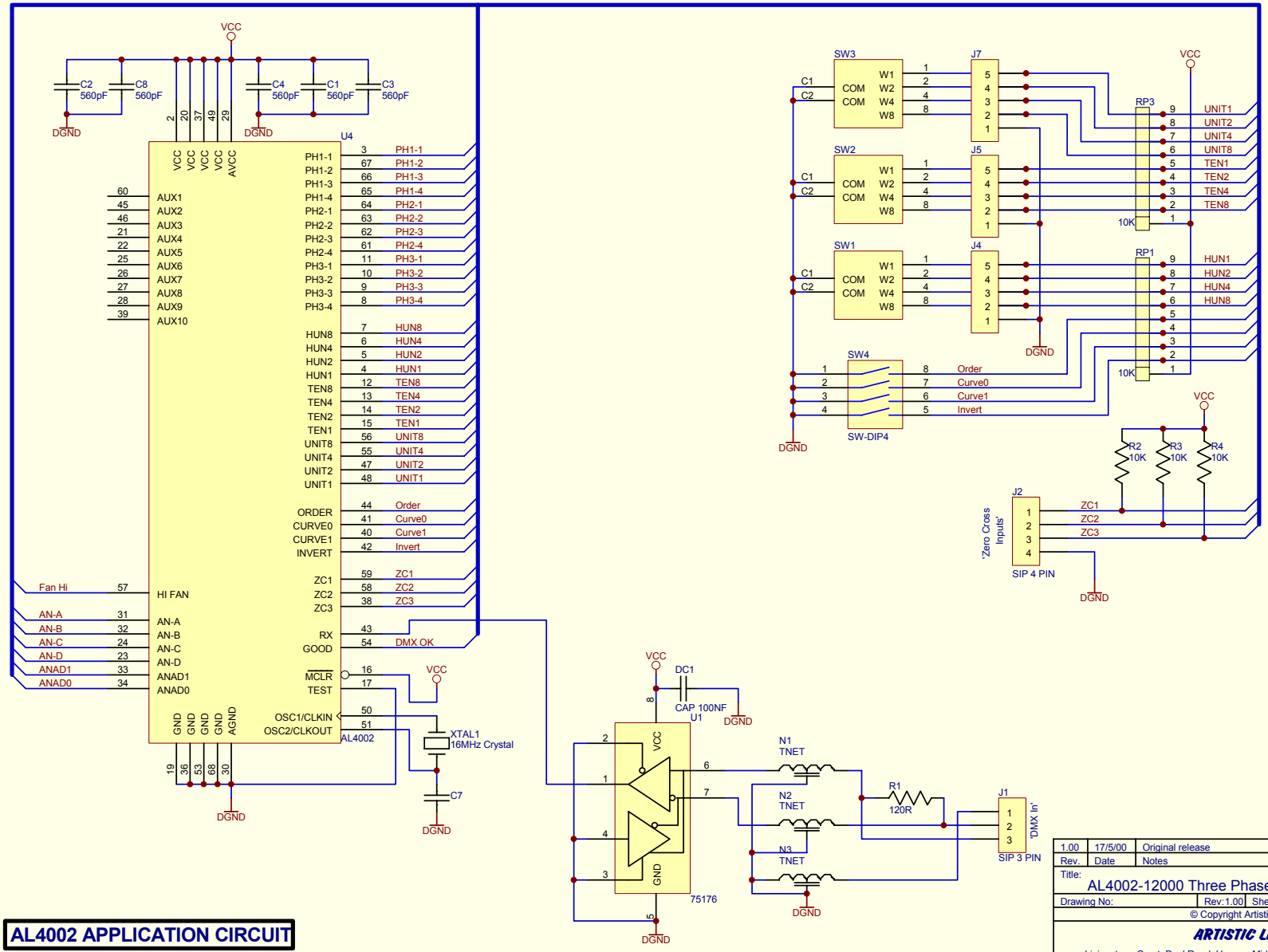
DMX512 Isolation

In order to ensure that DMX512 works reliably (this relates to all

systems) it is very important to ensure that the cable screen is connected to pin 1 and that pin 1 is not connected to mains earth. This simple precaution solves most DMX512 problems.
It is also a good idea to isolate the DMX512 input circuitry.

Application

The following circuit shows the AL4002 application drawing. This is also the circuit diagram for the evaluation PCB.



AL4002 APPLICATION CIRCUIT

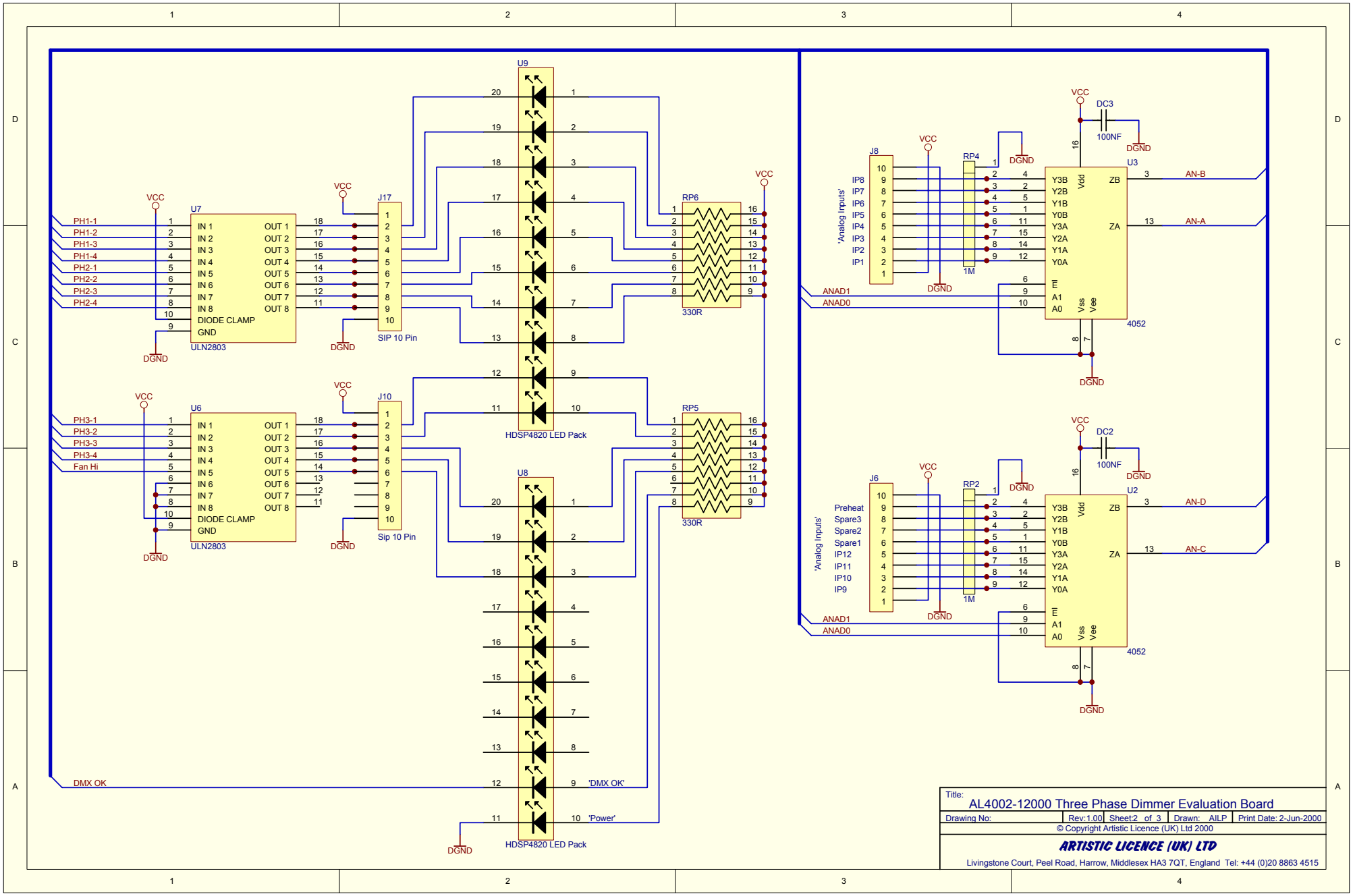
Rev.	Date	Notes
1.00	17/5/00	Original release

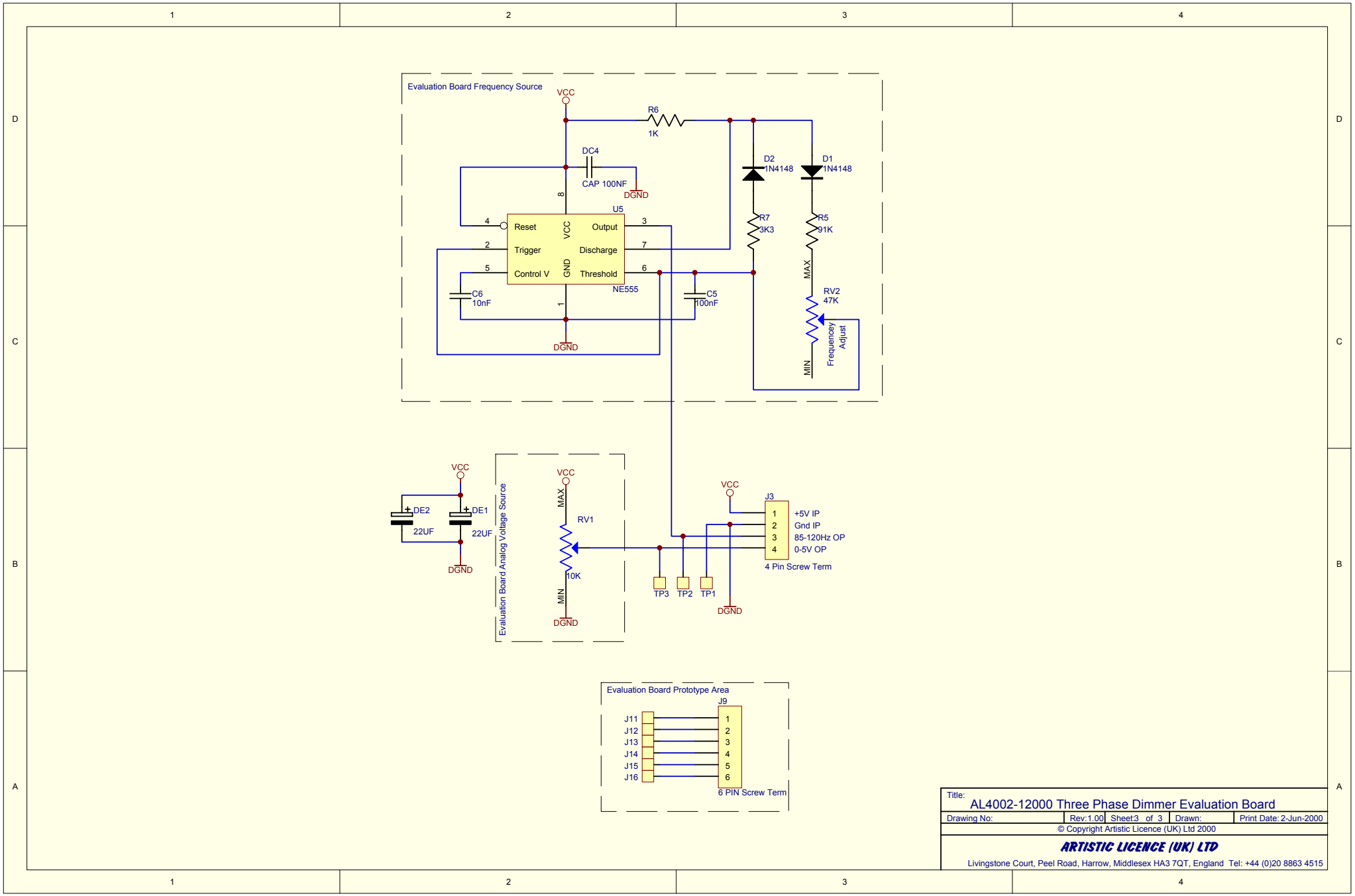
Title: AL4002-12000 Three Phase Dimmer Evaluation Board

Drawing No: | Rev:1.00 | Sheet:1 of 3 | Drawn: AILP | Print Date: 2-Jun-2000

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U9
U7
U6
U8
J17
J10
RP6
RP5
VCC
DGND
PH1-1
PH1-2
PH1-3
PH1-4
PH2-1
PH2-2
PH2-3
PH2-4
PH3-1
PH3-2
PH3-3
PH3-4
Fan Hi
IN 1
IN 2
IN 3
IN 4
IN 5
IN 6
IN 7
IN 8
OUT 1
OUT 2
OUT 3
OUT 4
OUT 5
OUT 6
OUT 7
OUT 8
DIODE CLAMP
GND
ULN2803
SIP 10 Pin
Sip 10 Pin
330R
HDSP4820 LED Pack
DMX OK
'DMX OK'
'Power'
J8
J6
RP4
RP2
DC3
DC2
100NF
1M
U3
U2
'Analog Inputs'
IP8
IP7
IP6
IP5
IP4
IP3
IP2
IP1
Preheat
Spare3
Spare2
Spare1
IP12
IP11
IP10
IP9
ANAD1
ANAD0
Y3B
Y2B
Y1B
Y0B
Y3A
Y2A
Y1A
Y0A
Vdd
Vss
Vee
ZB
ZA
AN-B
AN-A
AN-D
AN-C
4052
Title:
AL4002-12000 Three Phase Dimmer Evaluation Board
Drawing No:
Rev:1.00
Sheet:2 of 3
Drawn: AILP
Print Date:2-Jun-2000
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Evaluation Board Frequency Source
VCC
R6
1K
DC4
CAP 100NF
DGND
D2
1N4148
D1
1N4148
U5
Reset
Trigger
Control V
VCC
GND
Output
Discharge
Threshold
NE555
R7
3K3
R5
91K
MAX
RV2
47K
Frequency Adjust
MIN
C6
10nF
C5
100nF
DE2
22UF
DE1
22UF
Evaluation Board Analog Voltage Source
RV1
10K
J3
+5V IP
Gnd IP
85-120Hz OP
0-5V OP
4 Pin Screw Term
TP3
TP2
TP1
Evaluation Board Prototype Area
J9
J11
J12
J13
J14
J15
J16
6 PIN Screw Term
Title:
AL4002-12000 Three Phase Dimmer Evaluation Board
Drawing No:
Rev:1.00
Sheet3 of 3
Drawn:
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Bill Of Materials

The following components are required for the evaluation PCB. Please note that some component values may require modification subject to the intended application.

Designator	Part Type
C1	560pF
C2	560pF
C3	560pF
C4	560pF
C5	100nF
C6	10nF
C7	
C8	560pF
D1	1N4148
D2	1N4148
DC1	CAP 100NF
DC2	100NF
DC3	100NF
DC4	CAP 100NF
DE1	22UF
DE2	22UF
J1	SIP 3 PIN
J10	Sip 10 Pin
J11	TESTPOINT
J12	TESTPOINT
J13	TESTPOINT
J14	TESTPOINT
J15	TESTPOINT
J16	TESTPOINT
J17	SIP 10 Pin
J2	SIP 4 PIN
J3	4 Pin Screw Term
J4	CONN SIP 5PIN
J5	CONN SIP 5PIN
J6	CONN SIP 10PIN
J7	CONN SIP 5PIN
J8	CONN SIP 10PIN
J9	6 PIN Screw Term
N1	TNET
N2	TNET
N3	TNET
R1	120R
R2	10K
R3	10K
R4	10K
R5	91K
R6	1K
R7	3K3
RP1	10K
RP2	1M
RP3	10K
RP4	1M
RP5	330R
RP6	330R

Designator	Part Type
RV1	10K
RV2	47K
SW1	BCD-HOZ
SW2	BCD-HOZ
SW3	BCD-HOZ
SW4	SW-DIP4
TP1	TESTPOINT
TP2	TESTPOINT
TP3	TESTPOINT
U1	75176
U2	4052
U3	4052
U4	AL4002
U5	NE555
U6	ULN2803
U7	ULN2803
U8	HDSP4820 LED Pack
U9	HDSP4820 LED Pack
XTAL1	16MHz Crystal

THE AL4005 DEMUX PROCESSOR

Overview

The AL4005 is designed to receive the DMX512 protocol and provide all the control signals necessary to generate 96 channels of analogue control signals.

The AL4005 is suitable for the development of DMX512 to Analogue Decoders and also provides an elegant solution to the DMX512 upgrade of existing analogue dimmers.

The AL4005 is supplied as a 40 pin DIL device.

Key features:

- DMX512 input
- Good data received indicator
- Parallel output for external Digital to Analogue conversion
- Individual channel test at 50% and 100%
- DMX base address is set by external thumbwheel inputs

Pin Functions

Pin	Function	In/Out
1	VCC 5V	-
2	Chip Select 0	O
3	Chip Select 1	O
4	Chip Select 2	O
5	Chip Select 3	O
6	Chip Select 4	O
7	Chip Select 5	O
8	DAC Select	O
9	TRIG - Scope Trigger	O
10	GND	-
11	BCD Unit 1	I
12	BCD Unit 2	I
13	BCD Unit 4	I
14	BCD Unit 8 (msb)	I
15	BCD Ten 1	I
16	BCD Ten 2	I
17	BCD Ten 4	I
18	BCD Ten 8 (msb)	-
19	Osc1 (16mhz crystal)	-
20	Osc2 (16mhz crystal)	-
21	NC	-
22	DMX512 TTL Receive	I
23	NC	-
24	Good Rx (Open Collector, Lo=Good)	O
25	NC	-
26	NC	-
27	Test (Connect to GND)	I
28	BCD Hundred 4	I
29	BCD Hundred 2	I
30	BCD Hundred 1	I
31	GND	-
32	MCLR (Connect to VCC)	I
33	D7 (msb dac data)	O
34	D6	O
35	D5	O
36	D4	O
37	AD3 (msb channel address)	O
38	AD2	O
39	AD1	O
40	AD0	O

Circuit Operation

The AL4005 chip selects and address outputs are used to address a single demultiplex channel from the 96 available. The Analogue multiplex output is then routed through that channel to the sample and hold circuit.

The application drawings that follow show this process using 4067 16 channel de-multiplexors and TL084 op-amps. The

TL084 has been chosen in order to provide a very high input impedance for the sample and hold circuit.

It should be noted that the address and chip select signals are all converted to CMOS voltages, in order to drive the demultiplex circuitry at a 15V supply.

The analogue output of the DAC is amplified to produce the analogue level required by the demultiplexors.

DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.

Test Mode

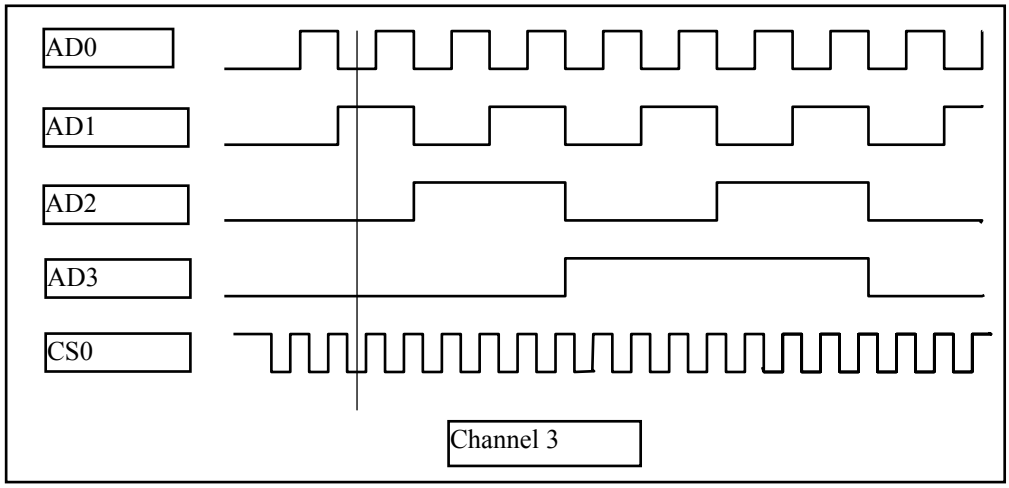
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When Test mode is active, the DMX512 input is disabled.
When the Hundred inputs are set to either '6' or '7' Test mode is active. The Hundred input sets the intensity, while the Ten and Units inputs select the channel.

The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input
0	0	0	All Off	Off
0	0	1	Controlled by inputs	On
5	1	2	Controlled by inputs	On
7	0	1	Output 1 @ 100%	Off
6	0	1	Output 1 @ 50%	Off
7	6	9	Output 69 @ 100%	Off
6	1	2	Output 12 @ 50%	Off

Output timing

The signals AD0-AD3 and CS0-CS5 are used to address the demultiplexor circuits. The signals continue to operate independent of whether the DMX512 input exists.
The cycle time will vary depending on the refresh rate of the received DMX512.



DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Power Supply

The AL4005 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

The AL4005 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

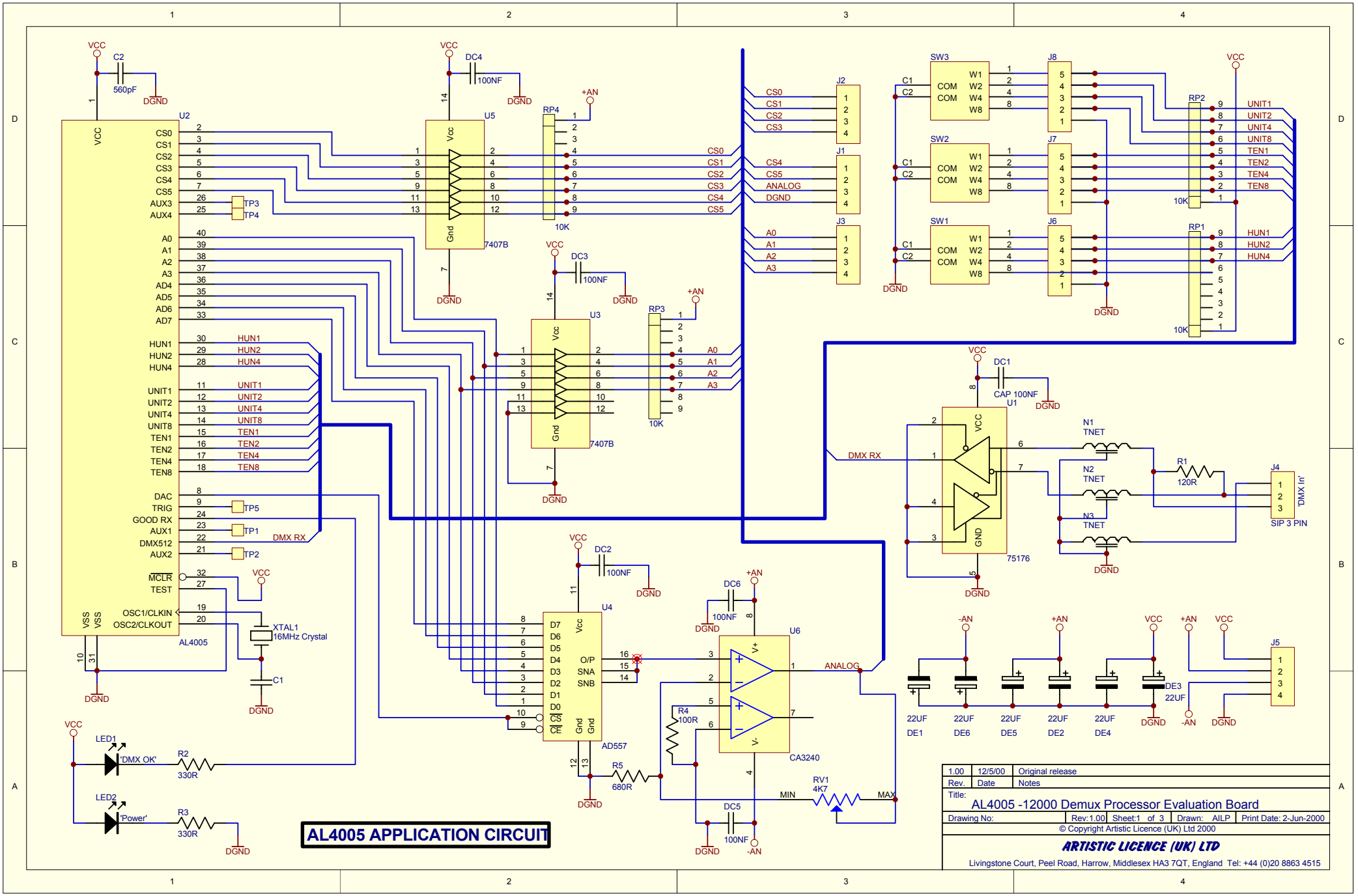
EMC

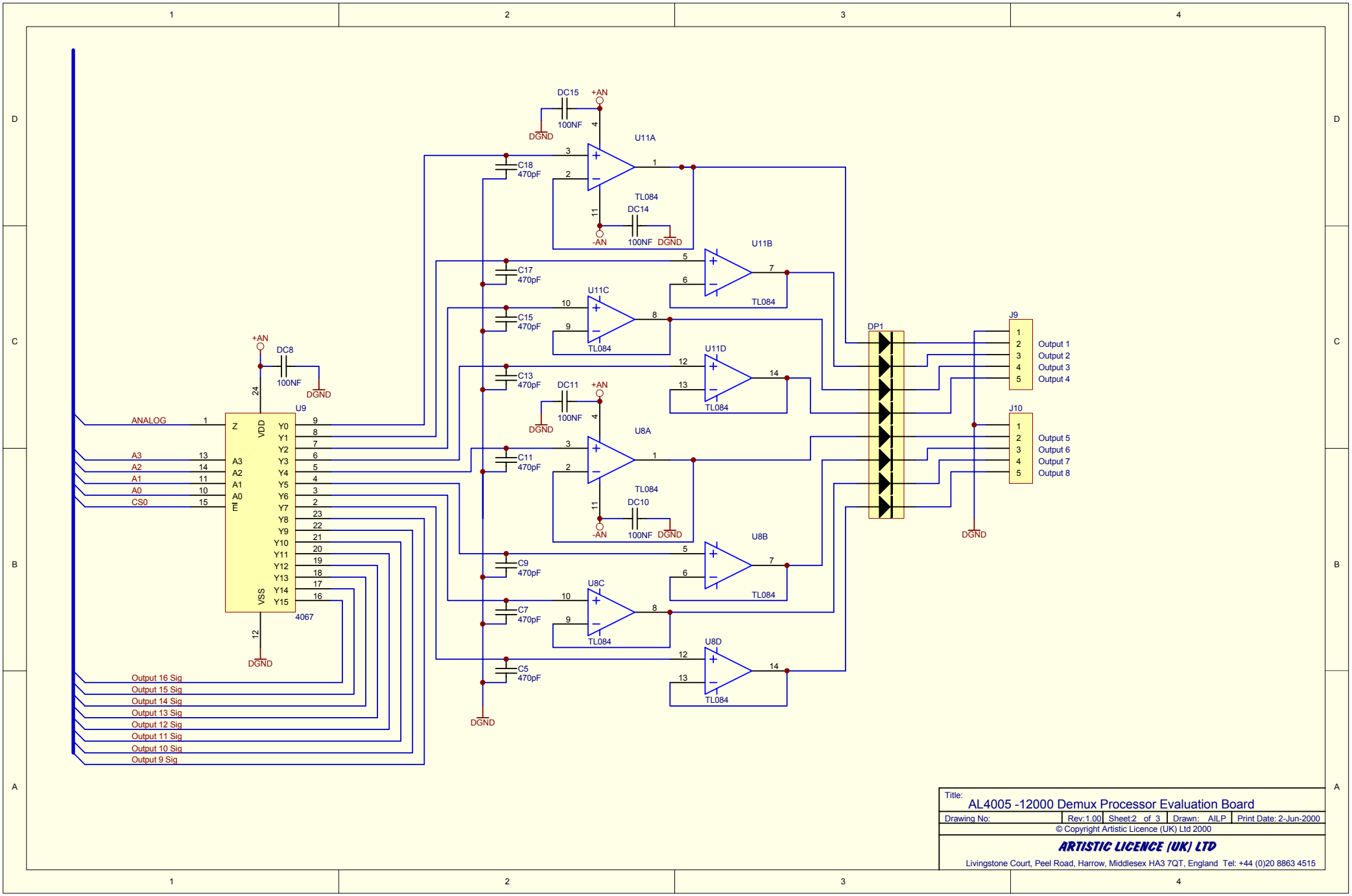
There are two major considerations in reducing the electromagnetic radiation generated by the AL4005.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4005 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

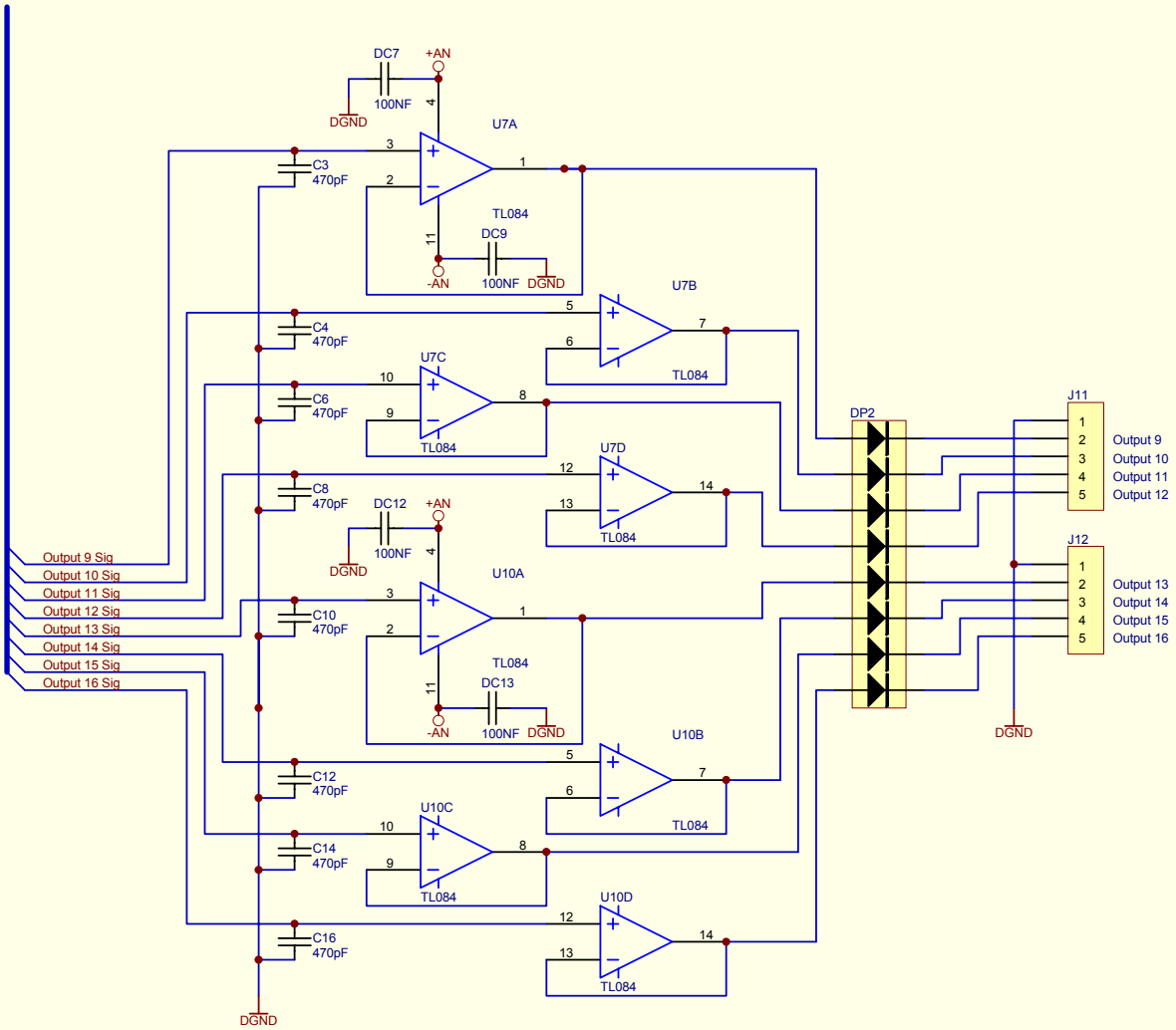
Application

The following circuit shows the AL4005 application drawing. This is also the circuit diagram for the evaluation PCB.

The evaluation card is designed to output 16 channels, however further PCB's can be added to extend the number of outputs.


AL4005 APPLICATION CIRCUIT
AL4005
7407B
AD557
CA3240
75176
SIP 3 PIN
'DMX In'
'DMX OK'
'Power'
XTAL1
16MHz Crystal
DMX RX
ANALOG
MIN
MAX
1.00 | 12/5/00 | Original release
Rev. | Date | Notes
Title: AL4005 -12000 Demux Processor Evaluation Board
Drawing No: | Rev:1.00 | Sheet:1 of 3 | Drawn: AILP | Print Date: 2-Jun-2000
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DC15
+AN
100NF
DGND
U11A
C18
470pF
TL084
DC14
-AN
100NF
DGND
U11B
C17
470pF
U11C
C15
470pF
TL084
+AN
DC8
100NF
DGND
U11D
C13
470pF
DC11
+AN
100NF
DGND
U8A
C11
470pF
TL084
DC10
-AN
100NF
DGND
U8B
C9
470pF
U8C
C7
470pF
TL084
U8D
C5
470pF
TL084
DGND
U9
ANALOG
A3
A2
A1
A0
CS0
Z
VDD
VSS
E
A3
A2
A1
A0
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
Y8
Y9
Y10
Y11
Y12
Y13
Y14
Y15
4067
DGND
Output 16 Sig
Output 15 Sig
Output 14 Sig
Output 13 Sig
Output 12 Sig
Output 11 Sig
Output 10 Sig
Output 9 Sig
DP1
J9
Output 1
Output 2
Output 3
Output 4
J10
Output 5
Output 6
Output 7
Output 8
DGND
Title:
AL4005 -12000 Demux Processor Evaluation Board
Drawing No:
Rev:1.00
Sheet:2 of 3
Drawn: AILP
Print Date: 2-Jun-2000
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Title: AL4005 -12000 Demux Processor Evaluation Board

Drawing No:	Rev:1.00	Sheet:3 of 3	Drawn: AILP	Print Date: 2-Jun-2000

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Bill Of Materials

The following components are required for the evaluation PCB. Please note that some component values may require modification subject to the intended application.

Designator	Part Type
C1	
C10	470pF
C11	470pF
C12	470pF
C13	470pF
C14	470pF
C15	470pF
C16	470pF
C17	470pF
C18	470pF
C2	560pF
C3	470pF
C4	470pF
C5	470pF
C6	470pF
C7	470pF
C8	470pF
C9	470pF
DC1	CAP 100NF
DC10	100NF
DC11	100NF
DC12	100NF
DC13	100NF
DC14	100NF
DC15	100NF
DC2	100NF
DC3	100NF
DC4	100NF
DC5	100NF
DC6	100NF
DC7	100NF
DC8	100NF
DC9	100NF
DE1	22UF
DE2	22UF
DE3	22UF
DE4	22UF
DE5	22UF
DE6	22UF
DP1	Diode Pack
DP2	Diode Pack

Designator	Part Type
J1	CONN SIP 4PIN
J10	CONN SIP 5PIN
J11	CONN SIP 5PIN
J12	CONN SIP 5PIN
J2	CONN SIP 4PIN
J3	CONN SIP 4PIN
J4	CONN SIP 3PIN
J5	CONN SCREW 4PIN
J6	CONN SIP 5PIN
J7	CONN SIP 5PIN
J8	CONN SIP 5PIN
J9	CONN SIP 5PIN
LED1	LED 3MM
LED2	LED 3MM
N1	Tnet 150uHNET
N2	Tnet 150uHNET
N3	Tnet 150uHNET
R1	120R
R2	330R
R3	330R
R4	100R
R5	680R
RP1	10K
RP2	10K
RP3	10K
RP4	10K
RV1	4K7
SW1	BCD-HOZ
SW2	BCD-HOZ
SW3	BCD-HOZ
TP1	TESTPOINT
TP2	TESTPOINT
TP3	TESTPOINT
TP4	TESTPOINT
TP5	TESTPOINT
U1	75176
U10	TL084
U11	TL084
U2	AL4005
U3	7407B
U4	AD557
U5	7407B
U6	CA3240
U7	TL084
U8	TL084
U9	4067
XTAL1	16MHz Crystal

THE AL4006/9 ANALOGUE AND RELAY PROCESSOR

Overview

The AL4006 is designed to receive the DMX512 protocol and provide two analogue outputs and sixteen switch outputs.

Applications include:

- Single and dual channel dimmer control
- Smoke machine remote control
- Colour scroller control
- Motor speed control
- Robotics

The AL4006 - AL4009 are supplied as 40 pin DIL devices.

Key features:

- DMX512 input
- Good data received indicator
- 16 relay drive outputs
- Individual channel test at 50% and 100%
- DMX base address is set by external thumbwheel inputs

Pin Functions

Pin	Function	In/Out
1	VCC 5V	-
2	SW 1	O
3	SW 2	O
4	SW 3	O
5	SW 4	O
6	SW 5	O
7	SW6	O
8	SW 7	O
9	SW 8	O
10	GND	-
11	BCD Unit 1	I
12	BCD Unit 2	I
13	Analogue 1	O
14	Analogue 2	O
15	BCD Ten 1	I
16	BCD Ten 2	I
17	BCD Ten 4	I
18	BCD Ten 8 (msb)	I
19	Osc1 (16mhz crystal)	-
20	Osc2 (16mhz crystal)	-
21	NC	-
22	DMX512 TTL Receive	I
23	NC	-
24	Good Rx (Open Collector, Lo=Good)	O
25	BCD Unit 8	I
26	BCD Unit 4	I
27	Test (Connect to GND)	I
28	BCD Hundred 4	I
29	BCD Hundred 2	I
30	BCD Hundred 1	I
31	GND	-
32	MCLR (Connect to VCC)	I
33	SW 16	O
34	SW 15	O
35	SW14	O
36	SW 13	O
37	SW 12	O
38	SW11	O
39	SW 10	O
40	SW 9	O

Circuit Operation

The AL4006 provides sixteen outputs (SW1-SW16) for driving relays or other binary devices. The relay outputs are controlled by the levels of sixteen consecutive DMX512 channels, starting at the base address.
The SW0 output will switch low when the base address channel exceeds
50%.
(The AL4009 will switch high when the base address channel exceeds 50%.)
The maximum sink current of the relay outputs is 35mA. For most relays
an external darlington driver package will be required (See Application
drawing). An output level of 0V represents relay on. (The AL4009 has reversed polarity relay outputs, so an output level of 5V represents relay on).

The AL4006 provides two analogue outputs which represent the DMX512 levels of two consecutive DMX512 channels, starting at the base address.

The outputs are TTL Pulse width modulated at a frequency of 15.625khz. These outputs can be used directly for motor speed control, or integrated and amplified for an analogue 0-10V output. The maximum sink current of the analogue outputs is 35mA.

DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.
The relationship between DMX512 channels and outputs is shown below:

DMX512 Channel	Relay controlled	Analogue Output controlled
DMX Base + 0	SW 1	Analogue 1
DMX Base + 1	SW 2	Analogue 2
DMX Base + 2	SW 3	N/A
DMX Base + 3	SW 4	N/A
DMX Base + 4	SW 5	N/A
DMX Base + 5	SW 6	N/A
DMX Base + 6	SW 7	N/A
DMX Base + 7	SW 8	N/A
DMX Base + 8	SW 9	N/A
DMX Base + 9	SW 10	N/A
DMX Base + 10	SW 11	N/A
DMX Base + 11	SW 12	N/A
DMX Base + 12	SW 13	N/A
DMX Base + 13	SW 14	N/A
DMX Base + 14	SW 15	N/A
DMX Base + 15	SW 16	N/A

Test Mode

The BCD inputs are also used to enter test mode.
When Test mode is active, the DMX512 input is disabled.
When the Hundred inputs are set to either '6' or '7' Test mode is active. The Hundred input sets the intensity, while the Ten and Units inputs select the channel.

The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input
0	0	0	All Off	Off
0	0	1	Controlled by inputs	On
5	1	2	Controlled by inputs	On
7	0	1	SW 1 On, Analogue 1 @ 100%	Off
6	0	1	Analogue 1 @ 50%	Off
7	0	9	SW On	Off

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Power Supply

The AL4006/9 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

The AL4006/9 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

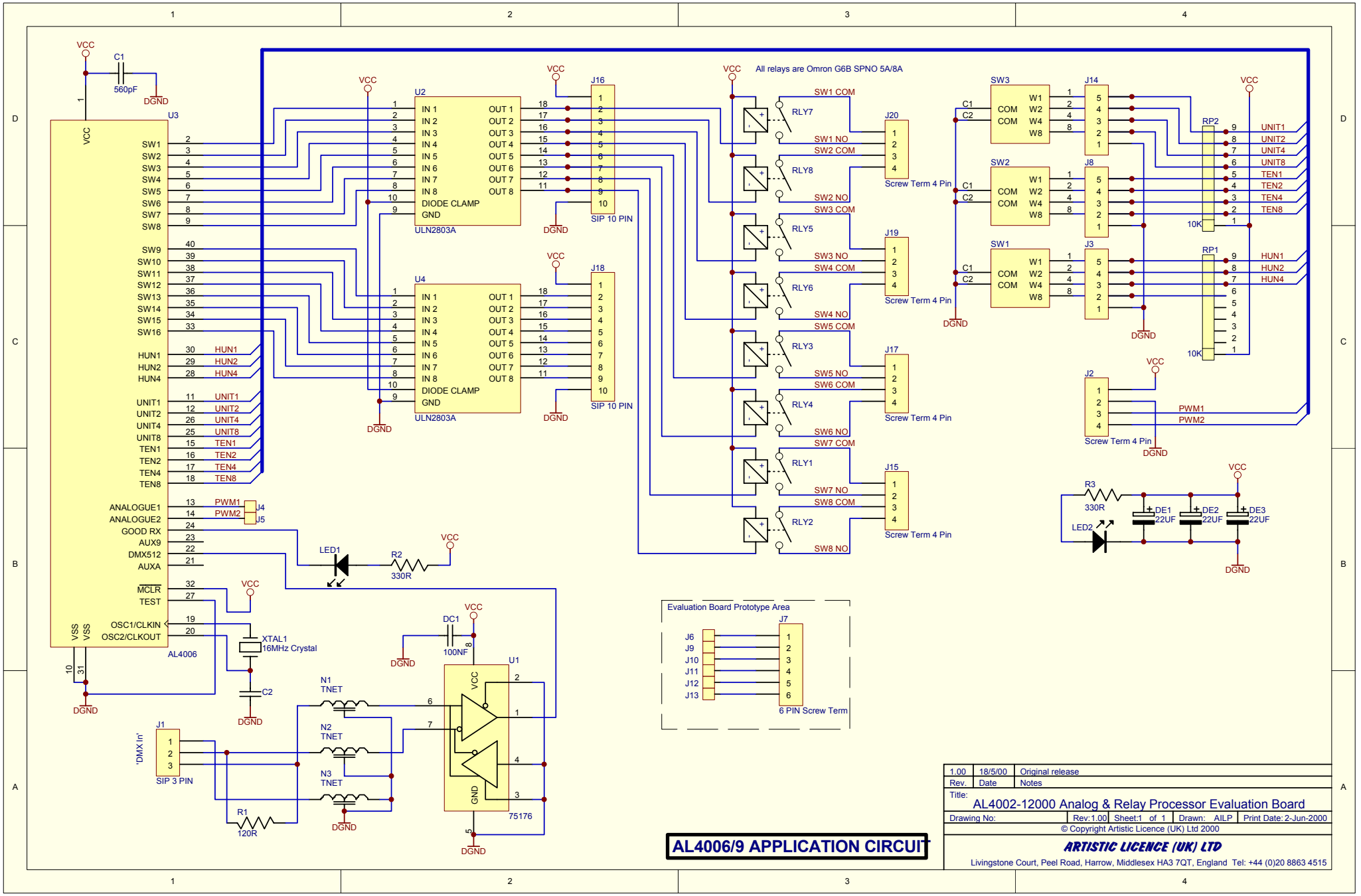
EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4006/9.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4005 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

Application

The following circuit shows the AL4006/9 application drawing. This is also the circuit diagram for the evaluation PCB.

The evaluation card provides 8 relays on board, plus an 8 channel darlington drive output for an additional 8 relays.


AL4006/9 APPLICATION CIRCUIT
All relays are Omron G6B SPNO 5A/8A
Evaluation Board Prototype Area
AL4002-12000 Analog & Relay Processor Evaluation Board
1.00 18/5/00 Original release
Rev. Date Notes
Drawing No: Rev:1.00 Sheet:1 of 1 Drawn: AILP Print Date: 2-Jun-2000
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Bill Of Materials

The following components are required for the evaluation PCB.

Designator	Part Type
C1	560pF
C2	
DC1	100NF
DE1	22UF
DE2	22UF
DE3	22UF
J1	SIP 3 PIN
J10	TESTPOINT
J11	TESTPOINT
J12	TESTPOINT
J13	TESTPOINT
J14	CONN SIP 5PIN
J15	Screw Term 4 Pin
J16	SIP 10 PIN
J17	Screw Term 4 Pin
J18	SIP 10 PIN
J19	Screw Term 4 Pin
J2	Screw Term 4 Pin
J20	Screw Term 4 Pin
J3	CONN SIP 5PIN
J4	TESTPOINT
J5	TESTPOINT
J6	TESTPOINT
J7	CONN SCREW 6PIN
J8	CONN SIP 5PIN
J9	TESTPOINT
LED1	LED 3MM
LED2	LED 3MM
N1	TNET
N2	TNET
N3	TNET
R1	120R
R2	330R
R3	330R
RLY1	RLY OMRON G6B SPNO
RLY2	RLY OMRON G6B SPNO
RLY3	RLY OMRON G6B SPNO
RLY4	RLY OMRON G6B SPNO
RLY5	RLY OMRON G6B SPNO
RLY6	RLY OMRON G6B SPNO
RLY7	RLY OMRON G6B SPNO
RLY8	RLY OMRON G6B SPNO
RP1	10K
RP2	10K
SW1	BCD-HOZ
SW2	BCD-HOZ
SW3	BCD-HOZ
U1	75176
U2	ULN2803A
U3	AL4006
U4	ULN2803A
XTAL1	16MHz Crystal

THE AL4007 DEMUX PROCESSOR

Overview

The AL4007 is designed to receive the DMX512 protocol and provide all the control signals necessary to generate 72 channels of analogue control signals.

The AL4007 is suitable for the development of DMX512 to Analogue Decoders and also provides an elegant solution to the DMX512 upgrade of existing analogue dimmers.

The AL4007 is supplied as a 40 pin DIL device.

Key features:

- DMX512 input
- Good data received indicator
- Parallel output for external Digital to Analogue conversion
- Individual channel test at 50% and 100%
- DMX base address is set by external thumbwheel inputs

Pin Functions

Pin	Function	In/Out
1	VCC 5V	-
2	CS Address 0	O
3	CS Address 1	O
4	CS Address 2	O
5	CS Enable	O
6	CS 8	O
7	NC	-
8	DAC Select	O
9	TRIG - Scope Trigger	O
10	GND	-
11	BCD Unit 1	I
12	BCD Unit 2	I
13	BCD Unit 4	I
14	BCD Unit 8 (msb)	I
15	BCD Ten 1	I
16	BCD Ten 2	I
17	BCD Ten 4	I
18	BCD Ten 8 (msb)	-
19	Osc1 (16mhz crystal)	-
20	Osc2 (16mhz crystal)	-
21	NC	-
22	DMX512 TTL Receive	I
23	NC	-
24	Good Rx (Open Collecter, Lo=Good)	O
25	NC	-
26	NC	-
27	Test (Connect to GND)	I
28	BCD Hundred 4	I
29	BCD Hundred 2	I
30	BCD Hundred 1	I
31	GND	-
32	MCLR (Connect to VCC)	I
33	D7 (msb dac data)	O
34	D6	O
35	D5	O
36	D4	O
37	AD3 (msb channel address)	O
38	AD2	O
39	AD1	O
40	AD0	O

Circuit Operation

The AL4007 and 74LS138 provide chip selects and address outputs, used to address a single demultiplex channel from the 72 available.

The analogue multiplex output is then routed through that channel to the sample and hold circuit.

The application drawings which follow show this process using 4051, 8 channel de-multiplexors and TL084 op-amps. The TL084 has been chosen in order to provide a very high input impedance for the sample and hold circuit.

It should be noted that the analogue multiplex bus and the 4051's are all run at 5V supply in order to avoid converting the address and chip select signals to CMOS voltages.

The final conversion to a 10V output is produced by using a gain stage in the output amplifier.

DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.

Test Mode

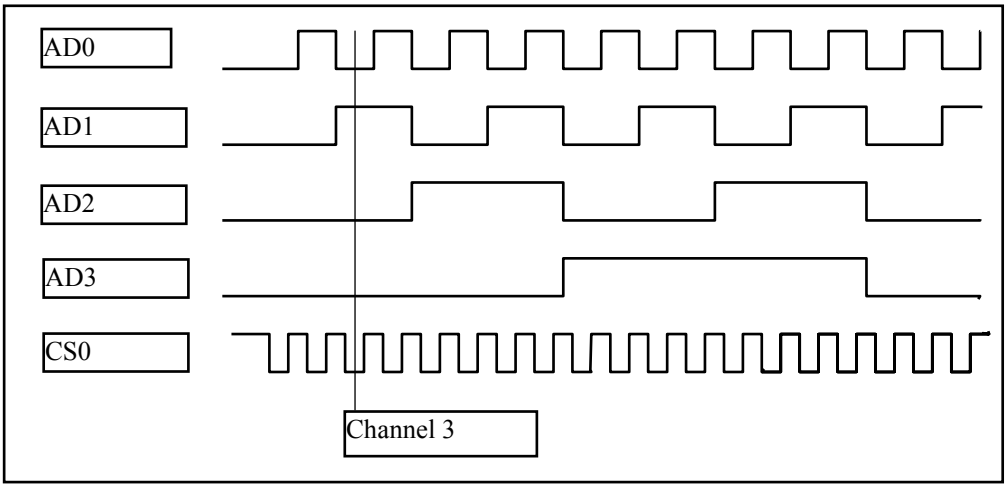
The BCD inputs are also used to enter test mode.
When Test mode is active, the DMX512 input is disabled.
When the Hundred inputs are set to either '6' or '7' Test mode is active. The Hundred input sets the intensity, while the Ten and Units inputs select the channel.

The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input
0	0	0	All Off	Off
0	0	1	Controlled by inputs	On
5	1	2	Controlled by inputs	On
7	0	1	Output 1 @ 100%	Off
6	0	1	Output 1 @ 50%	Off
7	6	9	Output 69 @ 100%	Off
6	1	2	Output 12 @ 50%	Off

Output timing

The signals AD0-AD3 and CS are used to address the demultiplexor circuits. The signals continue to operate independent of whether the DMX512 input exists.
The cycle time will vary depending on the refresh rate of the received DMX512.



DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Power Supply

The AL4007 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

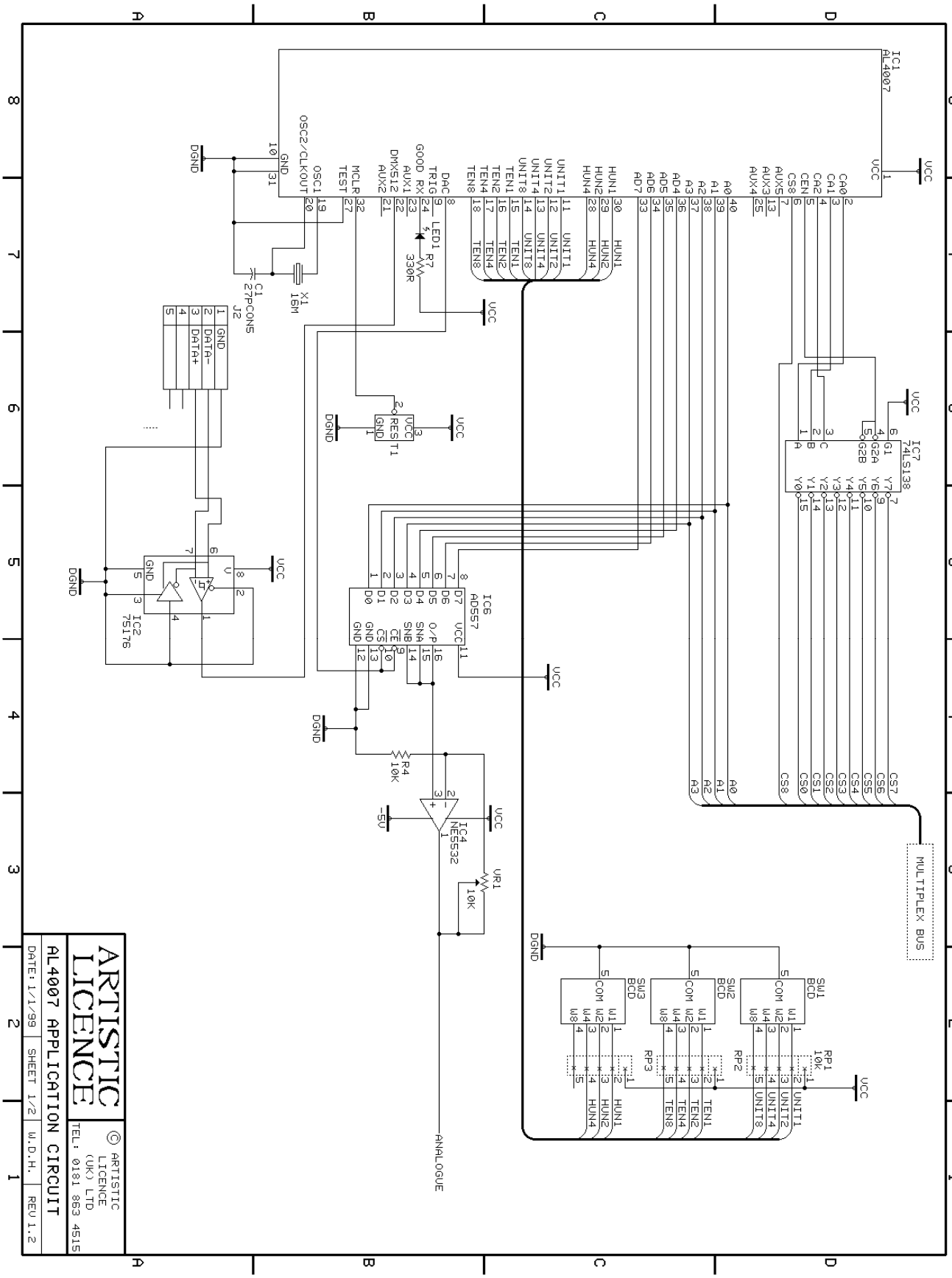
The AL4007 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

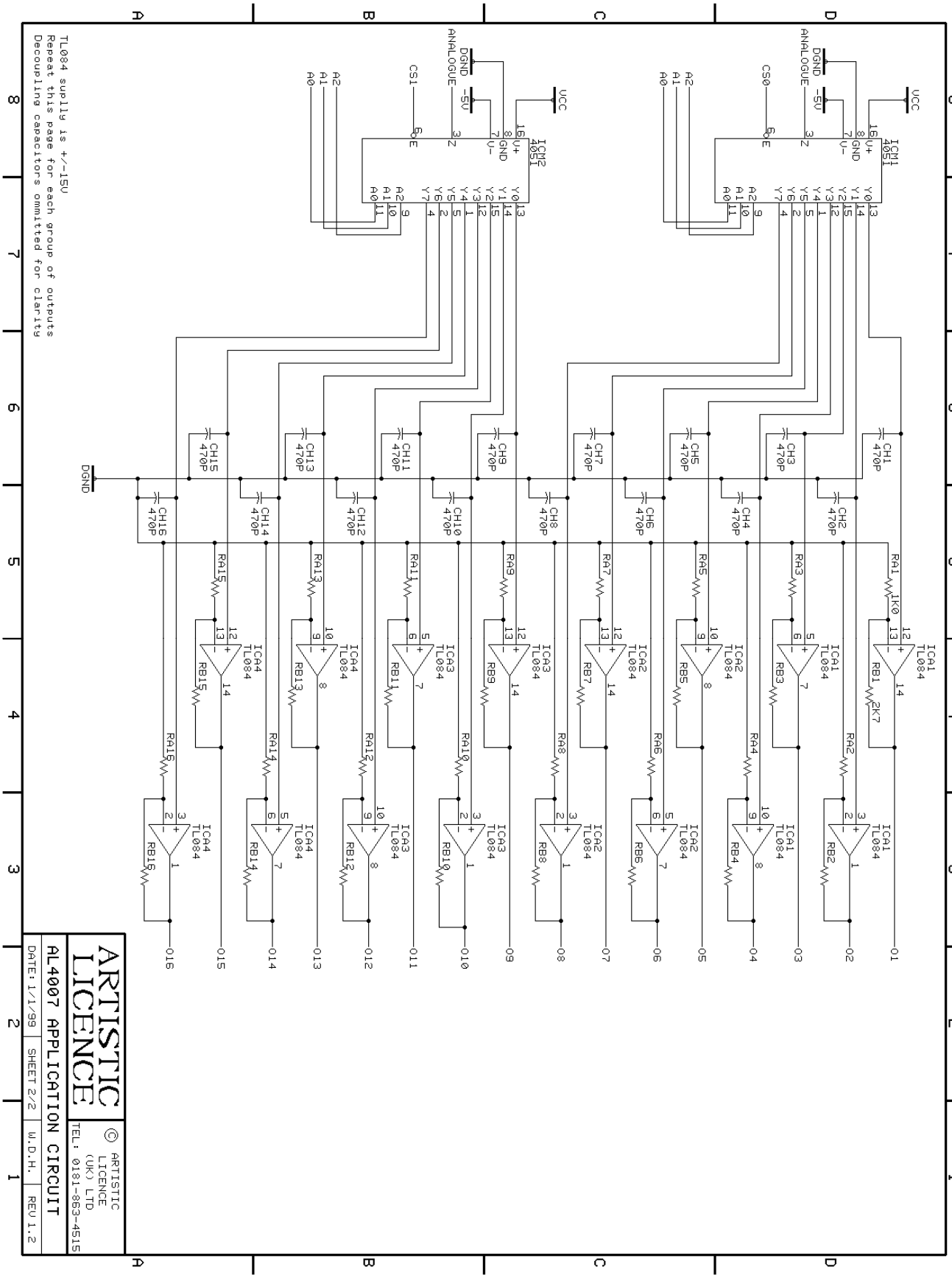
EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4007.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4005 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

Application

The following circuit shows the AL4007 application drawing.


IC1 AL4007
IC7 74LS138
IC6 AD557
IC4 NE5532
IC2 75176
MULTIPLEX BUS
ANALOGUE
SW1 BCD
SW2 BCD
SW3 BCD
RP1 10k
RP2
RP3
VR1 10K
R4 10K
R7 330R
LED1
C1 27PCON5
X1 16M
J2
RESET
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AL4007 APPLICATION CIRCUIT
DATE: 1/1/99
SHEET 1/2
W.D.H.
REV 1.2


TL084 supply is +/-15V
Repeat this page for each group of outputs
Decoupling capacitors ommitted for clarity
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AL4007 APPLICATION CIRCUIT
DATE: 1/1/99
SHEET 2/2
W.D.H.
REV 1.2

THE AL4008 RELAY PROCESSOR

Overview

The AL4008 is designed to receive the DMX512 protocol and provide the control signals necessary to drive 24 relays or other switched devices.

The AL4008 is suitable for the development of DMX512 Switch Packs.

The AL4008 is supplied as a 40 pin DIL device.

Key features:

- DMX512 input
- Good data received indicator
- Multiplexed relay drive
- Individual channel test
- DMX base address is set by external thumbwheel inputs

Pin Functions

Pin	Function	In/Out
1	VCC 5V	-
2	CS Address 0	O
3	CS Address 1	O
4	CS Address 2	O
5	NC	-
6	NC	-
7	NC	-
8	NC	-
9	TRIG - Scope Trigger	O
10	GND	-
11	BCD Unit 1	I
12	BCD Unit 2	I
13	BCD Unit 4	I
14	BCD Unit 8 (msb)	I
15	BCD Ten 1	I
16	BCD Ten 2	I
17	BCD Ten 4	I
18	BCD Ten 8 (msb)	I
19	Osc1 (16mhz crystal)	-
20	Osc2 (16mhz crystal)	-
21	NC	-
22	DMX512 TTL Receive	I
23	NC	-
24	Good Rx (Open Collecter, Lo=Good)	O
25	NC	-
26	NC	-
27	Test (Connect to GND)	I
28	BCD Hundred 4	I
29	BCD Hundred 2	I
30	BCD Hundred 1	I
31	GND	-
32	MCLR (Connect to VCC)	I
33	D7	O
34	D6	O
35	D5	O
36	D4	O
37	D3	O
38	D2	O
39	D1	O
40	D0	O

Circuit Operation

The AL4008 provides chip selects and a data bus which is used to address a single switched output from the 24 available.
The output devices used are 74HCT374 octal latches.
Please not the 470R pull down network attached to the data bus. This device ensures that the relays do not momentarily energise during the power on sequence.

DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.

Test Mode

The BCD inputs are also used to enter test mode.
When Test mode is active, the DMX512 input is disabled.
When the Hundred inputs are set to '7' Test mode is active. The Ten and Units inputs select the channel.
The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input
0	0	0	All Off	Off
0	0	1	Controlled by inputs	On
5	1	2	Controlled by inputs	On
7	0	1	Output 1 On	Off
6	1	9	Output 19 On	Off

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Power Supply

The AL4008 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

The AL4008 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal

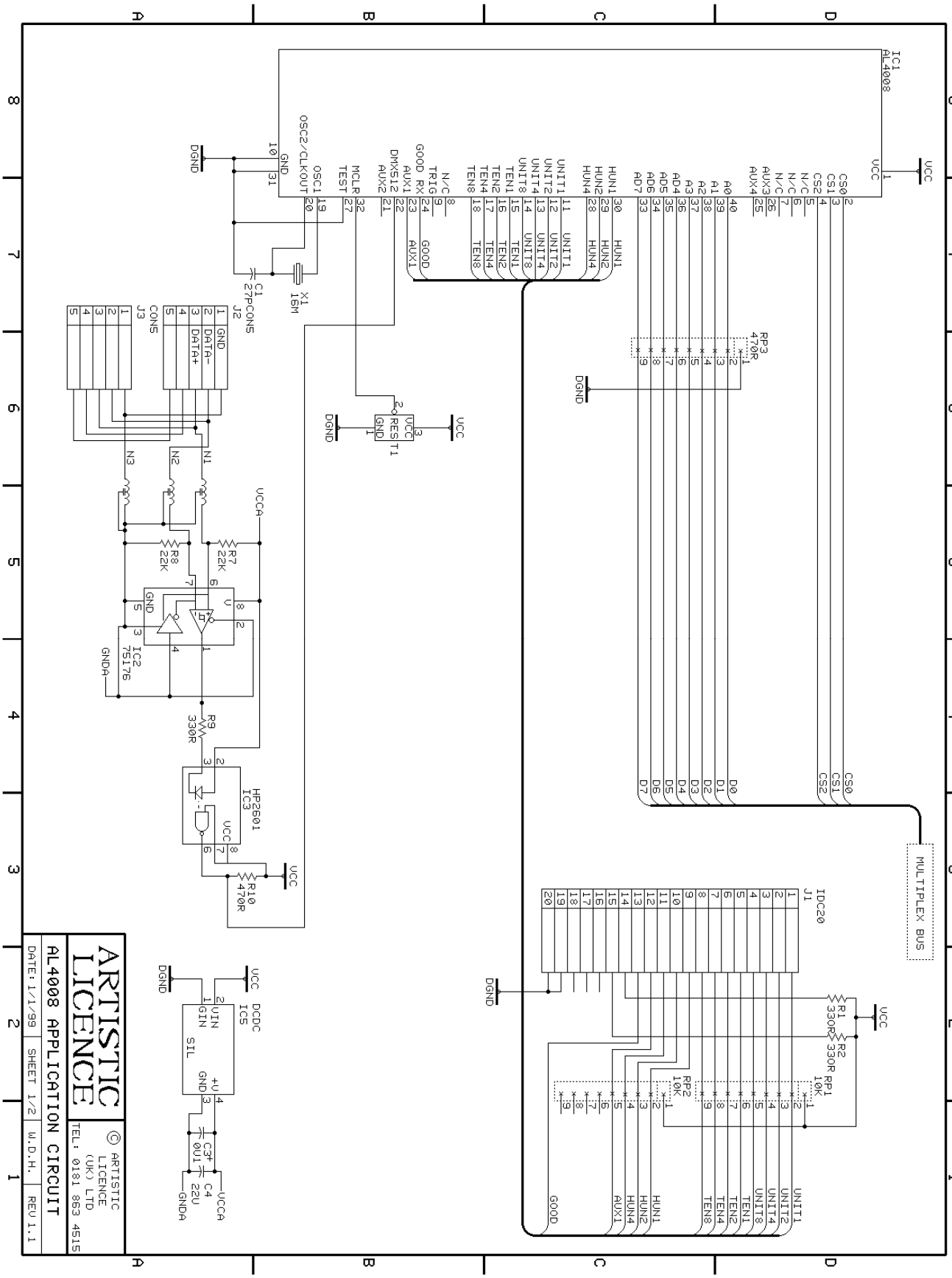
connection and Ground. The inherent pcb and lead capacitance must be included in this.

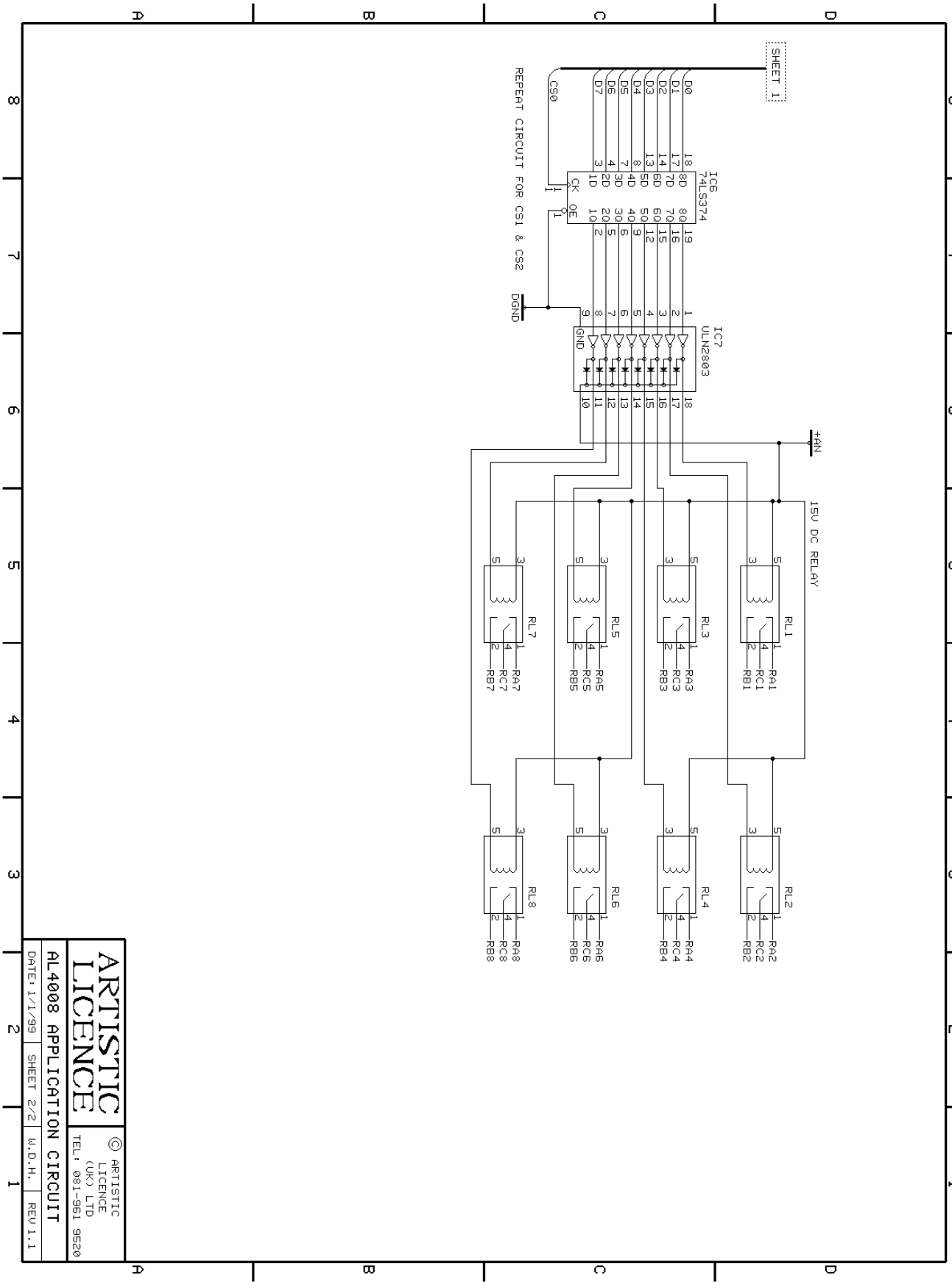
EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4008.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4005 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

Application

The following circuit shows the AL4008 application drawing.


IC1
AL4008
MULTIPLEX BUS
IDC20
J1
DCDC
IC5
HP2601
IC3
IC2
75176
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AL4008 APPLICATION CIRCUIT
DATE: 1/1/99
SHEET 1/2
W.D.H.
REV 1.1


SHEET 1
CS0
REPEAT CIRCUIT FOR CS1 & CS2
IC6
74LS374
IC7
ULN2803
DGND
+9V
15V DC RELAY
RL1
RL2
RL3
RL4
RL5
RL6
RL7
RL8
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REV 1.1

THE AL4016 SINGLE PHASE DIMMER

Overview

The AL4016 is designed to receive the DMX512 protocol and provide 16 phase position modulated outputs in order to switch thyristor or triac loads.

The AL4016 provides all the control processes required by modern digital lighting dimmers

Key Features include:

- DMX512 input
- Good data received indicator
- 16 phase position modulated outputs
- Automatic mains frequency and phase tracking
- Anti-Thump turn on circuitry
- Power Linear dimmer curve
- Pre-heat selection
- Individual channel test at 50% and 100%

The AL4016 is a 40 pin DIL device.

Pin Functions

Pin	Function	In/Out
1	VCC 5V	-
2	Phase Output 1	O
3	Phase Output 2	O
4	Phase Output 3	O
5	Phase Output 4	O
6	Phase Output 5	O
7	Phase Output 6	O
8	Phase Output 7	O
9	Phase Output 8	O
10	GND	-
11	BCD Unit 1	I
12	BCD Unit 2	I
13	BCD Unit 4	I
14	BCD Unit 8 (msb)	I
15	BCD Ten 1	I
16	BCD Ten 2	I
17	BCD Ten 4	I
18	BCD Ten 8 (msb)	-
19	Osc1 (16MHz crystal)	-
20	Osc2 (16MHz crystal)	-
21	Preheat (On=Hi)	I
22	DMX512 TTL Receive	I
23	Select 1	I
24	Good Rx (Open Collector, Lo=Good)	O
25	Select 0	I
26	Zero Cross (Active Lo)	I
27	Test (Connect to GND)	I
28	BCD Hundred 4	I
29	BCD Hundred 2	I
30	BCD Hundred 1	I
31	GND	-
32	MCLR (Connect to VCC)	I
33	Phase Output 16	O
34	Phase Output 15	O
35	Phase Output 14	O
36	Phase Output 13	O
37	Phase Output 12	O
38	Phase Output 11	O
39	Phase Output 10	O
40	Phase Output 9	O

Phase Outputs

The AL4016 phase outputs are synchronised to the zero cross input signal over a frequency range of 45Hz to 65Hz. The output phase trigger position is defined by the DMX512 input level and modified by a square law curve. When driving a thyristor load, this arrangement provides a linear variation in the load power for a linear variation of the DMX512 input level.

The Zero Cross input provides synchronisation to the mains frequency.
An active low pulse at the mains zero cross point (i.e. twice the mains frequency) is required. The pulse width should be in the order of 100uS. It is imperative that this input be a 'clean', fast rise time pulse. Any jitter at the Zero Cross input will produce flickering on the outputs.
The application circuit shows a suitable circuit for use with a low voltage AC output from a transformer.

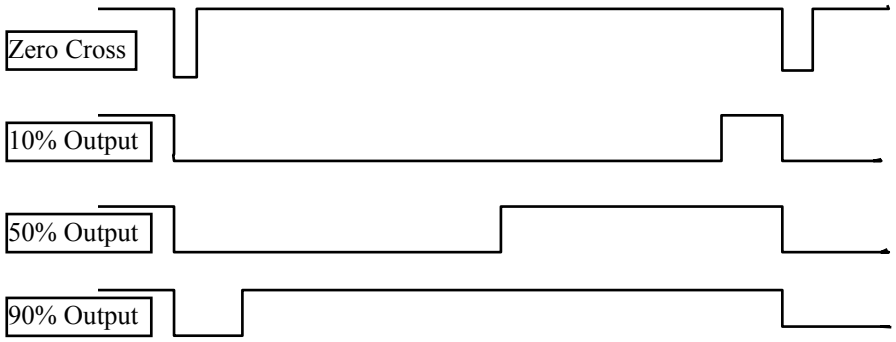
Dimmer Patch Select

Two inputs Select 0 and Select 1 are used to define the relationship between the DMX512 input and the Phase outputs. In normal single phase operation these two inputs should be connected to 0V. The following table details the other options. It can be seen that options 1,2 & 3 are used when three AL4016 processors are used for three phase control, but use a common DMX512 Base Address.

Phase Output	Controlled by DMX Channel			
Select 0	0	1	0	1
Select 1	0	0	1	1
1	1	1	2	3
2	2	4	5	6
3	3	7	8	9
4	4	10	11	12
5	5	13	14	15
6	6	16	17	18
7	7	19	20	21
8	8	22	23	24
9	9	25	26	27
10	10	28	29	30
11	11	31	32	33
12	12	34	35	36
13	13	37	38	39
14	14	40	41	42
15	15	43	44	45
16	16	46	47	48

Output timing

The following examples show the relationship between Output, Zero cross for three input values.



DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.

Test Mode

The BCD inputs are also used to enter test mode.
When Test mode is active, the DMX512 input is disabled.
When the Hundred inputs are set to either '6' or '7' Test mode is active. The Hundred input sets the intensity, while the Ten and Units inputs select the channel.
The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input
0	0	0	All Off	Off
0	0	1	Controlled by inputs	On
5	1	2	Controlled by inputs	On
7	0	1	Output 1 @ 100%	Off
6	0	1	Output 1 @ 50%	Off
7	0	9	Output 9 @ 100%	Off
6	1	2	Output 12 @ 50%	Off

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Power Supply

The AL4016 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the Zero Cross and Crystal connections.

Crystal

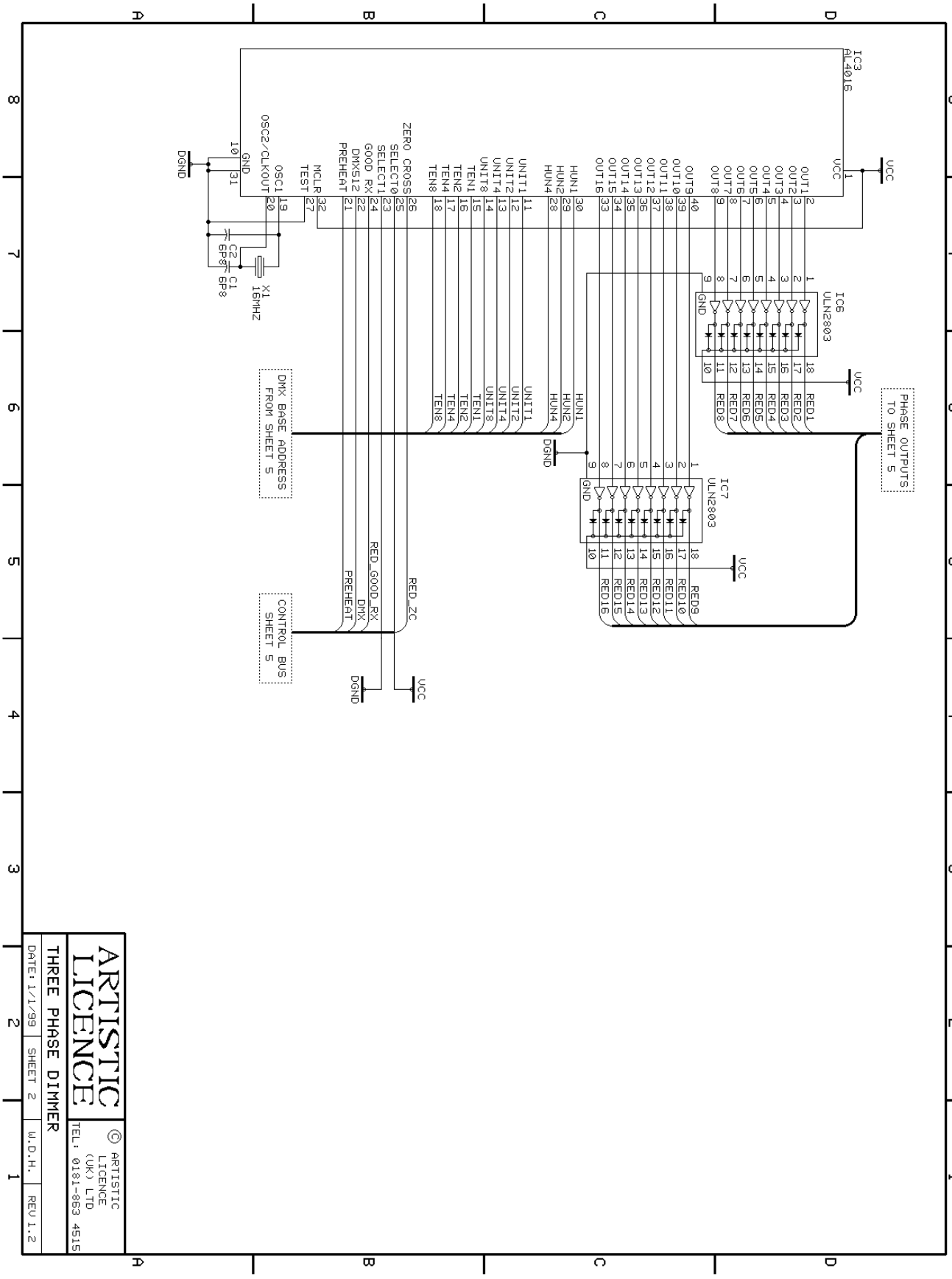
The AL4016 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

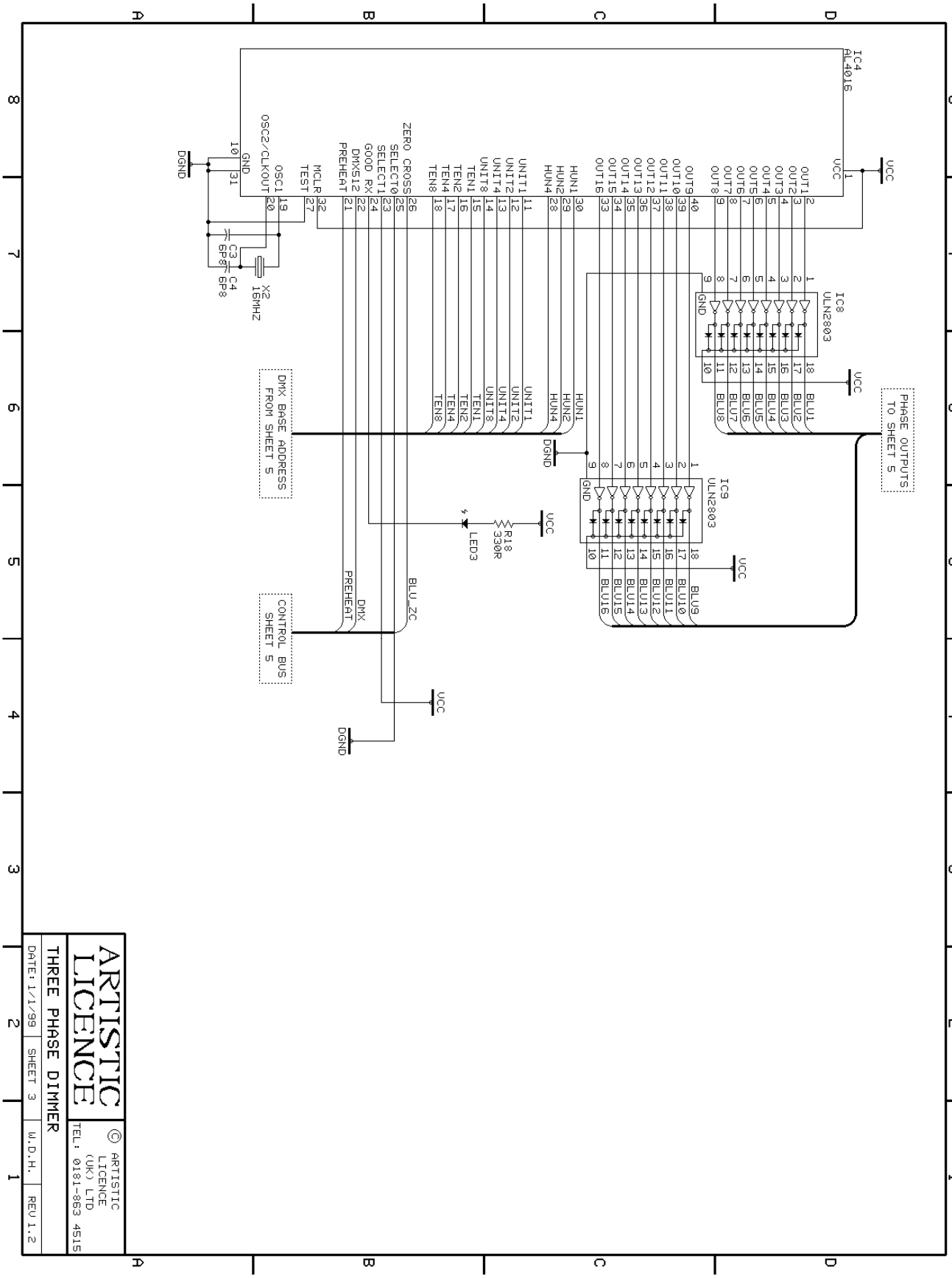
EMC

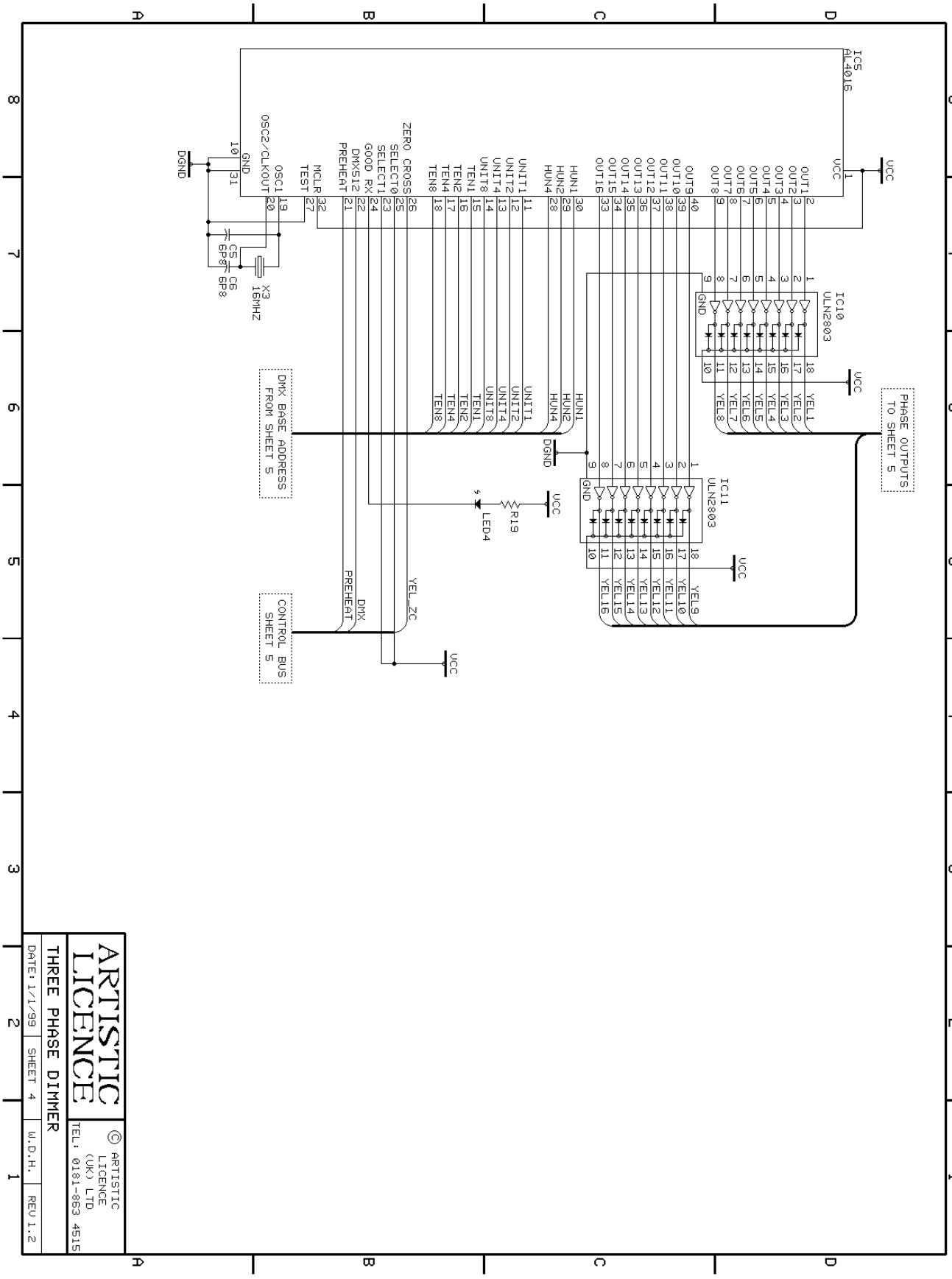
There are two major considerations in reducing the electromagnetic radiation generated by the AL4016.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4002 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

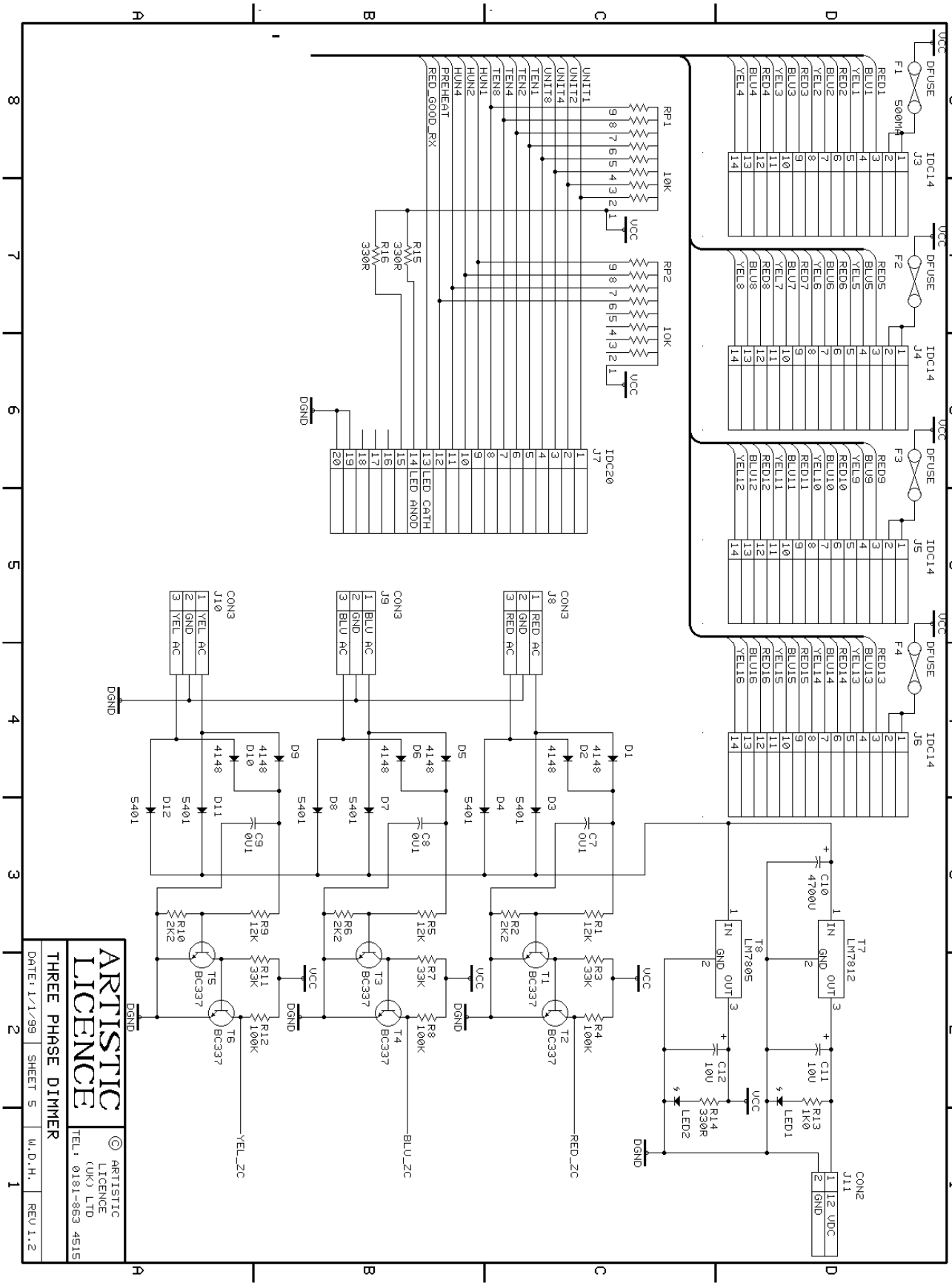
Application

The following circuit shows the AL4016 application drawing.


IC3
AL4016
VCC
OUT1
OUT2
OUT3
OUT4
OUT5
OUT6
OUT7
OUT8
OUT9
OUT10
OUT11
OUT12
OUT13
OUT14
OUT15
OUT16
GND
OSC2/CLKOUT
OSC1
MCLR
TEST
ZERO CROSS
SELECT0
SELECT1
GOOD RX
DMXS12
PREHEAT
UNIT1
UNIT2
UNIT4
UNIT8
TEN1
TEN2
TEN4
TEN8
HUN1
HUN2
HUN4
DGND
C2
6P8
C1
6P8
X1
16MHZ
IC6
ULN2803
IC7
ULN2803
RED1
RED2
RED3
RED4
RED5
RED6
RED7
RED8
RED9
RED10
RED11
RED12
RED13
RED14
RED15
RED16
PHASE OUTPUTS
TO SHEET 5
DMX BASE ADDRESS
FROM SHEET 5
CONTROL BUS
SHEET 5
RED_GOOD_RX
DMX
PREHEAT
RED_ZC
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SHEET 2
W.D.H.
REV 1.2


IC4
AL4016
VCC
OUT1
OUT2
OUT3
OUT4
OUT5
OUT6
OUT7
OUT8
OUT9
OUT10
OUT11
OUT12
OUT13
OUT14
OUT15
OUT16
GND
OSC2/CLKOUT
OSC1
MCLR
TEST
PREHEAT
DMXS12
GOOD RX
SELECT1
SELECT0
ZERO CROSS
TEN8
TEN4
TEN2
TEN1
UNIT8
UNIT4
UNIT2
UNIT1
HUN4
HUN2
HUN1
DGND
C3
6P8
C4
6P8
X2
16MHZ
IC8
ULN2803
BLU1
BLU2
BLU3
BLU4
BLU5
BLU6
BLU7
BLU8
IC9
ULN2803
BLU9
BLU10
BLU11
BLU12
BLU13
BLU14
BLU15
BLU16
PHASE OUTPUTS
TO SHEET 5
DMX BASE ADDRESS
FROM SHEET 5
R18
330R
LED3
CONTROL BUS
SHEET 5
DMX
PREHEAT
BLU_ZC
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IC5
AL4016
VCC
OUT1
OUT2
OUT3
OUT4
OUT5
OUT6
OUT7
OUT8
OUT9
OUT10
OUT11
OUT12
OUT13
OUT14
OUT15
OUT16
GND
OSC2/CLKOUT
OSC1
MCLR
TEST
PREHEAT
DMXS12
GOOD RX
SELECT1
SELECT0
ZERO CROSS
TEN8
TEN4
TEN2
TEN1
UNIT8
UNIT4
UNIT2
UNIT1
HUN4
HUN2
HUN1
DGND
C5
6P8
C6
6P8
X3
16MHZ
IC10
ULN2803
IC11
ULN2803
YEL1
YEL2
YEL3
YEL4
YEL5
YEL6
YEL7
YEL8
YEL9
YEL10
YEL11
YEL12
YEL13
YEL14
YEL15
YEL16
PHASE OUTPUTS
TO SHEET 5
DMX BASE ADDRESS
FROM SHEET 5
CONTROL BUS
SHEET 5
R19
LED4
YEL_ZC
DMX
PREHEAT
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THREE PHASE DIMMER
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SHEET 5
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REV 1.2
IDC14
IDC20
CON3
CON2
DFUSE
F1 500MA
F2
F3
F4
RP1 10K
RP2 10K
R15 330R
R16 330R
R17 330R
LED ANOD
LED CATH
RED_GOOD_RX
PREHEAT
UNIT1
UNIT2
UNIT4
UNIT8
TEN1
TEN2
TEN4
TEN8
HUN1
HUN2
HUN4
VCC
DGND
T7 LM7812
T8 LM7805
C10 4700U
C11 10U
C12 10U
R13 1K0
R14 330R
LED1
LED2
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
4148
5401
C7 0U1
C8 0U1
C9 0U1
R1 12K
R2 2K2
R3 33K
R4 100K
R5 12K
R6 2K2
R7 33K
R8 100K
R9 12K
R10 2K2
R11 33K
R12 100K
T1 BC337
T2 BC337
T3 BC337
T4 BC337
T5 BC337
T6 BC337
RED_ZC
BLU_ZC
YEL_ZC

T H E A L 4 0 2 0 MULTIPLEX - MERGE PROCESSOR

Overview

The AL4020 is designed to receive both a DMX512 input and up to 72 analogue inputs. The two input types are htp merged to form a DMX512 output.

The start channel of the analogue data on the DMX512 output can be selected with the BCD switch inputs.

Setting the BCD inputs to "000" disable the DMX512 input and allows the device to operate as a simple analogue to DMX512 converter without any user controls.

Multiple devices can be cascaded using the DMX512 inputs and outputs to produce products with up to 512 channel capability. The insertion delay of the AL4020 is less than 60uS which ensures that cascaded device designs do not introduce a DMX512 response delay.

Pin Functions

Pin	Function	In/Out
1	VCC +5V	-
2	ANIN (analogue multiplex input)	I
3	Not Used (connect to GND)	I
4	Not Used (connect to GND)	I
5	Not Used (connect to GND)	I
6	Not Used (connect to GND)	I
7	Not Used (connect to GND)	I
8	BCD Hundred 1	I
9	BCD Hundred 2	I
10	BCD Hundred 4	I
11	VCC 5V	-
12	GND	-
13	Osc1 (16MHz crystal)	-
14	Osc2 (16MHz crystal)	-
15	Good Rx (Open Collector, Lo=Good)	O
16	NC	-
17	Scope Trigger	O
18	NC	-
19	A0	O
20	A1	O
21	A2	O
22	CS Address 0	O
23	NC	-
24	NC	O
25	DMX512 TTL Transmit	O
26	DMX512 TTL Receive	I
27	CS Address 1	O
28	CS Address 2	O
29	CS Enable	O
30	Chip Select 8	O
31	GND	-
32	VCC (5V)	-
33	BCD Unit 1	I
34	BCD Unit 2	I
35	BCD Unit 4	I
36	BCD Unit 8 (msb)	I
37	BCD Ten 1	I
38	BCD Ten 2	I
39	BCD Ten 4	I
40	BCD Ten 8 (msb)	I

Circuit Operation

The AL4020 and 74LS138 provide chip selects and address outputs which are used to address a single analogue input from the 72 available.
The level is converted to an 8 bit digital value and buffered internally.
The level buffer is passed through a digital filter to reduce analogue noise and bit jitter prior to forming the final output.

The application drawings which follow show this process using 4051 8 channel multiplexers. It should be noted that the analogue multiplex bus and the 4051's are all run at a 5V supply in order to avoid converting the address and chip select signals to CMOS voltages. This necessitates the potential divider at the input for each channel, such that the 10V input can be reduced in level. Components D1, D2 and R3 provide protection against over voltage or reverse polarity inputs.

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.

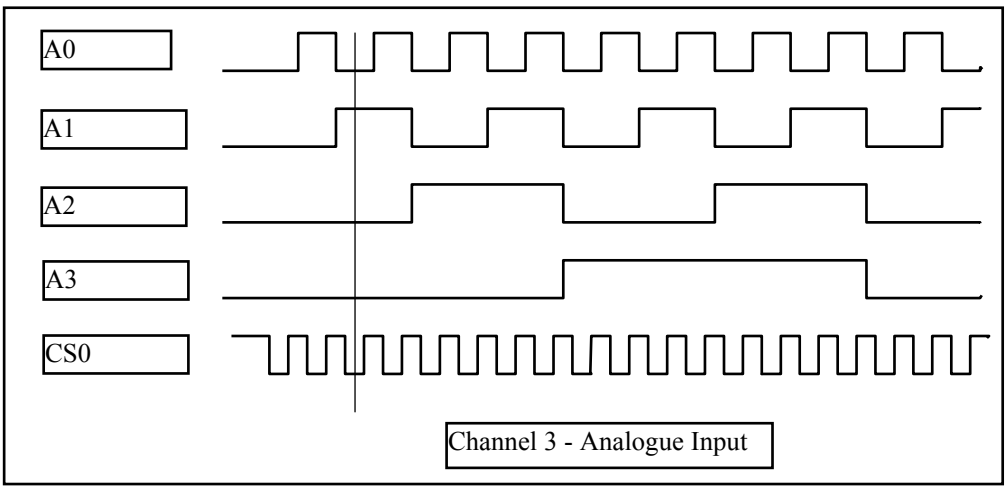
Test Mode

The BCD inputs are also used to enter test mode.
When Test mode is active, the DMX512 input is disabled.
When the Hundred inputs are set to either '6' or '7' Test mode is active. The Hundred input sets the intensity, while the Ten and Units inputs select the channel.
The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input	Analogue Inputs
0	0	0	Analogue only	Off	On
0	0	1	Analogue & DMX Merge	On	On
5	1	2	Analogue & DMX Merge	On	On
7	0	1	Output 1 @ 100%	Off	Off
6	0	1	Output 1 @ 50%	Off	Off
7	6	9	Output 69 @ 100%	Off	Off
6	1	2	Output 12 @ 50%	Off	Off

Output timing

The signals A0-A3 and CSxx are used to address the multiplexor circuits.
The cycle time will vary depending on the refresh rate of the received DMX512.



Device Operation

The AL4020 operates in two distinct modes:

- Analogue to DMX512 Conversion.
- Analogue and DMX512 Merge.

Analogue to DMX512 Conversion

The address wheels are set to 000 in order to enable Analogue to DMX512 conversion. In this mode, the DMX512 input is ignored and the received data indicator does not illuminate.
The DMX512 output transmits 512 channels of which the first 72 represent the analogue inputs.

Analogue and DMX512 Merge

The BCD inputs are set to a value in the range 001 to 512 in order to enable Analogue and DMX512 merge. In this mode, the DMX512 input and the analogue inputs are combined in a highest takes precedence form.
The BCD inputs set the channel address at which the analogue inputs are added to the DMX512 input.

Power Supply

The AL4020 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

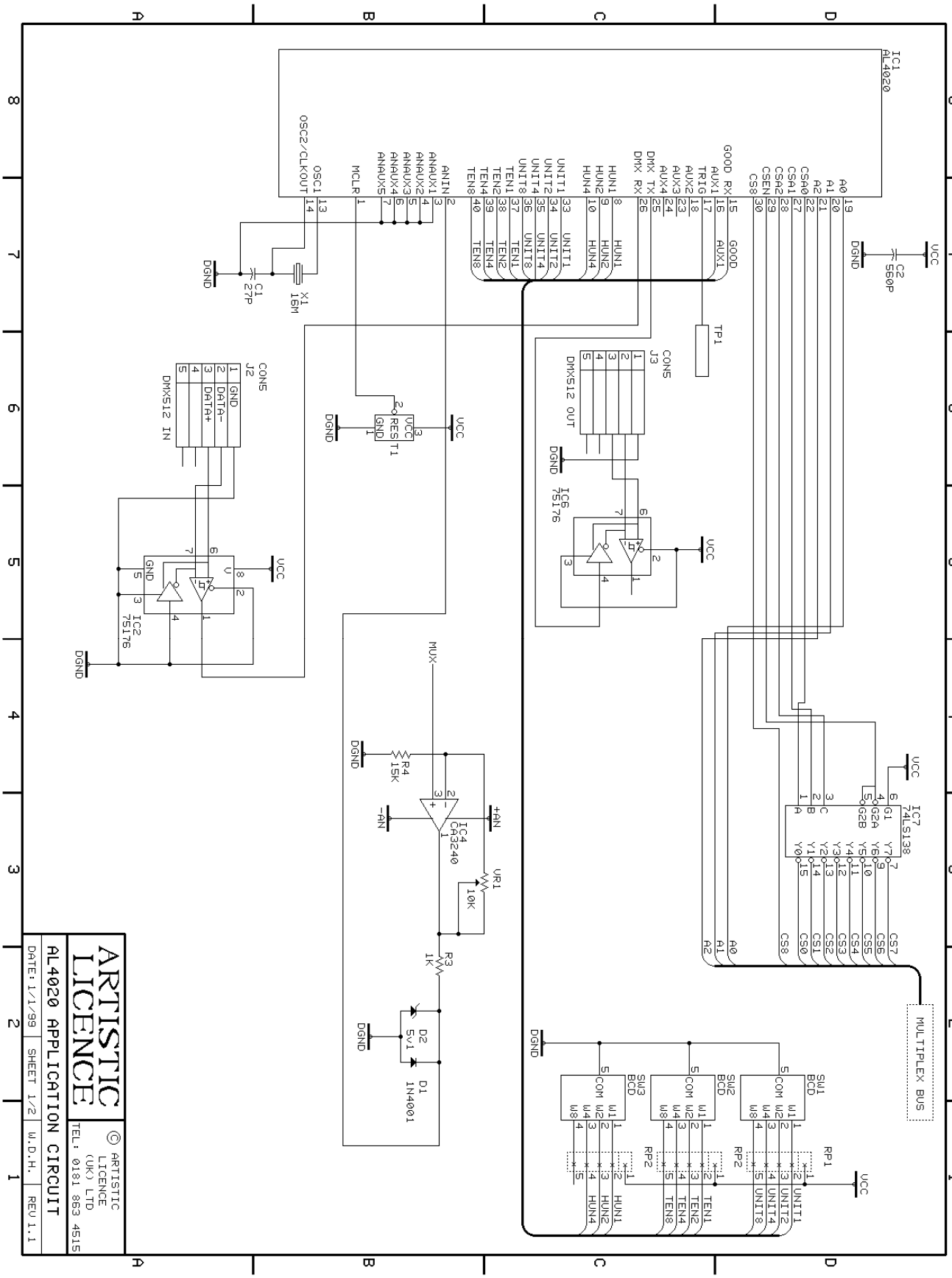
The AL4020 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

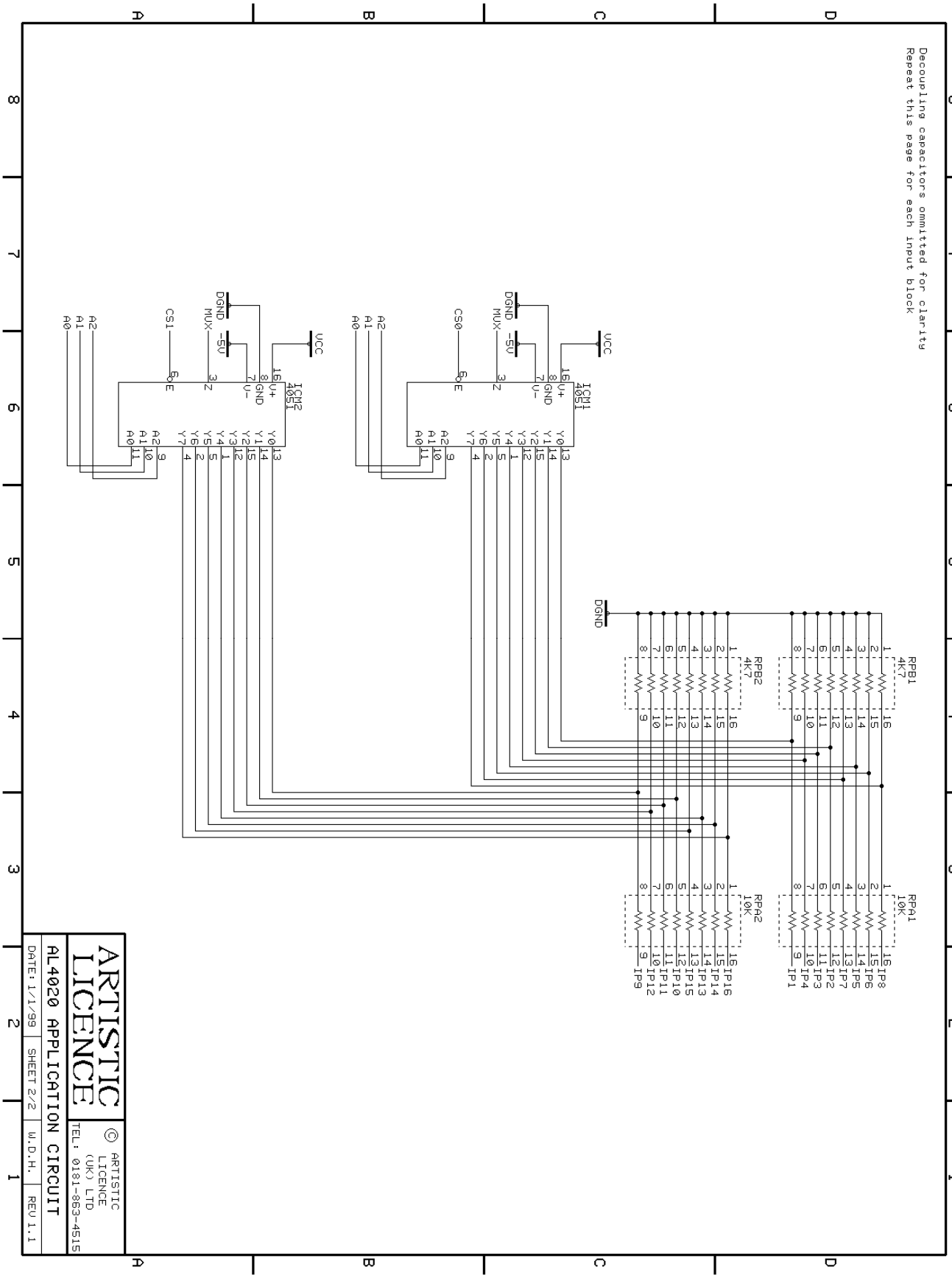
EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4020.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4005 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

Application

The following circuit shows the AL4020 application drawing.


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AL4020 APPLICATION CIRCUIT
DATE: 1/1/99
SHEET 1/2
W.D.H.
REV 1.1
IC1 AL4020
IC2 75176
IC3 CA3240
IC4 75176
IC7 74LS138
SW1 BCD
SW2 BCD
SW3 BCD
RP1
RP2
CON5 J2 DMX512 IN
CON5 J3 DMX512 OUT
MULTIPLEX BUS
MUX
VCC
DGND
C1 27P
C2 560P
X1 16M
R3 1K
R4 15K
VR1 10K
D1 1N4001
D2 5v1
TP1


Decoupling capacitors ommitted for clarity
Repeat this page for each input block
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SHEET 2/2
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THE AL4100 SERVO-CONTROL PROCESSOR

Overview

The AL4100 is designed to receive the DMX512 protocol and control two DC Motor servo circuits.

Applications include:

- Colour Changers
- Pan & Tilt Yokes
- Automated Luminaries
- Moving Faders
- Special Effects

Pin Functions

Pin	Function	In/Out
1	VCC (5V)	-
2	SERVO A (Shaft posn channel A)	I Analogue
3	SERVO B (Shaft posn channel B)	I Analogue
4	TRIM A (Max posn set - channel A)	I Analogue
5	TRIM B (Max posn set - channel B)	I Analogue
6	Not Used (connect to ground)	I Analogue
7	MAX SPEED (Max speed set - A & B)	I Analogue
8	FERR (Following error set - A & B)	I Analogue
9	STALL SPEED (Stall speed set - A & B)	I Analogue
10	TRAP (Trapezium point set - A & B)	I Analogue
11	VCC (5V)	-
12	GND	-
13	Osc1 (16MHz crystal)	I
14	Osc2 (16MHz crystal)	I
15	DIR A (Direction / Brake - channel A)	O
16	PWM A (Speed - channel A)	O
17	PWM B (Speed - channel B)	O
18	DIR B (Direction / Brake - channel B)	O
19	SERVO (set as, Servo=Lo, Speed=Hi)	I
20	Good Rx (Lo= good DMX512 data)	O
21	BAS0 (DMX base offset)	I
22	BAS1 (DMX base offset)	I
23	!DIR A (Direction / Brake - channel A)	O
24	!DIR B (Direction / Brake - channel B)	O
25	Not Used (connect to ground)	I Analogue
26	DMX (TTL DMX receive)	I
27	BCD Hundred 1	I
28	BCD Hundred 2	I
29	BCD Hundred 4	I
30	BCD Hundred 8	I
31	GND	-
32	VCC (5V)	-
33	BCD Unit 1	I
34	BCD Unit 2	I
35	BCD Unit 4	I
36	BCD Unit 8 (msb)	I
37	BCD Ten 1	I
38	BCD Ten 2	I
39	BCD Ten 4	I
40	BCD Ten 8 (msb)	I

Circuit Operation

The AL4100 uses two consecutive channels of DMX data to position two DC servo motors.

The DMX512 base address is set by the BCD thumbwheel switches. Additional offset inputs BAS0, BAS1 allow up to four AL4100 processors to connect to the same thumbwheels with contiguous

addresses.

Each motor is position controlled using a programmable trapezoid speed profile. Motor speed is controlled with a pulse width modulated output, whilst two binary outputs control direction, de-clutching and motor breaking.

Motor position is sensed by an analogue potentiometer input. The AL4100 provides inputs for control of the maximum allowed position, maximum desired speed, motor stall speed, required positional accuracy and acceleration.

Additionally the AL4100 may be switched to speed control mode, in which case the DMX512 input controls motor speed not position.

Pin Functions

The motion control pins function as detailed below:

PWM

The PWM output is a TTL pulse width modulation of the required motor speed. Output frequency is 64KHz. Output high represents maximum speed.

DIR !DIR

The DIR & !DIR outputs control the motor operation. Their primary function is to set the motor direction.
Coding is as follows:

DIR	!DIR	Motion
0	0	Dynamic Breaking
0	1	Rotate CW
1	0	Rotate CCW
1	1	Dynamic Breaking

TRIM SERVO

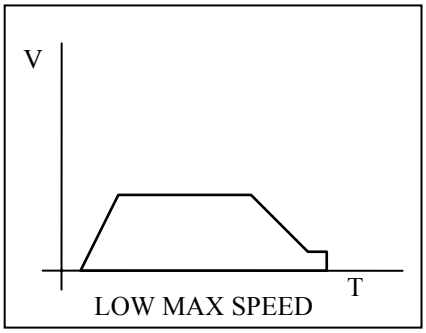
The TRIM input sets the maximum allowed position of the motor. The AL4100 will scale the incoming DMX data to fit the range set by TRIM as shown by the following table:

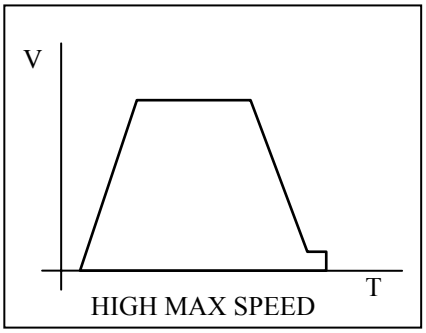
TRIM Voltage	DMX Level	Position as percentage of TRIM	SERVO Voltage
5V	0	0%	0V
5V	127	50%	2.5V
5V	255	100%	5.0V
4V	0	0%	0V
4V	127	50%	2.0V
4V	255	100%	4.0V

The TRIM input is most important in applications such as colour changers where it is used to calibrate the number of gels fitted.

MAX

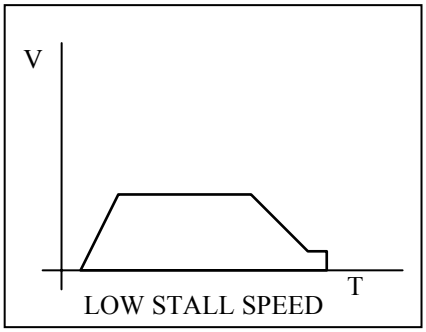
The MAX input is used to set the maximum speed allowed during a position search. The voltage at this input will normally be set by a divider
chain. It is not intended as a user control



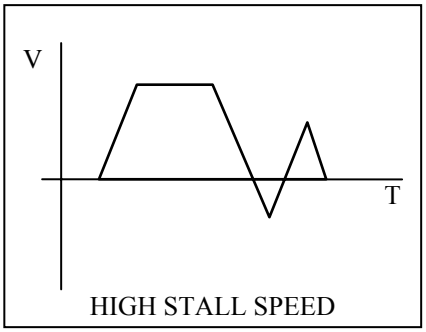


STALL

The STALL input sets the minimum speed below which the motor will stall. This value is significant in ensuring that the motor reaches its required position. If the STALL speed is set too high, the motor will overshoot and then hunt back to the correct position. If set too low, the motor will stall out before the required position is reached.
The voltage at this input will normally be set by a divider chain.
It is not intended as a user control.



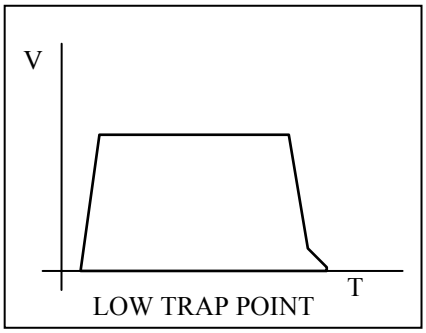
LOW STALL SPEED



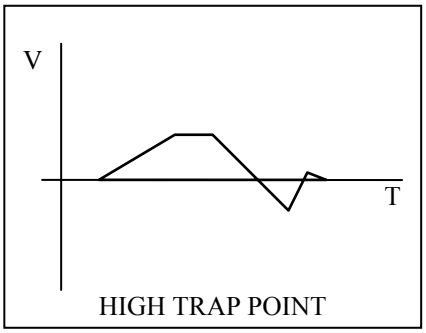
HIGH STALL SPEED

TRAP

The TRAP (Trapezium Point) input is used to set the distance from the search position at which the motor will start to decelerate. This value is significant in ensuring that the motor reaches its required position. If the TRAP is set too high, the motor will decelerate too slowly, overshoot and then hunt back to the correct position. If set too low, the motor and gearbox will be unnecessarily stained by harsh deceleration.
The voltage at this input will normally be set by a divider chain.
It is not intended as a user control.



LOW TRAP POINT



HIGH TRAP POINT

FERR

The FERR (Following Error) input is used to set the allowed error between the requested position and the motor position. Clearly the lowest value is preferred as it equates to the highest accuracy. However issues such as noise pick-up and variable load make it impractical to attain a zero following error. If FERR is set to a value lower than a particular application can attain, the servo loop will become unstable and hunting will occur. The FERR input is linearly calibrated to 16 bit resolution. That is:
FERR=0V equates to 0 bit allowed inaccuracy.
FERR=5V equates to 16 bits allowed inaccuracy.

SERVO

This control input should be connected to GND for servo operation.

MODE

When connected to VCC, the motors will rotate continuously as follows:

SERVO MODE	DMX Address	MOTOR A	MOTOR B
Lo	1	Position Request	N/A
Lo	2	N/A	Position Request
Hi	1	Speed	N/A
Hi	2	0-127 = Off 128-191 = CW 192-255 = CCW	NA
Hi	3	N/A	Speed
Hi	4	N/A	0-127 = Off 128-191 = CW 192-255 = CCW

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.
The BAS inputs provide an offset to the DMX Base Address. The purpose
of this is to allow up to four devices to be connected to the same thumbwheel switches. It is then possible to produce an eight channel subsystem.

BAS1	BAS0	Add to DMX Base Address
0	0	0
0	1	2
1	0	4
1	1	6

Test Mode

The BCD inputs are also used to enter test mode.
When Test mode is active, the DMX512 input is disabled.
When the Hundred inputs are set to either '6' or '7' Test mode is active. The Hundred input sets the motor channel, while the Ten and Units inputs select the position.
The following examples show the operation:

Hundreds	Tens	Units	Output	DMX Input
0	0	0	No Function	Off
0	0	1	Normal Operation	On
5	1	2	Normal Operation	On
6	0	0	Motor A Min Position	Off
6	5	0	Motor A Centre Position	Off
6	9	9	Motor A Max Position	Off
7	0	0	Motor B Min Position	Off
7	5	0	Motor B Centre Position	Off
7	9	9	Motor B Max Position	Off

Power Supply

The AL4100 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

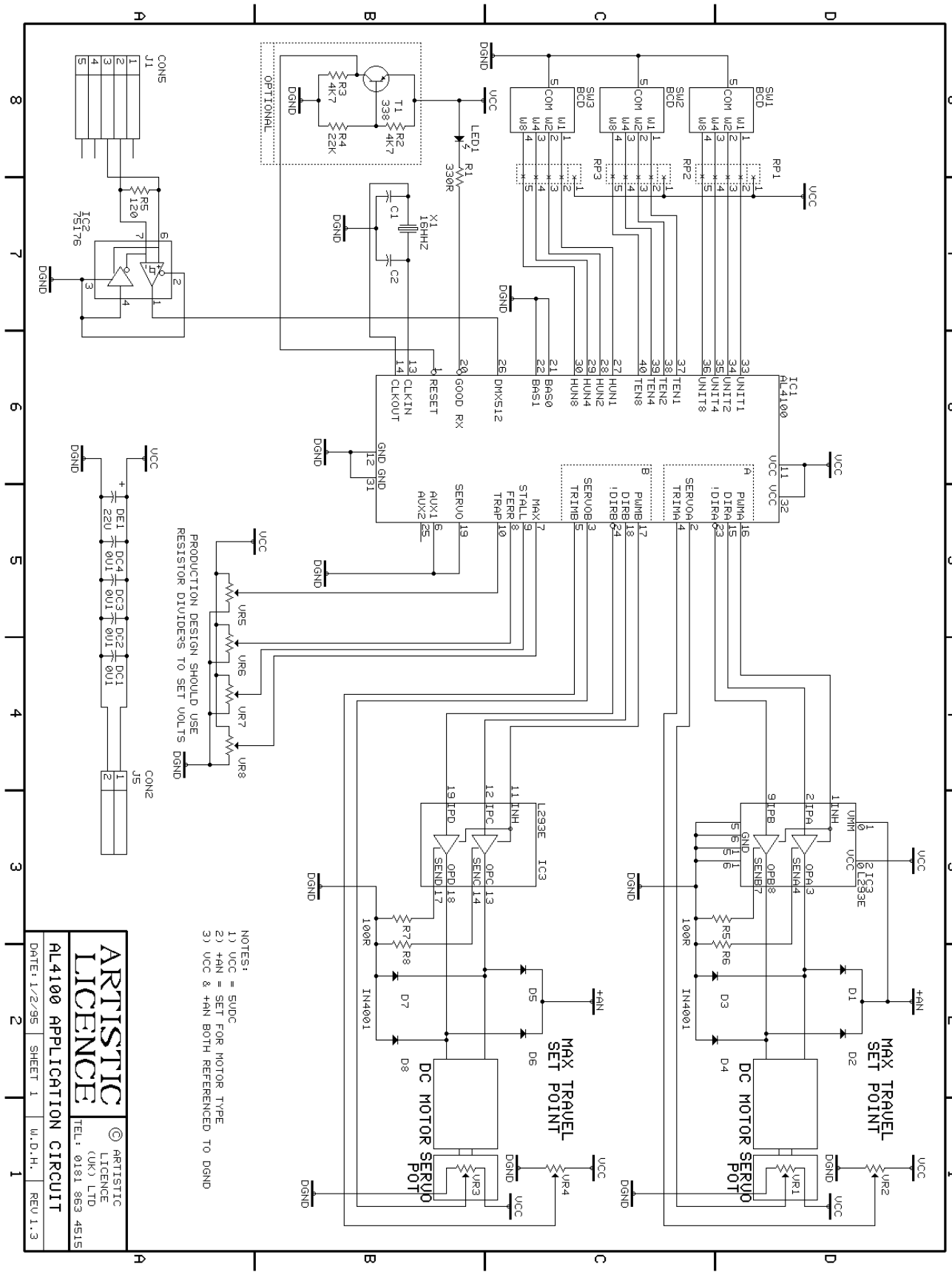
The AL4100 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

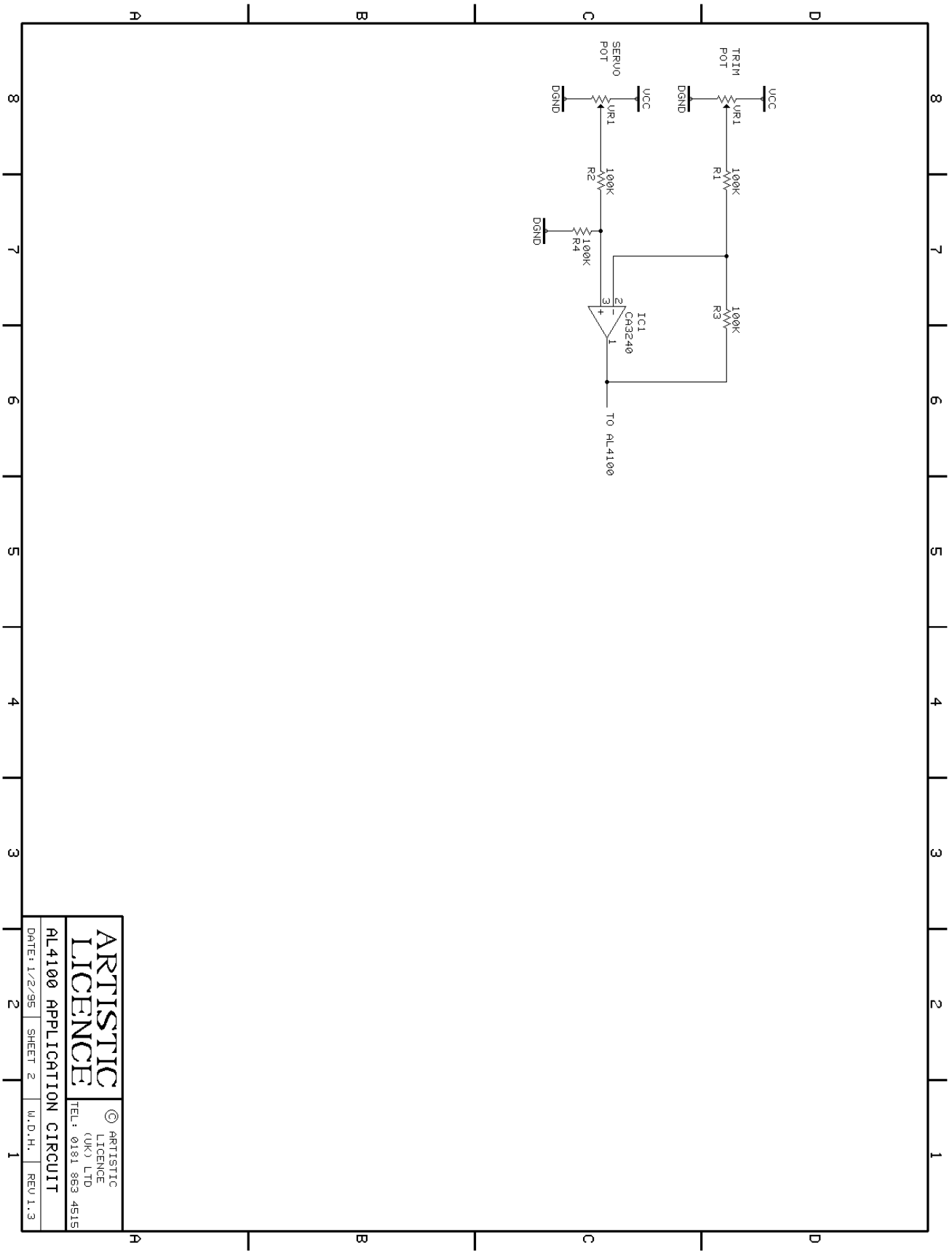
EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4100.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4005 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

Application

The following circuit shows the AL4100 application drawing.


AL4100 APPLICATION CIRCUIT
ARTISTIC LICENCE
© ARTISTIC LICENCE (UK) LTD
TEL: 0181 863 4515
DATE: 1/2/95
SHEET 1
W.D.H.
REV 1.3
NOTES:
1) VCC = 5VDC
2) +AN = SET FOR MOTOR TYPE
3) VCC & +AN BOTH REFERENCED TO DGND
PRODUCTION DESIGN SHOULD USE RESISTOR DIVIDERS TO SET VOLTS
OPTIONAL
MAX TRAVEL SET POINT
DC MOTOR SERVO POT
IC1 AL4100
IC2 7S176
IC3 L293E
SW1 BCD
SW2 BCD
SW3 BCD
RP1
RP2
RP3
CON5 J1
CON2 J5
X1 16MHZ
LED1
T1 338
R1 330R
R2 4K7
R3 4K7
R4 22K
R5 120
R7 100R
R8 100R
R5 100R
R6 100R
D1
D2
D3
D4
D5
D6
D7
D8
IN4001
DE1 22U
DC1 0U1
DC2 0U1
DC3 0U1
DC4 0U1
VR1
VR2
VR3
VR4
VR5
VR6
VR7
VR8
VCC
DGND
+AN


SERVO POT
TRIM POT
VCC
DGND
VR1
R1
R2
R3
R4
100K
IC1
CA3240
TO AL4100
ARTISTIC LICENCE
© ARTISTIC LICENCE (UK) LTD
TEL: 0181 863 4515
AL4100 APPLICATION CIRCUIT
DATE: 1/2/95
SHEET 2
W.D.H.
REV 1.3

THE AL4400 PWM PROCESSOR

Overview

The AL4400 provides all the control features required for a high specification pulse width modulated dimmer.

The AL4400 is suitable for a wide range of low voltage dimming applications. The device is optimised for control of RGB or CMY colour mixing devices such as LED (Light Emitting Diode), LEP (Light Emitting Polymer) or dichroic filtered lamps.

The AL4400 forms the core of the Artistic Licence Power-Pipe. Indeed, this document should be read in conjunction with the Power-Pipe User Guide.

The AL4400 provides four pulse width modulated outputs that are generally connected to MOSFET devices for high power switching.

Control can be either internal or external. In External mode, the DMX512 protocol is used to control each drive signal. Internal mode provides a range of pre-programmed effects.

Four rotary wheels provide the user interface. Please refer to the Power-Pipe User Guide for details of the user interface.

The AL4400 is supplied as a 40 pin DIL device.

Pin Functions

Pin	Name	Function	In/Out
1	VCC 5V	Power supply	-
2	AUX1	Not used – no connection	O
3	AUX2	Not used – no connection	O
4	TCL	Reserved for future version supporting the LM74 temperature sensor.	O
5	TADR	Reserved for future version supporting the LM74 temperature sensor.	O
6	AUX3	Not used – no connection	O
7	TDA	Reserved for future version supporting the LM74 temperature sensor. Pull to 5V via 3K3 resistor.	I/O
8	SCL	Reserved for future version supporting the 24LC256 flash drive.	O
9	SDA	Reserved for future version supporting the 24LC256 flash drive. Pull to 5V via 3K3 resistor.	I/O
10	TXENAB	Active high signal enables 75176 DMX512 transmit.	O
11	VCC 5V	Power supply	-
12	GND	Power supply	-
13	CLKIN	16 MHz Crystal	I
14	CLKOUT	16 MHz Crystal	O
15	RED	PWM Output	O
16	GREEN	PWM Output	O
17	BLUE	PWM Output	O
18	AMBER	PWM Output	O
19	MODE1		I
20	MODE2		I
21	MODE4		I
22	MODE8		I
23	POW	Active low system good LED drive.	O
24	RXENAB	Active low signal enables 75176 DMX512 receive.	O
25	DMXTX	DMX512 transmit	O
26	DMXRX	DMX512 receive	I
27	HUN1		I
28	HUN2		I
29	HUN4		I
30	HUN8		I
31	GND	Power supply	-
32	VCC 5V	Power supply	-
33	TEN1		I
34	TEN2		I
35	TEN4		I
36	TEN8		I
37	UNIT1		I
38	UNIT2		I
39	UNIT4		I
40	UNIT8		I

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

POW LED

The GOOD signal is driven low to show that the system is operating normally. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Power Supply

The AL4800 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

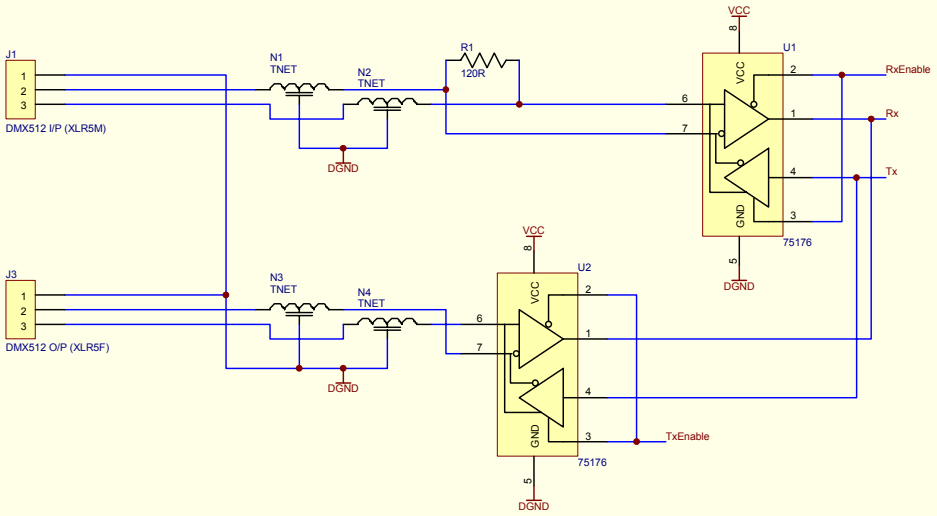
The AL4400 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4400.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4800 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

Application

The following circuit shows the AL4400 application drawing.

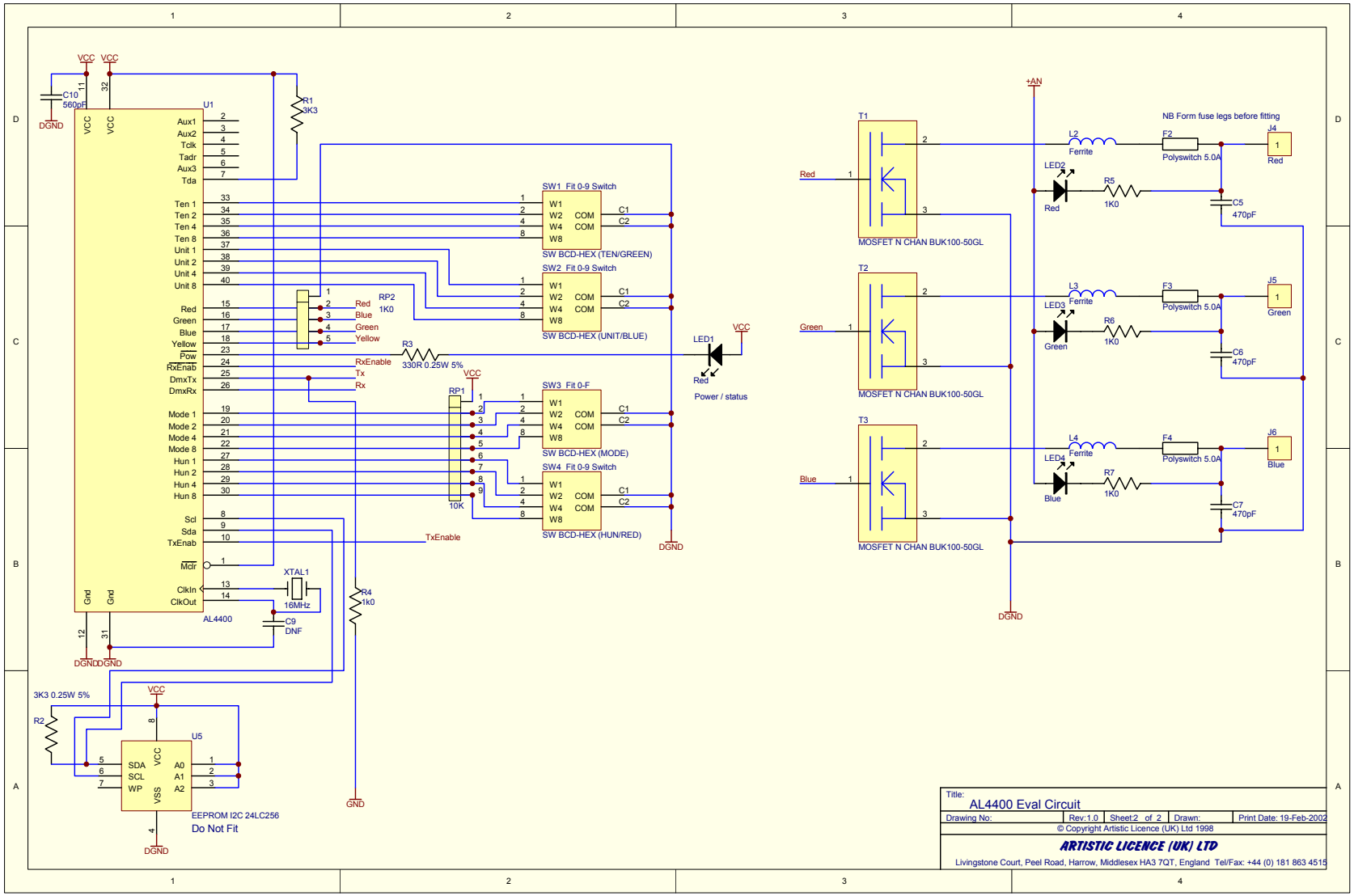


Title:
AL4400 Eval Circuit

Drawing No:	Rev:1.0	Sheet:1 of 2	Drawn:	Print Date: 19-Feb-2002

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Livingstone Court, Peel Road, Harrow, Middlesex HA3 7QT, England Tel/Fax: +44 (0) 181 863 4515


U1
AL4400
U5
EEPROM I2C 24LC256
Do Not Fit
SW1 Fit 0-9 Switch
SW BCD-HEX (TEN/GREEN)
SW2 Fit 0-9 Switch
SW BCD-HEX (UNIT/BLUE)
SW3 Fit 0-F
SW BCD-HEX (MODE)
SW4 Fit 0-9 Switch
SW BCD-HEX (HUN/RED)
T1
T2
T3
MOSFET N CHAN BUK100-50GL
NB Form fuse legs before fitting
Power / status
XTAL1
16MHz
Title:
AL4400 Eval Circuit
Drawing No:
Rev:1.0
Sheet:2 of 2
Drawn:
Print Date: 19-Feb-2002
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THE AL4800 MOVING LAMP PROCESSOR

Overview

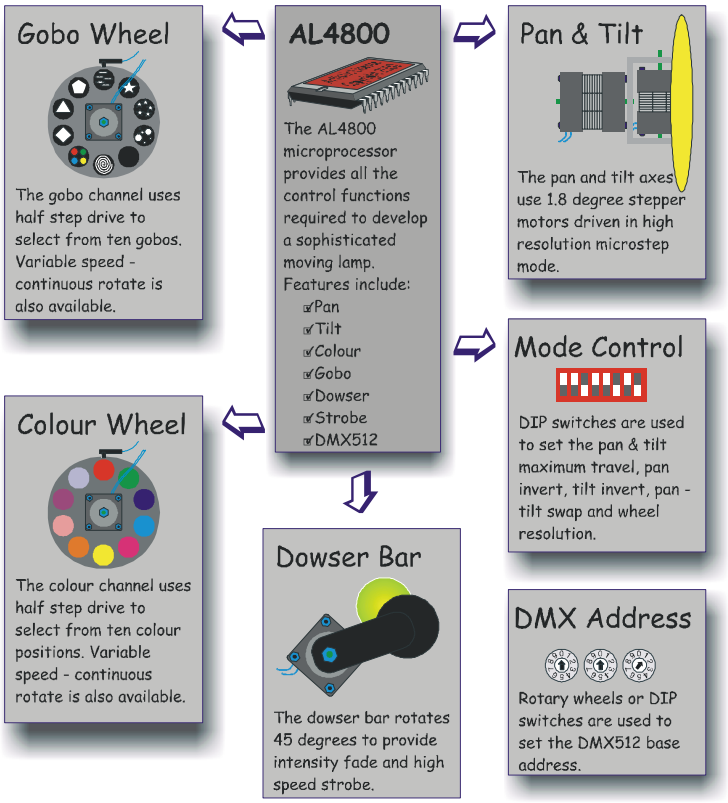
The AL4800 provides all the control features required for a medium specification moving lamp or moving yoke.

The basic configuration provides micro-stepping (high resolution) outputs for pan and tilt. Pin inputs allow 2 settings for pan motor type and 4 settings for tilt motor type.

The colour wheel and gobo wheel motors are also high resolution micro-stepping motors and assumed to be continuously rotating with inputs for a home sensor.

The dowser motor is also a wave drive stepper motor channel with end stop sensor.

The AL4800 is supplied as a 40 pin DIL device.



Pin Functions

Pin	Function	In/Out
1	VCC 5V	-
2	Pan Select 1	I
3	Pan Select 2	I
4	Tilt Select 1	I
5	Tilt Select 2	I
6	Good DMX	O
7	DMX Direction	O
8	BCD Hun 1	I
9	BCD Hun 2	I
10	BCD Hun 4	I
11	VCC 5V	-
12	GND	-
13	Clk In	I
14	Clk Out	O
15	DAC Wr	O
16	DAC A0	O
17	DAC A1	O
18	DAC LDAC	O
19	D0	I/O
20	D1	I/O
21	D2	I/O
22	D3	I/O
23	O/P1 Clk	O
24	O/P2 Clk	O
25	DMX Transmit	O
26	DMX Receive	I
27	D4	I/O
28	D5	I/O
29	D6	I/O
30	D7	I/O
31	GND	-
32	VCC 5V	-
33	Audio Trigger	I
34	Colour Index	I
35	Gobo Index	I
36	Mode	I
37	Pan Invert	I
38	Tilt Invert	I
39	Pan / Tilt Swap	I
40	IP Clk	O

Circuit Operation

The AL4800 uses two additional 8 bit latches (U8 & U9) and an 8 bit receiver (U6) in order to extend the effective pin count of the circuit.
Pin functions are described below.

Motors

The AL4800 is designed to use 1.8° per step motors. The reference motor is a 12V stepper motor: Japan Servo Co Ltd KH42HM2B013. (Farnell 586-389)

The motor control channels are defined as follows:

Motor	Function	Type	Drive	Mode
1	Pan	Bi-Polar	Micro-Step	Constrained
2	Tilt	Bi-Polar	Micro-Step	Constrained
3	Colour	Uni-Polar	Micro-Step	Continuous
4	Gobo	Uni-Polar	Micro-Step	Continuous
5	Dowser	Uni-Polar	Half-Step	Constrained

Sensors

The motor channels use the following sensing methods to detect end stops or zero positions:

Motor	Function	Sensor
1	Pan	Overdrive motor into end stop
2	Tilt	Overdrive motor into end stop
3	Colour	Hall Effect or Mechanical Switch
4	Gobo	Hall Effect or Mechanical Switch
5	Dowser	Overdrive motor into end stop

DMX512 Control

The five motors are position controlled based on the DMX512 input levels. Two operating modes are available and programmed via the Mode input. Mode 1 is selected with the Mode pin hi or floating. Mode 2 when the pin is connected to ground.

DMX Channel	**Name**	**Level**	**Mode 1**	**Mode 2**
1	Pan	0-255	Full range of motor as set by PanSel1/2. Full range is calibrated on power up by driving the motor from end to end.	
2	Tilt	0-255	Full range of motor as set by TiltSel1/2. Full range of motor. Full range is calibrated on power up by driving the motor from end to end.	

DMX Channel	Name	Level	Mode 1	Mode 2
3	Colour	0-12	Colour 1 (Open White)	Colour 1 (Open White)
		13-24		Forward spin fastest
		25-37	Colour 2	Forward spin fast
		38-49		Forward spin slow
		50-62	Colour 3	Forward spin slowest
		63-74		Reverse spin slowest
		75-89	Colour 4	Reverse spin slow
		80-99		Reverse spin fast
		100-112	Colour 5	Reverse spin faster
		113-119		Colour 1
		120-124		Colour 1 – 2
		125-129	Colour 6	Colour 2
		130-137		Colour 2 – 3
		138-143		Colour 3
		144-149		Colour 3 – 4
		150-156	Colour 7	Colour 4
		157-163		Colour 4 - 5
		164-169		Colour 5
		170-174		Colour 5 – 6
		175-180	Colour 8	Colour 6
		181-187		Colour 6 – 7
		188-194		Colour 7
		195-199		Colour 7 – 8
		200-206	Colour 9	Colour 8
		207-212		Colour 8 – 9
		213-219		Colour 9
		220-224		Colour 9 – 10
		225-230	Colour 10	Colour 10
		231-249		Colour 10 - 1
		250-255	Colour 1 (Open White)	Colour 1 (Open White)

DMX	Name	Level	Mode 1	Mode 2
4	Gobo	0-12	Gobo 1 (Open)	Gobo 1 (Open)
		13-24		Forward spin fastest
		25-37	Gobo 2	Forward spin fast
		38-49		Forward spin slow
		50-62	Gobo 3	Forward spin slowest
		63-74		Reverse spin slowest
		75-89	Gobo 4	Reverse spin slow
		80-99		Reverse spin fast
		100-112	Gobo 5	Reverse spin faster
		113-119		Gobo 1
		120-124		Gobo 1 – 2
		125-129	Gobo 6	Gobo 2
		130-137		Gobo 2 – 3
		138-143		Gobo 3
		144-149		Gobo 3 – 4
		150-156	Gobo 7	Gobo 4
		157-163		Gobo 4 - 5
		164-169		Gobo 5
		170-174		Gobo 5 – 6
		175-180	Gobo 8	Gobo 6
		181-187		Gobo 6 – 7
		188-194		Gobo 7
		195-199		Gobo 7 – 8
		200-206	Gobo 9	Gobo 8
		207-212		Gobo 8 – 9
		213-219		Gobo 9
		220-224		Gobo 9 – 10
		225-230	Gobo 10	Gobo 10
		231-249		Gobo 10 - 1
		250-255	Gobo 1 (Open)	Gobo 1 (Open)
5	Effects	Dimmer channel 6 must be above zero.		
		0-25	No Action	
		26-225	Slow to Fast Strobe	
		226-255	No Action	
6	Dimmer	0-255	Dimmer closed to open	

Redirect Control

A further three input pins alter the operation of the Pan and Tilt motors:

Pin	State	Function
Pan Invert	Off (+5V)	Pan operates normally.
	On (GND)	Pan movement relative to DMX channel level is inverted
Tilt Invert	Off (+5V)	Tilt operates normally.
	On (GND)	Tilt movement relative to DMX channel level is inverted
Pan / Tilt Swap	Off (+5V)	Pan and Tilt operate as described above.
	On (GND)	Pan and Tilt motors are swapped.

Pan Axis

The Pan motor is driven in high resolution micro-stepping mode. In order to cater for variations in final product design, four different ranges of total travel are available.
The travel settings are selected by PanSel1 & PanSel2. These pins are intended to be set at design time. They are not user controls.
The pan axis calibrates by driving the motor into a mechanical limit. It is important to ensure that the mechanical stop is set to coincide with one of the natural steps of the motor. This will ensure that the calibration phase provides maximum accuracy.

PanSel2	PanSel1	Travel
0	0	360 degrees
0	1	240 degrees
1	0	180 degrees
1	1	120 degrees

Tilt Axis

The Tilt motor is driven in high resolution micro-stepping mode. In order to cater for variations in final product design, four different ranges of total travel are available.
The travel settings are selected by TiltSel1 & Tilt2. These pins are intended to be set at design time. They are not user controls.
The tilt axis calibrates by driving the motor into a mechanical limit. It is important to ensure that the mechanical stop is set to coincide with one of the natural steps of the motor. This will ensure that the calibration phase provides maximum accuracy.

TiltSel2	TiltSel1	Travel
0	0	360 degrees
0	1	240 degrees
1	0	180 degrees
1	1	120 degrees

DAC Selection

The pan and tilt channels use a DAC to provide high resolution micro-stepping. The pin LDAC is driven to ground potential by the AL4800 allowing a number of compatible parts to be used. The AD7305 is the design reference part. The Maxim MAX506 can also be used.

Colour & Gobo

The colour and gobo channels use a sensor pin to provide a position reference to calibrate the wheel. The Colour (Gobo) Index pin should be pulsed low to indicate the reference position.
The reference position should preferably be generated by an opto slot sensor. A low resistance pull-up of approximately 1K0 is advised tin order to reduce the possibility of spurious triggering.
The following table shows the position of each wheel stop in both stepper counts and degrees.

Position	Count (0-200)	Angle (0-360)
Index	1	-7.2
Position 1	5	0
Position 1-2	15	18
Position 2	25	36
Position 2-3	35	54
Position 3	45	72
Position 3-4	55	90
Position 4	65	108
Position 4-5	75	126
Position 5	85	144
Position 5-6	95	162
Position 6	105	180
Position 6-7	115	198
Position 7	125	216
Position 7-8	135	234
Position 8	145	252
Position 8-9	155	270
Position 9	165	288
Position 9-10	175	306
Position 10	185	324
Position 10-1	195	342

Dowser

The dowser channel moves a total angle of 22.5°. A resolution of 50 steps is provided within this range.

Power On Calibration

At power on, the following events occur:

- Colour Wheel rotates until index position is found.
- Gobo Wheel rotates until index position is found.
- Dowser drives into closed position.
- Pan & Tilt drive into end stops.

- DMX Data accepted.

DMX512 Base Address

The BCD inputs are used to set the DMX base address. The inputs are set up for connection of three standard Binary Coded Decimal Thumbwheel switches. The BCD inputs are active low, i.e. to select a base address of 001, connect UNIT1 to ground and pull up all other BCD inputs. Valid addresses range from 001 to 512.

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485 or RS422 receiver such as the 75176.

Good Data

The GOOD signal is driven low to show that valid DMX512 data is being received. The output is suitable for direct connection (via a current limit resistor) to the cathode of a 20mA LED.

Power Supply

The AL4800 requires a 5V DC supply at approximately 6mA
All Ground and Vcc pins must be connected.
The device is CMOS so normal handling precautions should be adhered to.
Good decoupling and pcb layout conventions (i.e. fill all unused pcb areas with grounded copper planes) should be followed.
Particular attention should be paid to routing the crystal connections.

Crystal

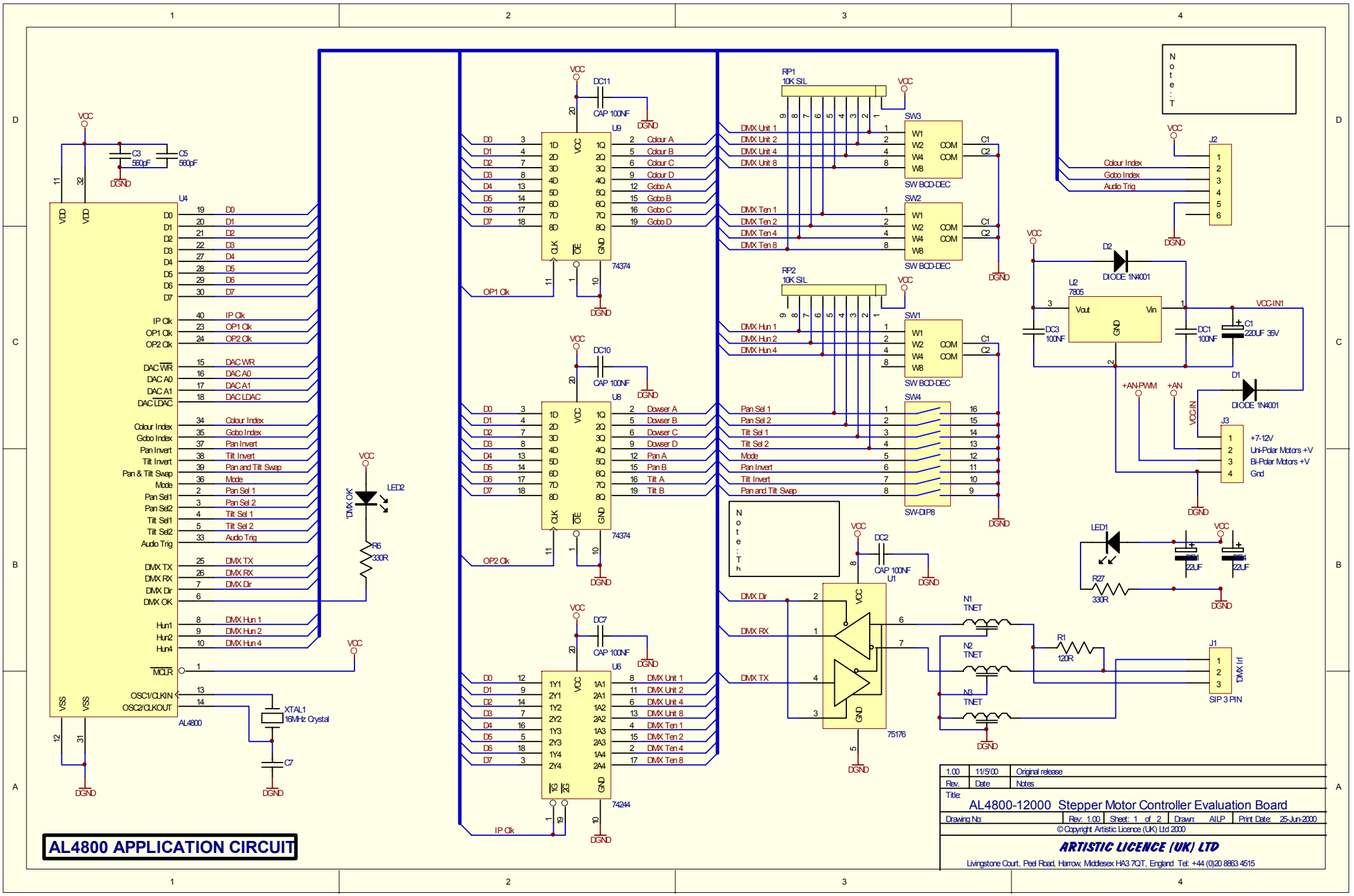
The AL4800 requires a 16MHz, fundamental mode crystal. A tuning capacitance of approximately 12pF is suggested between each crystal connection and Ground. The inherent pcb and lead capacitance must be included in this.

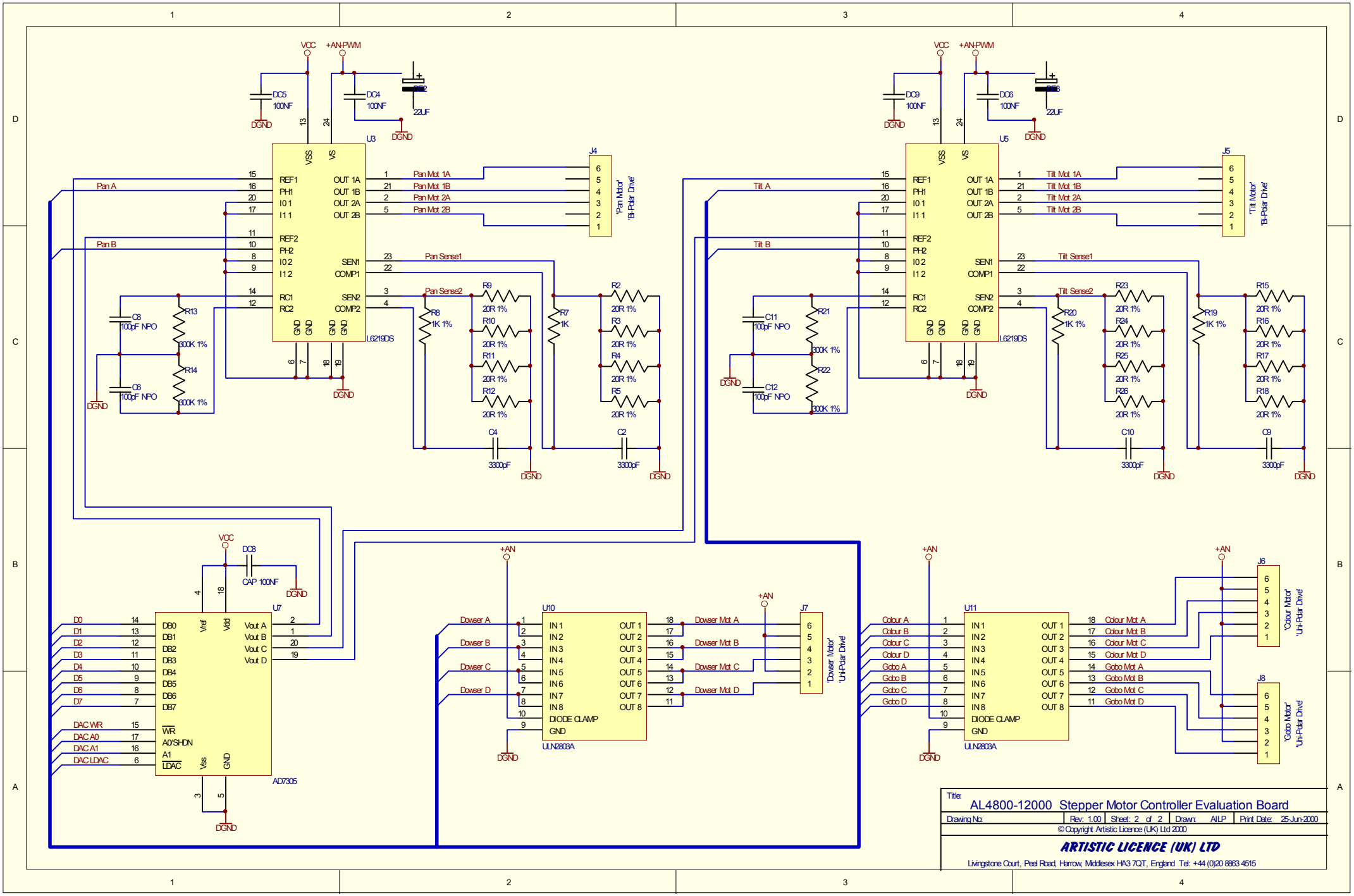
EMC

There are two major considerations in reducing the electromagnetic radiation generated by the AL4800.
The device tends to generate noise at the third harmonic of the crystal (48MHz). Adding the crystal capacitors detailed above reduces this to acceptable limits.
Broad band noise can be reduced by good decoupling. A range of 22uF aluminium capacitors and 0u1 ceramic capacitors should be used on the circuit board. A 560pF ceramic capacitor should also be connected across the AL4800 power pins as close as possible to the device.
The use of 150uH inductors in series with all three DMX512 signals is also an effective way to reduce conducted emissions.

Application

The following circuit shows the AL4800 application drawing. This is also the circuit diagram for the evaluation PCB.


AL4800 APPLICATION CIRCUIT
AL4800-12000 Stepper Motor Controller Evaluation Board
ARTISTIC LICENCE (UK) LTD
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1.00 11/5/00 Original release
Rev. Date Notes
Drawing No: Rev: 1.00 Sheet: 1 of 2 Drawn: AILP Print Date: 25-Jun-2000
AL4800
74374
74244
75176
SW BCD-DEC
SW-DIP8
SIP 3 PIN
DIODE 1N4001
7805
Colour Index
Gobo Index
Audio Trig
DMX Unit 1
DMX Unit 2
DMX Unit 4
DMX Unit 8
DMX Ten 1
DMX Ten 2
DMX Ten 4
DMX Ten 8
DMX Hun 1
DMX Hun 2
DMX Hun 4
Pan Sel 1
Pan Sel 2
Tilt Sel 1
Tilt Sel 2
Mode
Pan Invert
Tilt Invert
Pan and Tilt Swap
DMX Dir
DMX RX
DMX TX
DMX OK
OP1 Clk
OP2 Clk
IP Clk
+7-12V
Uni-Polar Motors +V
Bi-Polar Motors +V
Gnd



Title: AL4800-12000 Stepper Motor Controller Evaluation Board

Drawing No:	Rev: 1.00	Sheet: 2 of 2	Drawn: AILP	Print Date: 25-Jun-2000

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Bill Of Materials

The following components are required for the evaluation PCB. Please note that some component values may require modification subject to the intended application.

Designator	Part Type
C1	220UF 35V
C10	3300pF
C11	100pF NPO
C12	100pF NPO
C2	3300pF
C3	560pF
C4	3300pF
C5	560pF
C6	100pF NPO
C8	100pF NPO
C9	3300pF
D1	DIODE 1N4001
D2	DIODE 1N4001
DC1	100NF
DC10	CAP 100NF
DC11	CAP 100NF
DC2	CAP 100NF
DC3	100NF
DC4	100NF
DC5	100NF
DC6	100NF
DC7	CAP 100NF
DC8	CAP 100NF
DC9	100NF
DE1	22UF
DE2	22UF
DE3	22UF
DE4	22UF
J1	3 PIN
J2	SCREW 6PIN
J3	SCREW 4PIN
J4	SCREW 6PIN
J5	SCREW 6PIN
J6	SCREW 6PIN
J7	SCREW 6PIN
J8	SCREW 6PIN
LED1	LED 3MM
LED2	LED 3MM
N1	TNET 270PF
N2	TNET 270PF
N3	TNET 270PF

Designator	Part Type
R1	120R 1%
R10	20R 1%
R11	20R 1%
R12	20R 1%
R13	300K 1%
R14	300K 1%
R15	20R 1%
R16	20R 1%
R17	20R 1%
R18	20R 1%
R19	1K 1%
R2	20R 1%
R20	1K 1%
R21	300K 1%
R22	300K 1%
R23	20R 1%
R24	20R 1%
R25	20R 1%
R26	20R 1%
R27	330R
R3	20R 1%
R4	20R 1%
R5	20R 1%
R6	330R 1%
R7	1K 1%
R8	1K 1%
R9	20R 1%
RP1	10K SIL
RP2	10K SIL
SW1	BCD-DEC
SW2	BCD-DEC
SW3	BCD-DEC
SW4	DIP8
U1	75176
U10	ULN2803A
U11	ULN2803A
U2	7805
U3	L6219DS
U4	AL4800
U5	L6219DS
U6	74LS244
U7	AD7305
U8	74LS374
U9	74LS374
XTAL1	16MHz Crystal

THE AL2001/AL3001 DIGITAL DIMMER

Overview

The AL2001/AL3001 combination provide all the core functionality required to produce a sophisticated three phase digital dimming system.

Key Features include:

- LCD display and keypad
- DMX512 input
- Analogue input
- Three phase operation
- Good data received indicator
- 12 phase position modulated outputs
- Automatic mains frequency and phase tracking
- Anti-Thump turn on circuitry
- Four user selectable dimmer curves
- User programmable Pre-heat per channel

AL3001 Pin Functions

The AL3001 is supplied as a 40 pin plastic DIL package. All signal levels are TTL. The pin functions are as follows:

Pin	Function	In/Out
1	VCC	-
2	SDA (eeprom data)	I/O
3	SCLK (eeprom clock)	O
4	NC	I
5	NC	I
6	LCD D4 (lcd data)	I/O
7	LCD D5 (lcd data)	I/I
8	LCD D6 (lcd data)	I/O
9	LCD D7 (lcd data)	I/O
10	GND	-
11	SW5 (Enter)	I
12	SW4 (Down)	I
13	SW3 (Up)	I
14	SW2 (Right)	I
15	SW1 (Left)	I
16	LCDE (lcd E clock)	O
17	LCDWR (lcd write)	O
18	LCDRS (lcd register select)	O
19	Crystal Connection (OSC1)	I
20	Crystal Connection (OSC2)	I
21	NC	-
22	TTL DMX512 Input (RX)	I
23	Good Data, Lo=Good (GOOD)	O
24	FAN (drives low on hi-temp)	O
25	ACK (handshake to AL2001)	I/O
26	NC	-
27	TEST (connect to GND)	I
28	SYN (handshake to AL2001)	I/O
29	FRM (handshake to AL2001)	I/O
30	NC	-
31	GND	-
32	MCLR (connect to VCC)	I
33	T7 (data bus to AL2001)	I/O
34	T6 (data bus to AL2001)	I/O
35	T5 (data bus to AL2001)	I/O
36	T4 (data bus to AL2001)	I/O
37	T3 (data bus to AL2001)	I/O
38	T2 (data bus to AL2001)	I/O
39	T1 (data bus to AL2001)	I/O
40	T0 (data bus to AL2001)	I/O

AL2001 Pin Functions

The AL2001 is supplied as a 84 pin PLCC package. All signal levels are TTL. The pin functions are as follows:

Pin	Function	In/Out
1	NC	-
2	GND	-
3	NC	-
4	NC	-
5	ANAD0 (analogue mult address)	O
6	ANAD1 (analogue mult address)	O
7	ANAD2 (analogue mult address)	O
8	SIX (connect to VCC)	I
9	NC	-
10	NC	-
11	P86 (Phase fail detect)	I
12	RESET (active lo)	I
13	XTAL	I
14	EXTAL	O
15	MD1 (connect to VCC)	I
16	MD0 (connect to VCC)	I
17	REDZC2 (red zero cross)	I
18	STBY (connect to VCC)	I
19	VCC +5V	-
20	NC	-
21	NC	-
22	NC	-
23	GND	-
24	GND	-
25	T7 (transfer to AL3001)	I/O
26	NC	-
27	T5 (transfer to AL3001)	I/O
28	T4 (transfer to AL3001)	I/O
29	T3 (transfer to AL3001)	I/O
30	T2 (transfer to AL3001)	I/O
31	T1 (transfer to AL3001)	I/O
32	T0 (transfer to AL3001)	I/O
33	ACK (byte ack to AL3001)	O
34	NC	-
35	CYCLEI (PLL connect to CYCLEO)	I
36	REDZC1 (red zero cross input)	I
37	BLUZC (blue zero cross input)	I
38	YELZC (yellow zero cross input)	I
39	FRM (frame sync from AL3001)	I
40	SYN (byte sync from AL3001)	I

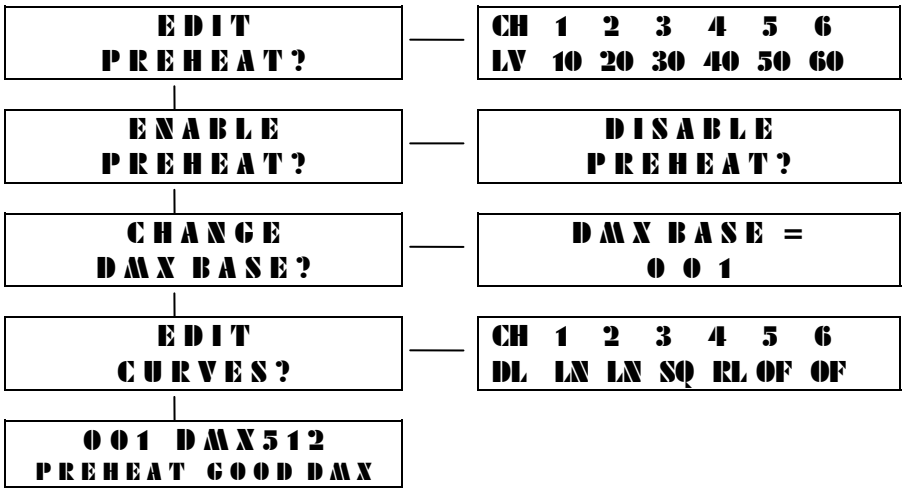
Pin	Function	In/Out
41	GND	-
42	AVCC (analogue +5V supply)	-
43	AN3 (connect to GND)	I
44	AN-BLU (analogue multiplex i/p)	I
45	AN-YEL (analogue multiplex i/p)	I
46	AN-RED (analogue multiplex i/p)	I
47	AN-TEMP (temperature sensor i/p)	I
48	AN5 (connect to GND	I
49	AN6 (connect to GND	I
50	AN7 (connect to GND	I
51	AVSS (analogue GND)	I
52	PHSEL1 (master zero cross select)	O
53	TRACKO (PLL connect to TRACKI)	O
54	PHSEL0 (master zero cross select)	O
55	TRACKI (PLL connect to TRACKO)	I
56	CYCLEO (PLL connect to CYCLEI)	O
57	REDZC0 (red zero cross input)	I
58	NC	-
59	NC	-
60	VCC +5V	-
61	NC	-
62	NC	-
63	NC	-
64	NC	-
65	NC	-
66	BLU3 (phase output)	O
67	BLU2 (phase output)	O
68	BLU1 (phase output)	O
69	BLU0 (phase output)	O
70	GND	-
71	NC	-
72	NC	-
73	NC	-
74	NC	-
75	RED3 (phase output)	O
76	RED2 (phase output)	O
77	RED1 (phase output)	O
78	RED0 (phase output)	O
79	YEL0 (phase output)	O
80	YEL1 (phase output)	O
81	YEL2 (phase output)	O
82	YEL3 (phase output)	O
83	NC	-
84	NC	-

System Operation

The AL2001 and AL3001 communicate via a bi-directional eleven bit parallel bus. The AL2001 contains all phase control timing circuits including the phase locked loops used to synchronise to mains frequency.
The AL2001 is also responsible for reading the analogue control voltage
inputs and the temperature sensor.
The AL3001 is the user interface processor, controlling the display, keyboard, DMX512 input and non-volatile data storage.

User Interface

The AL3001 controls the 2 by 20 character LCD and keypad to provide
the following menu structure:



Preheat

The Pre-heat levels are programmed by the user to define the minimum
allowed level for each channel. Preheat may be globally enabled or disabled. This data is stored in the eeprom.

Curves

The dimmer curves represent the rate of change of the dimmer output
with respect to both the DMX512 and analogue inputs. Four user programmable options are available, these are linear (LN), square (SQ), relay (RL) and off (OF). This data is stored in the eeprom.

DMX512 Input

The DMX512 input connects directly to the output of a standard RS485
or RS422 receiver such as the 75176. The base address of the received
block of twelve channels is set via the keypad and stored in eeprom.
The DMX512 data is merged with the analogue inputs on a highest takes
precedence basis.
The Good Rx indicator is an open collector output, suitable for connection
to a 20 mA LED. The output sinks current when DMX512 errors are detected.

Analogue Input

The Analogue inputs are routed through the AL2001 as shown in the application drawings. The DMX512 data is merged with the analogue inputs on a highest takes precedence basis. The circuitry must be calibrated such that the maximum voltage is 5V at the AN_xxx inputs.

Zero Cross

The AL2001 requires a zero cross for each phase. The circuit will operate on single phase in which case all zero cross inputs must be commoned. The Zero Cross input provides synchronisation to the mains
frequency. An active low pulse at the mains zero cross point (i.e. twice the mains frequency) is required. The pulse width should be in the order of 100uS. It is imperative that this input be a 'clean', fast rise time pulse. Any jitter at the Zero Cross input will produce flickering on the outputs.

The external 4539 multiplexor is used to ensure that the REDZC input
is always connected to a source of zero cross pulses. Should the red zero cross fail, the AL2001 will select one of the other phases to ensure continued operation.

Temperature Sensor

The temperature sensor is calibrated over a range of 5V. The following
bands are defined:

- 0V - 1.37V High temperature fault, causes output switch off and fan drive
- 1.56 - 2.15V High temperature warning, causes fan drive
- 2.2V - 5V Within normal tolerance.

The temperature input is disabled by connection to AVCC. The above voltages assume that AVCC=5V.

Errors

The AL3001 shows error conditions on the LCD screen and the GOOD Led.
The following conditions are displayed:

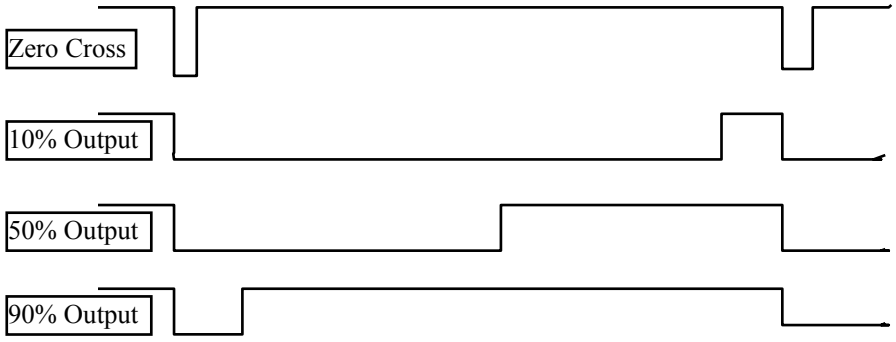
"GOOD DMX"	LED off		
"DMX ERRORS"	LED on		
"NO DMX"	LED off		
"HI TEMP"	LED on	FAN on	Outputs disabled
"TEMP > 70"	LED on	FAN on	

Power Supply

The AL2001/AL3001 requires a 5V power supply at a current of approximately
40mA. Good power supply decoupling is essential, and normal pcb layout
conventions should be followed (i.e. ground plane any and all empty areas of pcb).
All GND and VCC inputs must be connected.

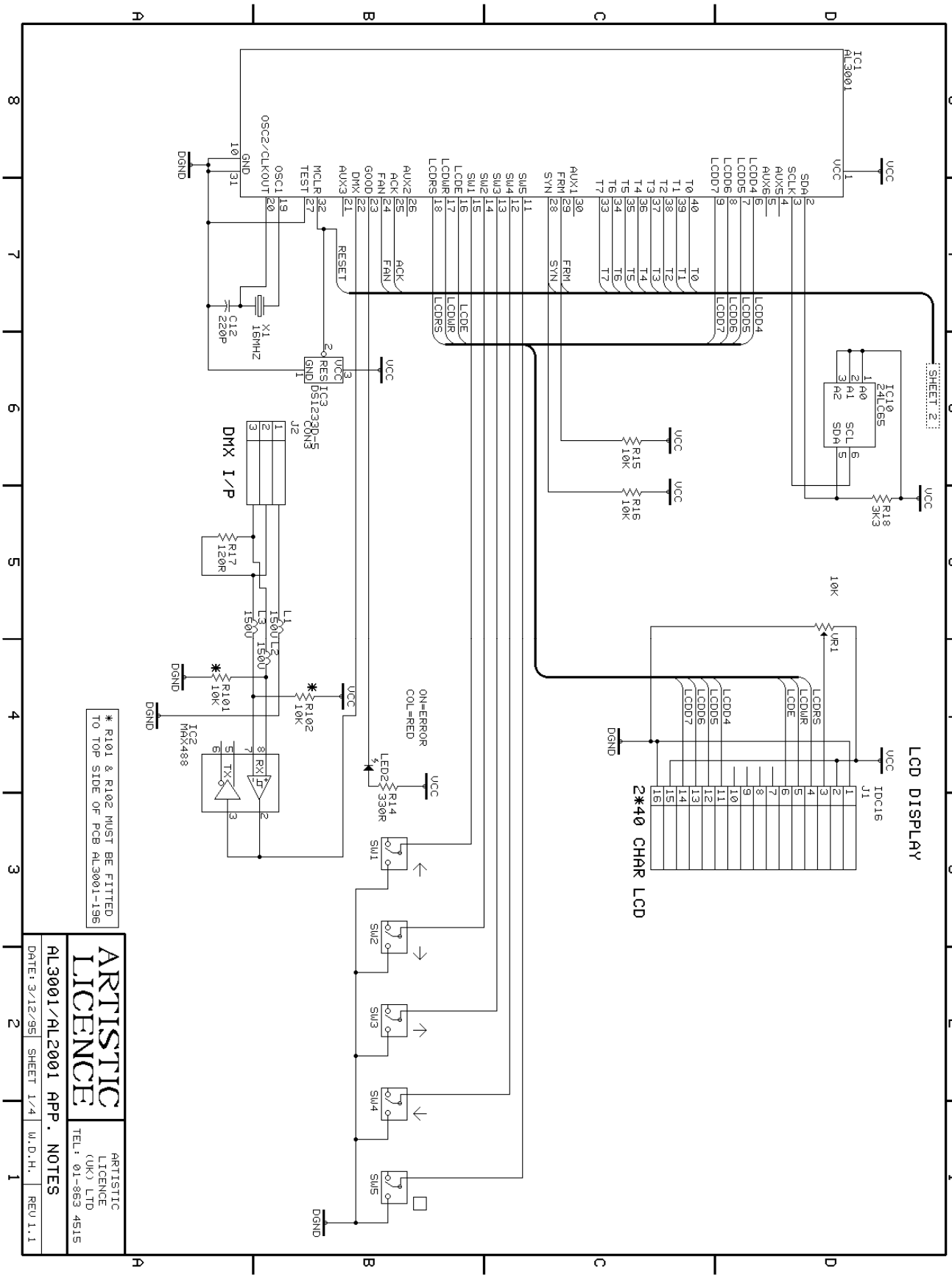
Output timing

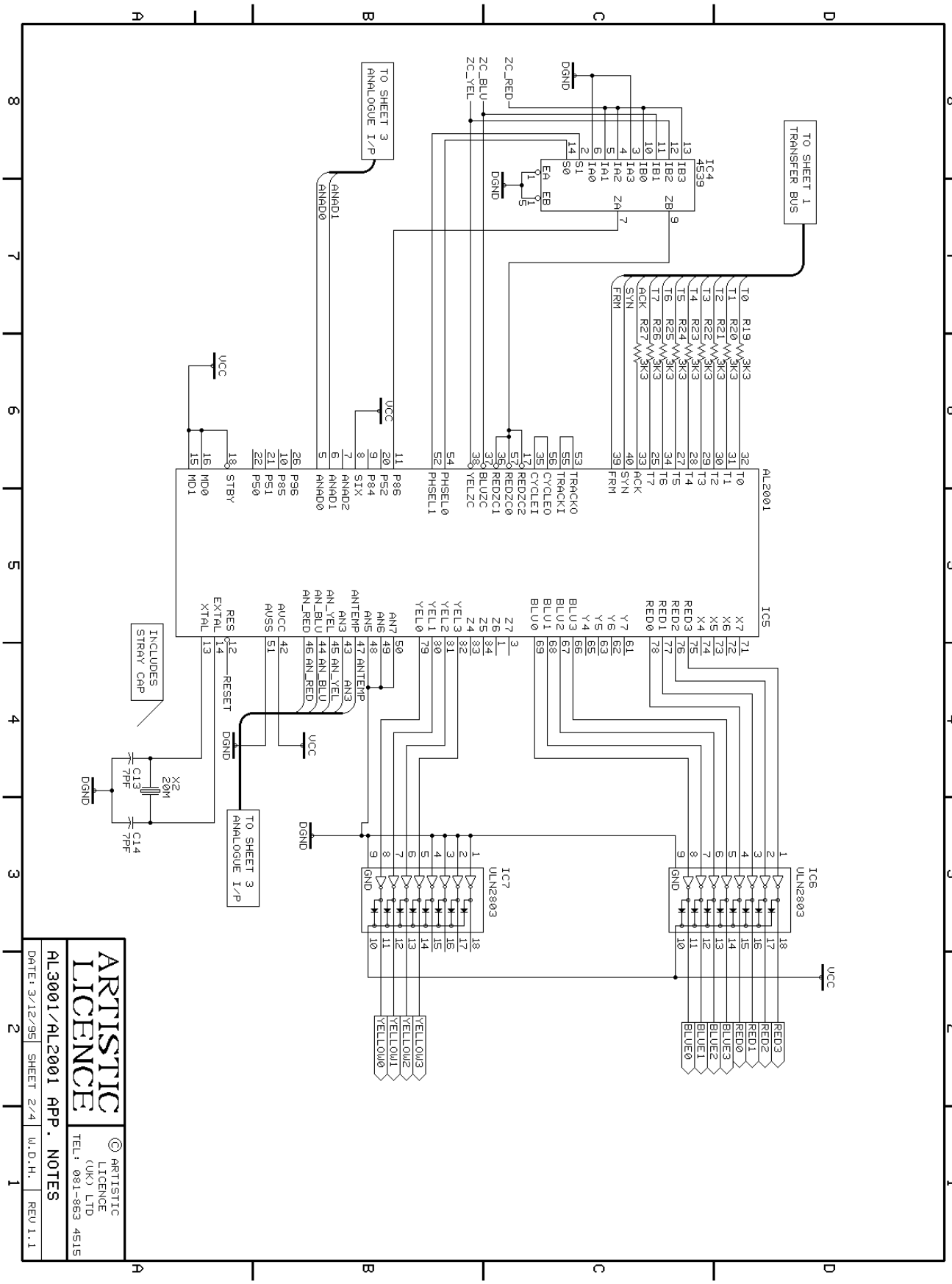
The following examples show the relationship between Output, Zero cross for three input values.

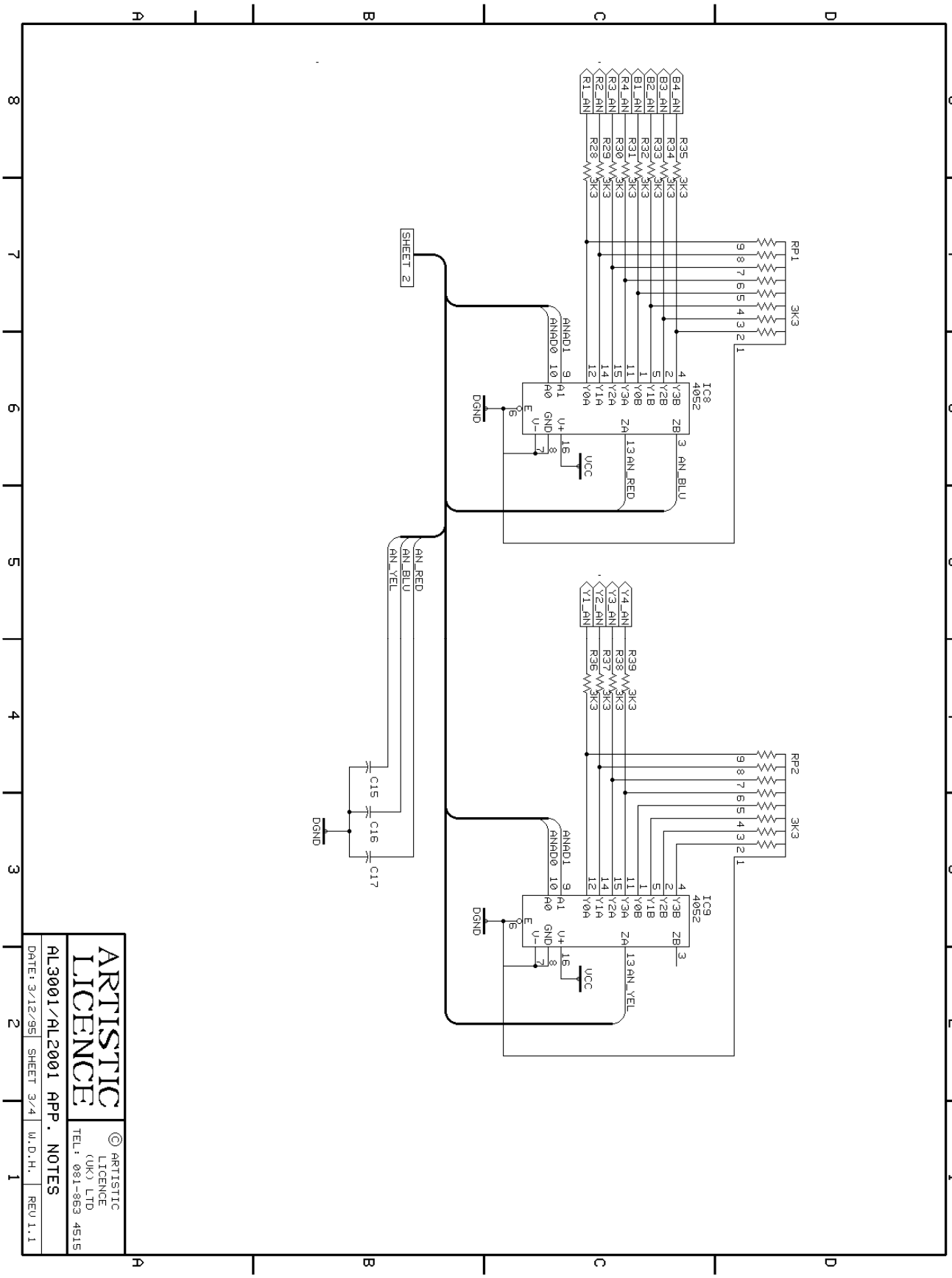


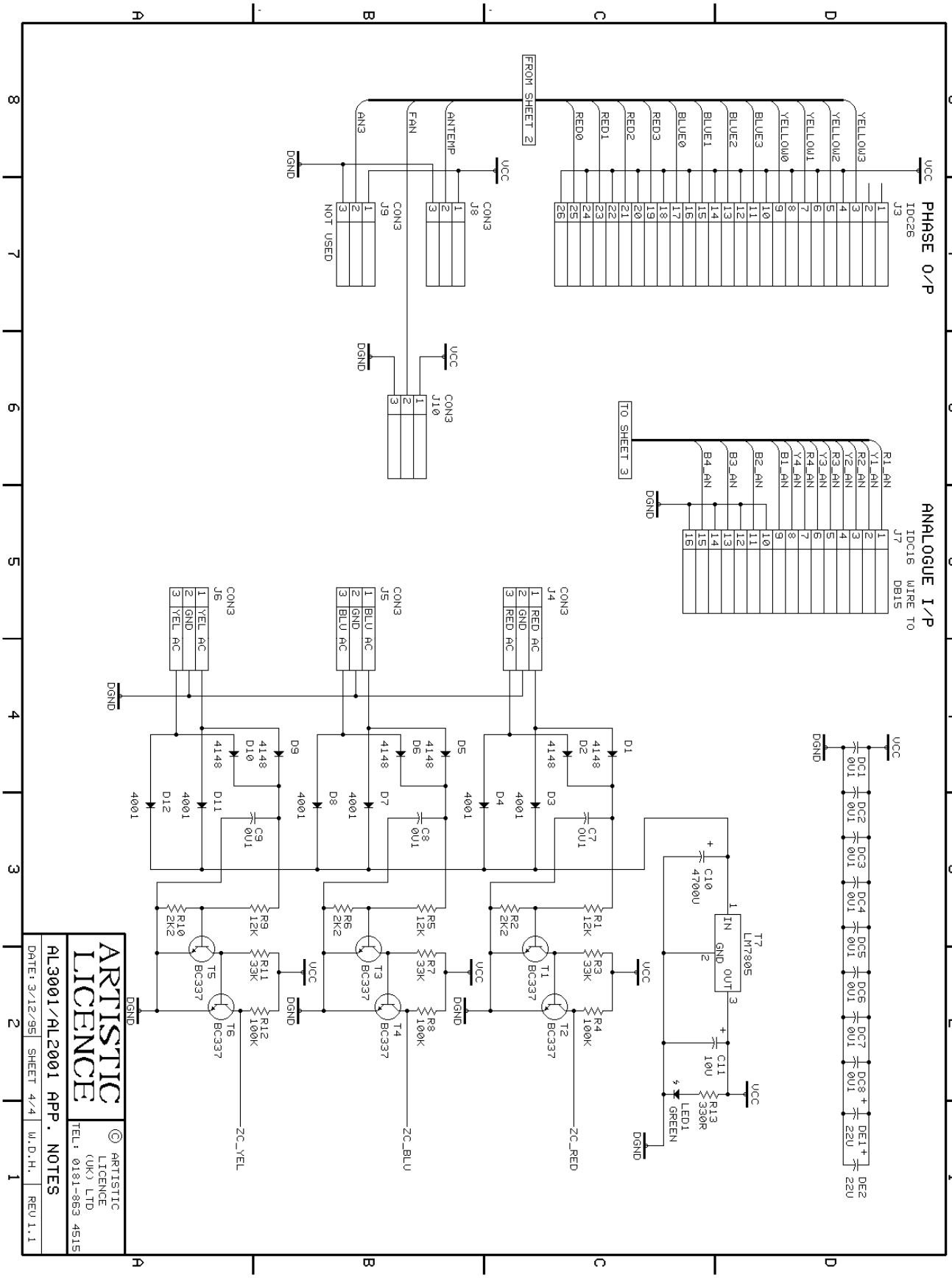
Application

The following circuit shows the AL4002 application drawing.


IC1
AL3001
SHEET 2
IC10
24LC65
R18
3K3
10K
VR1
LCD DISPLAY
2*40 CHAR LCD
IDC16
J1
DMX I/P
J2
IC3
DS1233D-5
R15
10K
R16
10K
X1
16MHZ
C12
220P
R17
120R
L1
150U
L2
150U
L3
150U
R101
10K
R102
10K
IC2
MAX488
LED2
R14
330R
ON=ERROR
COL=RED
SW1
SW2
SW3
SW4
SW5
* R101 & R102 MUST BE FITTED
TO TOP SIDE OF PCB AL3001-196
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LICENCE
(UK) LTD
TEL: 01-863 4515
AL3001/AL2001 APP. NOTES
DATE: 3/12/95
SHEET 1/4
W.D.H.
REV 1.1


TO SHEET 1
TRANSFER BUS
TO SHEET 3
ANALOGUE I/P
IC4
4539
AL2001
IC5
IC6
ULN2803
IC7
ULN2803
INCLUDES
STRAY CAP
X2
20M
C13
7PF
C14
7PF
RESET
VCC
DGND
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W.D.H.
REV 1.1


B4_AN
B3_AN
B2_AN
B1_AN
R4_AN
R3_AN
R2_AN
R1_AN
R35 3K3
R34 3K3
R33 3K3
R32 3K3
R31 3K3
R30 3K3
R29 3K3
R28 3K3
RP1
3K3
SHEET 2
ANAD1
ANAD0
IC8
4052
DGND
VCC
AN_BLU
AN_RED
Y4_AN
Y3_AN
Y2_AN
Y1_AN
R39 3K3
R38 3K3
R37 3K3
R36 3K3
RP2
3K3
IC9
4052
AN_YEL
C15
C16
C17
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W.D.H.
REV 1.1


PHASE O/P
IDC26
J3
FROM SHEET 2
YELLOW3
YELLOW2
YELLOW1
YELLOW0
BLUE3
BLUE2
BLUE1
BLUE0
RED3
RED2
RED1
RED0
VCC
DGND
AN3
FAN
ANTEMP
NOT USED
CON3
J9
J8
J10
ANALOGUE I/P
IDC16 WIRE TO DB15
J7
TO SHEET 3
R1_AN
Y1_AN
R2_AN
Y2_AN
R3_AN
Y3_AN
R4_AN
Y4_AN
B1_AN
B2_AN
B3_AN
B4_AN
CON3
J4
J5
J6
RED AC
GND
BLU AC
YEL AC
4148
4001
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
C7
C8
C9
0U1
C10
4700U
C11
10U
DC1
DC2
DC3
DC4
DC5
DC6
DC7
DC8
DE1
DE2
22U
T7
LM7805
IN
GND
OUT
LED1
GREEN
R13
330R
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
12K
2K2
33K
100K
T1
T2
T3
T4
T5
T6
BC337
ZC_RED
ZC_BLU
ZC_YEL
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DATE: 3/12/95
SHEET 4/4
W.D.H.
REV 1.1

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1. DEFINITIONS AND INTERPRETATION In these Conditions: a) "the Seller" means Artistic Licence (UK) Ltd.; b) "Order Acknowledgement" means the Seller's Order Acknowledgement; c) "the Buyer "means the person firm or company named as the Buyer in the Order Acknowledgement; d) "the Goods" means the Goods which are the subject of the Order Acknowledgement including, where the order so admits, a part of those Goods; e) the terms set out in the Order Acknowledgement and in these Conditions are together referred to in these Conditions as "the Contract".
2. Contract Terms a) These Conditions shall be deemed to be incorporated in all agreements from time to time entered into between the Seller and the Buyer which provide for the sale of any Goods by the Seller to the Buyer. b) These Conditions shall apply in place of and prevail over any terms or Conditions contained in or referred to in the Buyer's order or in correspondence or elsewhere or implied by trade custom, practice or course of dealing unless specifically agreed to in writing by the Seller and any purported provisions to the contrary are hereby excluded or extinguished. c) The Seller has drawn up the Seller's Conditions of Sale in the light of Unfair Contract Terms Act 1977 as amended and considers them to be fair and reasonable and its prices and insurance arrangements are based upon Contracts made on these Conditions. If the Buyer considers these terms unreasonable, it must inform the Seller in writing before any Contract is made - otherwise it will be deemed to have accepted that the Seller's Conditions are fair and reasonable. d) All orders placed by the Buyer by telephone shall be deemed to incorporate these Terms and Conditions. f) No alteration to the Contract or any of these Conditions shall be binding on the Seller unless agreed in writing.
3. SPECIFICATIONS AND INFORMATION Where the Seller is not the manufacturer the Seller relies on the specifications supplied by the manufacturer.
4. PRICES a) The Seller's catalogues, price lists and quotations do not constitute offers made by the Seller unless they are expressed to be fixed quotations remaining open for a specific period and the Seller reserves the right to withdraw or revise the same at any time prior to the Order Acknowledgement. b) Unless otherwise specified all prices are exclusive of: I) Value added Tax and any other United Kingdom tax or duty payable: ii) Custom and Excise duties, import or export duties and all other taxes, tariffs and surcharges of any nature at an time levied or imposed in any country or territory: iii) carriage, packing and insurance; and iv) any release certificates or certificates of conformity; c) The Seller reserves the right, by giving notice to the Buyer at any time before delivery, to increase the price of the Goods to reflect any increases in the cost to the Seller which is due to any factor beyond the control of the Seller, any change in delivery dates, quantities or specifications for the Goods which is requested by the Buyer, or any delay caused by any instructions of the Buyer or failure of the Buyer to give the Seller adequate information or instructions.
5. PAYMENT TERMS a) Except as otherwise specified in writing payment for the Goods shall be due in full, without deduction or set-off, within 14 days of the date of the Seller's invoice. b) In the event of any payment becoming overdue (and without prejudice to any other right the Seller may have) the Seller shall be entitled to charge interest at the rate of 3% per month over the base rate of the National Westminster Bank Plc on a daily basis to run from the due date for payment thereof until receipt by the Seller of the full amount, whether before or after judgement. c) If any matter referred to in Condition 12a has occurred, all monies unpaid by the Buyer to the Seller shall become immediately due and payable. d) The Seller reserves the right to suspend deliveries, to cancel any allowance of further credit or to require full or partial payment of the price of the Goods prior to delivery in the event of any payment not being made when due or if the Seller at its sole discretion at any time considers that the financial circumstances of the Buyer have ceased to justify the terms allowed. e) The Buyer shall not be entitled to any cash or other discount unless expressly agreed in writing. f) If the Seller is unable to deliver all the Goods ordered by the Buyer for reasons beyond the Seller's control (including, but not limited to those specified in Condition 15), the Buyer shall pay for such of the Goods as are delivered.
6. DELIVERY a) Any period or date for delivery stated in the Contract is intended as an estimate only and is not a contractual commitment and the Seller shall not be liable for any damage or losses arising out of the delay. b) Delivery of the Goods to the Buyer's address or any other place stipulated by the Buyer in writing shall constitute delivery. c) Where the Goods are to be delivered in instalments, each delivery shall constitute a separate Contract and failure by the Seller to deliver any one or more of the instalments shall not entitle the Buyer to treat the Contract as a whole as repudiated. d) Where orders are accepted by the Seller with delivery date(s) to be advised by the Buyer and the Buyer fails to confirm delivery requirements within 3 months or defers delivery the Seller reserves the right to treat the order as cancelled and make a cancellation charge in accordance with 9b below. e) If the Goods are lost or damaged in transit and before delivery the Seller will, at its sole discretion, refund the cost of, or repair or replace free of charge, the lost or damaged Goods.
7. PASSING OF PROPERTY AND RISK a) Property of the Goods sold shall remain in the Company until such time as ALL monies due from the Buyer to the Seller (upon whatever grounds and howsoever such liability shall have been incurred) have been paid in full. b) Notwithstanding the provisions in this of this Condition, the Seller shall be entitled to bring an action against the Buyer for the price of the Goods in the event of non-payment by the Buyer by the due date as if the property in the Goods had already passed to the Buyer. c) Until such time as the Property of the Goods passes to the Buyer (and provided the Goods are still in existence and have not been resold) the Seller shall be entitled at any time to require the Buyer to deliver up the Goods to the Seller and, if the Buyer fails to do so forthwith, to enter upon any premises of the Buyer where the Goods are stored with vehicles if necessary and repossess the Goods. e) The Buyer shall not be entitled to pledge or in any way charge by way of security for any indebtedness any of the Goods which remain the property of the Seller, but if the Buyer does so all monies owing by the Buyer to the Seller shall (without prejudice to any of the right or remedy of the Seller) become immediately due and payable. f) The Seller shall at its own risk take all necessary steps to protect the Seller's title to the Goods against claims by third parties. g) The risk in the Goods shall pass to the Buyer on delivery notwithstanding that the property may not have passed to it and the Buyer shall be responsible for insuring and seeing that the Goods are adequately packaged if returning them to the Seller for any reason. 8. Warranty And Acknowledgement a) A good title is warranted to the Goods sold by the Seller. b) i) the Seller guarantees at its' discretion either to refund the cost of, or replace free of charge any of the Goods found to its satisfaction to be defective owing to faulty design, materials or workmanship within six months of the date of delivery, provided that in the case of Goods or components not of the Seller's manufacture, its liability under this clause shall in no circumstances extend beyond any corresponding liability of the manufacture to the Seller of such Goods and components. ii) the Seller shall be under no liability under the above warranty (or any other warranty, Condition or guarantee) if the total price for the Goods has not been paid by the due date for payment. c) the Buyer relies on its own skill and judgement as to the sufficiency, capacity and performance of the Goods and as to their suitability for any purposes for which the Buyer may require them. d) The warranty in these Conditions are given in lieu of and replaces, exclude and extinguish all and every Condition, warranty, innominate term or representation whatsoever whether express or implied by statute, common law, trade usage, custom or otherwise in respect of the quality, fitness for purpose, merchantability, description of the Goods or otherwise.
9. Return Of Goods. a) The Buyer shall not return any of the Goods to the Seller or cancel any order without the Seller's written consent and the Seller shall be under no liability whatsoever for any Goods returned by the Buyer without such consent. b) The Seller reserves the right to make a cancellation charge for the cancellation of any order and/or the return of any Goods, which may include the cost of all loss (including loss of profit and losses incurred as the result of placing orders with third parties for supply of all or part of the Goods), costs (including the cost of all labour and materials used whether by the Seller or by a third party), damages, charges and expenses incurred by the Seller as a result of cancellation.
10.Limitation Of Damages a) Except as provided in these Conditions, the Seller shall be under no liability to the Buyer for any damages or losses, direct or indirect, resulting form defects in design, materials or workmanship or from any act or default of the Seller, whether negligent or otherwise. b) The Seller shall have no liability for any indirect or consequential losses or expenses suffered by the Buyer, however caused, including but not limited to loss of anticipated profits, goodwill, reputation, business receipts or Contracts, or losses or expenses resulting from third party claims. c) The Seller shall not be liable for the loss of, or damage to, any software programs occurring during or as the consequence of, the repair or upgrade of any of the Goods, whether under warranty or not. d) Subject to Condition 10e the Seller's aggregate liability to the Buyer whether for negligence, breach of Contract, misrepresentation or otherwise shall in no circumstances exceed the cost of the Goods which give rise to such liability in respect of any occurrence or series of occurrences. e) Nothing in these Conditions shall be interpreted as excluding or restricting any legal liability of the Seller for death or personal injury resulting from the negligence of the Seller, its employee, agents or sub-Contractors or restricting any other Seller's legal obligations arising under section 12 of the Supply of Goods and Services Act 1982 or the Consumer Protection Act 1987.
11. Termination a) This term applies if: i) The Buyer makes any voluntary arrangement with its creditors or becomes a subject to an administration order or (being an individual or firm) becomes bankrupt or (being a company) goes into liquidation (otherwise than for the purposes of amalgamation or reconstruction); or ii) an encumbrancer takes possession, or a receiver or an administrative receiver is appointed, of any of the property or assets of the Buyer; or iii) the Buyer ceases, or threatens to cease, to carry on business: or iv) the Buyer in breach of any term, Condition or provision of the Contract as required by law; or v) the Seller reasonably apprehends that any of the events mentioned above is about to occur in relation to the Buyer and notifies the Buyer accordingly. b) If this Condition applies then, without prejudice to any other right or remedy available to the Seller, the Seller shall be entitled to cancel the Contract or suspend any further deliveries under the Contract without any liability to the Buyer, and if the Goods have been delivered but not paid for price shall become immediately due and payable notwithstanding any previous agreement or agreement to the contrary.
12. Intellectual Property a) the Buyer acknowledges that all rights in respect of patents, copyrights, design rights, trade marks or other industrial or intellectual property rights connected with the Goods shall not pass to the Buyer. b) The Buyer shall indemnify the Seller against any and all liabilities, claims and costs incurred by or made against the Seller as a direct or indirect result of carrying out of any of the work required to be done or to the Goods in accordance with the requirements or specifications of the Buyer involving any infringement of any rights of any third party.
13. Software a) Where the Goods include software and the Buyer has been furnished with the developers licence, the Buyer shall sign and return it to the Seller within 7 days or as otherwise specified in the licence, and if the Buyer fails to sign and return the licence: i) the Seller reserves the right to withhold release of the software; and ii) the Seller shall nevertheless be entitled to payment in full for the software. b) In the absence of the developer's software licence being furnished, the Buyer shall accept a non-exclusive, non-transferable licence to use the software as provided in these Conditions and subject to the further provisions of Condition 13c and 13d. c) The Buyer shall not copy (other than for normal equipment operation), reproduce, translate, adapt, vary or modify the software or communicate any part of it to a third party without the prior written consent of the Seller. d) The licence granted by Condition 13b shall continue until or unless: i) either party gives to the other one month's prior written notice of termination, on or before the expiry of which the Buyer shall return or destroy the software as the Seller shall direct, which notice may only be given to the Seller if the continued use of or possession of the Software by the Buyer infringes the developer's third party rights or if the Seller is required to give notice by law; or ii) the Seller terminates the licence immediately if the Buyer fails, or has failed, to comply with any term or Condition of the Contract including (without limitation) breach of copyright, patent or confidentiality.
14. Force Majeure The Seller shall have no liability on respect of any failure or delay in fulfilling any of the Seller's obligations to the extent that fulfilment thereof if prevented, frustrated, impeded and/or delayed or rendered uneconomic as a consequence of any circumstance or event beyond the Seller's reasonable control.
15. Notices Any notice hereunder shall be deemed to have been given if delivered by hand or sent by prepaid first class post, telex or facsimile (confirmed by telephone and followed by notice by post) to the party concerned at its last known address, and deemed to have been received on the date of dispatch, if delivered by hand or sent by telex or facsimile, and on the third day after posting, if sent by post.
16. General a) The rights of the Seller shall not be prejudiced restricted by any indulgence or forbearance extended to the Buyer and no waiver of any breach shall operate as a waiver of any subsequent breach. b) If any of these Conditions or any part is held to be invalid for any purpose it shall for that purpose be deemed to have been omitted but shall not prejudice the effectiveness of the rest of these Conditions.
18. Proper Law The formation, construction and performance of the Contract shall be governed in all aspects by the Law of England and Wales and the parties hereto submit to the exclusive jurisdiction of the English courts.

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